

Weighted dilation graphs, smooth shifted primes and totient fibers

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Abstract

We prove Erdős's conjecture on the largest fibers of Euler's totient function: for every $\varepsilon > 0$, infinitely many positive integers n have more than $n^{1-\varepsilon}$ preimages. We also show that, for every fixed $\delta > 0$, there are at least $x^{1-o(1)}$ primes p in $2x < p \leq 5x$ whose predecessors have no prime factor exceeding x^δ .

1 Introduction

Euler's totient function can take the same value at many different integers. Write

$$g(n) = \#\{m \geq 1 : \varphi(m) = n\}$$

for the size of its fiber at n . We prove the following result.

Theorem 1.1. *For every real $\varepsilon > 0$, there are infinitely many positive integers n such that*

$$g(n) > n^{1-\varepsilon}.$$

[Theorem 1.1](#) resolves positively Erdős's conjecture on the largest fibers of Euler's totient function. Pomerance [19, p. 84] formulates the conjecture as $C = 1$, where C is the supremum of the exponents c for which $g(n) > n^c$ infinitely often, and attributes it to Erdős [7]. An elementary upper bound is $g(n) \ll_\eta n^{1+\eta}$ for every $\eta > 0$; we recall it in [Section 8](#). Thus the power exponent of n in the theorem is optimal.

The arithmetic input is a result about the prime factors of the predecessor of a prime. For $m > 1$, let $P^+(m)$ be its largest prime factor. Erdős's work on totient multiplicities and smooth shifted primes dates to [6, Section 3]. The underlying construction takes products of many primes whose predecessors have few available prime factors: many distinct products then have the same totient. The expectation that $p - 1$ can have all its prime factors smaller than every fixed power of p was recorded in [7]; see also [12, Introduction]. We prove a quantitative form.

Theorem 1.2. *For every fixed $0 < \delta < 1/4$, as $x \rightarrow \infty$,*

$$\#\{p : p \text{ prime}, 2x < p \leq 5x, P^+(p-1) \leq x^\delta\} \geq x^{1-o(1)}. \quad (1.1)$$

The quantity $o(1)$ may depend on δ .

In particular, for every $\varepsilon > 0$ there are infinitely many primes p with $P^+(p-1) \leq p^\varepsilon$. This resolves the smooth-predecessor conjecture positively as well. The count in [Equation \(1.1\)](#) also holds for every fixed $\delta > 0$, by using a smaller positive exponent when necessary. No uniformity as $\delta \rightarrow 0$ is asserted in [Theorem 1.2](#) or needed for [Theorem 1.1](#).

There is a substantial literature on fixed smoothness exponents. Baker and Harman [2, Section 1, Theorem 1 and Corollary 1] obtained exponent 0.2961 for shifted primes, with the corresponding

totient multiplicity exponent 0.7039. This improved Friedlander’s threshold $1/(2\sqrt{e}) + \varepsilon$; their introduction also describes preceding work of Pomerance, Balog, and Fouvry–Grupp. Lichtman [12, Theorem 1.1 and Corollary 1.3] proved a lower bound $x/(\log x)^C$ for $x < p \leq 2x$ and $P^+(p-1) \leq x^\beta$ whenever

$$\beta > \frac{15}{32\sqrt{e}} = 0.284311\dots,$$

and obtained multiplicity exponent 0.7156. These results concern fixed positive exponents. The expected Dickman law predicts a positive proportion of primes for every such exponent; see Granville [10, Section 5.3]. Bharadwaj and Rodgers [3, Theorem 7, Proposition 2, and Conjecture 5] prove the full Poisson–Dirichlet law for sequences satisfying their regularity and congruence-uniformity conditions together with level-one distribution. For shifted primes this gives the prediction under the Elliott–Halberstam conjecture; their Proposition 2 establishes distribution level one half and the other conditions unconditionally. Our unconditional bound $x^{1-o(1)}$ gives the full power exponent of the count for every fixed positive smoothness exponent. A positive limiting proportion is a stronger conclusion. The final passage from plentiful smooth shifted primes to large totient fibers belongs to the method of Erdős and Pomerance; Pomerance states the relevant transfer explicitly in [19, Theorem B]. We give the needed product-and-pigeonhole proof in full.

Smooth predecessors also enter the construction of Carmichael numbers by Alford, Granville and Pomerance [1], where their supply is combined with a separate distribution theorem for primes in arithmetic progressions. This is one reason that estimates for shifted smoothness have applications beyond the factorization problem itself. The contribution here includes the general graph and kernel theorems that produce the arithmetic lower bound, as well as the optimal exponent for totient fibers.

These graph and ideal-kernel estimates also serve as inputs to the companion paper [17, Theorem 1.1], where a separate determinant-graph estimate and prime-extraction argument yield the full Poisson–Dirichlet law for prime predecessors.

1.1 The analytic mechanism

The main obstacle is to detect primality in a sequence whose predecessors have a strongly constrained factorization. Congruence counts provide an initial sieve, and a further bilinear cancellation estimate permits the composite contribution to be removed. This division of work is familiar from prime-detecting sieves; compare Friedlander and Iwaniec [9, Section 1]. Our coefficient and range hypotheses are stated and verified below for the particular weight used here.

We count primes of the form $2u+1$, where u has a prescribed factorization over many logarithmic scales. Some factors lie in two separated ranges of subpower-sized primes, divided into small and big groups. Their weights mark distinct prime divisors. The functions attached to an integer depend only on the part of u outside these groups; the marked weights supply the divisibility labels for the dilation graph. Other factors occupy a geometric sequence of size bands, which permits a divisor of u to be chosen close to a specified size. One band contains a long factor ranging over rough integers, meaning integers with no prime factor below a prescribed cutoff.

The main new analytic ingredient is a transference theorem for weighted dilation graphs. A physical state consists of an integer and ordered lists of distinct group-prime divisors. An edge has shift kD for an integer $k \neq 0$, where D is a product of primes shared by its endpoint lists, together with independent free prime factors. The corresponding ideal operator acts on the big-group lists: its labels are independent, with probabilities proportional to $1/p$, and repetitions are allowed. The list length is sufficiently large but fixed as x tends to infinity. A small ideal operator norm yields an

averaged high moment of the physical graph, and hence cancellation in pairings whose first endpoint is independent of its mark list.

The transference proof averages the integer root by the Chinese remainder theorem. Repeated divisibility queries then pay a reciprocal prime factor only once; a memory retains their congruences between active uses. Averaging the omitted small marks makes transfers between memory and the active lists rare. A rank expansion restores distinct prime births while preserving cancellation on most edges: many independent birth equalities supply reciprocal-prime savings, whereas a small equality rank affects only a small fraction of the edges. The theorem applies to arbitrary signed kernels satisfying its explicit complexity and norm hypotheses.

Related divisibility-graph methods have a substantial history. Matomäki, Radziwiłł and Tao [14, Theorem 5.1] studied connectivity of a graph with prime-divisibility edges. Tao’s entropy-decrement argument [23] replaces such divisibility conditions by their mean at a suitable scale. Helfgott and Radziwiłł [11, Section 1.5] study centered adjacency operators through signed closed-walk moments, with repeated primes creating dependent congruences. Pila [18, Theorem 2.4 and Section 5] develops this approach for shifts that are products of primes. These works explain the roles of centering, long moments, and repeated-label bookkeeping. Our graph also carries ordered active lists, and its comparison with an independent-label operator requires the explicit lifespan and factorial-memory construction proved in Section 3.

The independent-label norm is made small by a comparison construction. Logarithmic localization and character kernels allow shared prime products to be replaced by independent products. The big groups are split into two blocks, with a bounded number of selected coordinates in each group. After averaging the list coordinates unused by the multiplier, a coordinate decomposition isolates components that have mean zero in all selected coordinates of one block. Permutation symmetry makes these components small, and signed comparison terms cancel the other components up to a controlled error. For each prescribed fixed accuracy, the construction gives kernels uniform in the additive frequency, with complexity fixed before the length of the marked lists is chosen.

Applying transference to the multiplicatively invariant endpoints reduces a fixed shifted correlation to averages over larger shifts. Products of free primes control the minor arcs. On the major arcs, a marked small prime and the compulsory long rough-integer factor control the local Fourier energy. A low-complexity character and Mellin discrepancy condition supplies the remaining cancellation. The argument permits arbitrary bounded residual coefficients.

These shifted correlations yield a Type II estimate for $mn = 2u + 1$. A probabilistic split of the geometric prime bands chooses a divisor e of u close to the scale of m . After Cauchy–Schwarz, off-diagonal factorizations become shifted endpoints by an exact determinant identity. The Type II estimate then permits primality and roughness tests on cofactors to be replaced by elementary rough proxies. Congruence estimates and a weighted sieve remove the composite values of $2u + 1$; a separate upper bound treats the small balanced range.

1.2 Organization and dependencies

Section 2 fixes the conventions and proves the elementary sieve used later. The central transference estimate is proved in Section 3, and the independent-label kernels are constructed in Section 4. Together they lead to the shifted-correlation theorem in Section 5. The Type II reduction occupies Section 6. The prime extraction and all its parameter choices are completed in Section 7. Section 8 finishes the proof of Theorem 1.1. The main implications are summarized in Figure 1.

The sieve geometry is fixed first, the required analytic accuracies next, and the discrepancy precision of the cofactor proxies last. The relevant theorems state this choice order, and the prime extraction verifies it. In particular, the needed discrepancy is established independently of the

estimate that uses it.

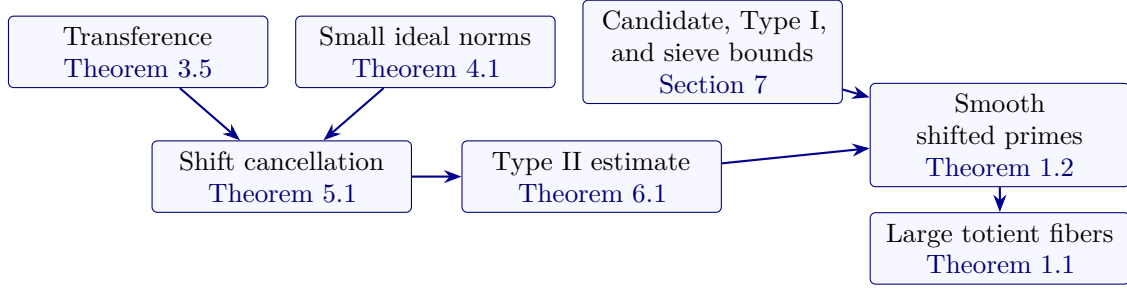


Figure 1: Main result dependencies. The auxiliary branch includes candidate mass, Type I distribution, proxy discrepancy, the elementary sieve, and the separate balanced-range bound. Arrows record the implications used in the proof; the order of choosing constants is described in the text.

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2 Notation and preliminary estimates

All factor variables are positive integers unless another domain is specified. We write $P^+(n)$ and $P^-(n)$ for the largest and least prime factors of $n > 1$, with $P^+(1) = 1$ and $P^-(1) = \infty$. An integer is y -smooth if $P^+(n) \leq y$, and is rough above y if $P^-(n) > y$. We use φ for Euler's function, τ for the divisor function, and

$$e(t) = \exp(2\pi it), \quad (z)_j = z(z-1) \cdots (z-j+1), \quad (z)_0 = 1.$$

We write $\mu(n) = 0$ if n is divisible by a prime square, and $\mu(n) = (-1)^r$ if n is a product of r distinct primes. The symbol ω will count distinct prime divisors in the specified prime groups; an unrestricted distinct-prime count will be identified when it occurs. In particular, multiplicities in an integer do not increase its group count.

Throughout the analytic argument,

$$L = \log x, \quad T = \log L, \quad W = \exp(\sqrt{L}), \quad (2.1)$$

and x tends to infinity. A dyad is an interval $[Y, 2Y)$, or a fixed constant enlargement of one; subinterval restrictions will be allowed explicitly. We write $n \asymp Y$ when n is bounded above and below by fixed positive multiples of Y .

Convention 2.1 (Parameters and uniformity). Every constant is fixed before $x \rightarrow \infty$. A bound $L^{O(1)}$ has an exponent independent of x . Its dependence on earlier fixed parameters is permitted unless explicitly excluded. For a two-sided size statement $Y = xL^{O(1)}$, we mean $|\log(Y/x)| = O(T)$. Arbitrary logarithmic accuracy means a bound $O_A(L^{-A})$ for every desired fixed A , with the auxiliary choices made in the specified order.

The geometric parameters used in the final sieve application are chosen first. The required Type II accuracy is chosen next. Inside the analytic argument the physical and ideal norm targets precede the comparison kernels; the mark length J is chosen after those kernels. The strength of the character and Mellin discrepancy is chosen after the graph and Fourier parameters, and the precision of the cofactor proxies is chosen last. Each result below specifies the uniformity needed to respect this order.

All Hilbert spaces are complex. An operator defined by transitions acts on a function at the target and sums or integrates its weighted values at the input. Symmetrization of a list means the orthogonal projection that averages all permutations of its coordinates.

A guide to recurring notation. The following table locates the objects used across sections. Their full definitions and hypotheses appear at the indicated references.

Notation	Role	Definition
L, T, W	Logarithmic scales and the roughness cutoff.	Equation (2.1)
$\mathcal{P}_g, V_g, \mu_g$	Prime groups, harmonic masses, and probability laws; the number of groups is $s \asymp T$.	Theorem 3.1
q, ω_g, ω	Fixed damping parameter and distinct group-prime counts.	Theorem 3.1
$J, \ell, M = J + \ell$	Fixed slot counts; J is chosen after the comparison kernels.	Section 3
m , later m_{probe}	Fixed bound on probes per big group, independent of J .	Theorem 4.1
$W_{\mathbf{t}}, W_\ell$	Marked divisor weights; in W_ℓ every group has ℓ marks.	Equation (5.1)
n_*, F_0, G_0	Group-free part and endpoint cores invariant under multiplication by group primes.	Section 5 Equation (5.3)

2.1 Classical prime estimates

We use the prime number theorem with an error smaller than any fixed negative power of the logarithm [22, Corollary 39 and Exercise 40], and Mertens' estimates [21, Theorems 15 and 26]

$$\sum_{p \leq y} \frac{1}{p} = \log \log y + O(1), \quad \prod_{p \leq y} \left(1 - \frac{1}{p}\right) \sim \frac{e^{-\gamma_E}}{\log y}. \quad (2.2)$$

Here and below sums or products indexed by p are over primes. The following forms of Siegel–Walfisz and the multiplicative large sieve will be used. The former follows from its von Mangoldt formulation by partial summation; see [22, Exercise 64]. For the latter see [15, Theorem 19.16].

Theorem 2.2 (Siegel–Walfisz). *For fixed $A, C > 0$, uniformly for $(a, r) = 1$ and $1 \leq r \leq (\log y)^C$,*

$$\#\{p \leq y : p \equiv a \pmod{r}\} = \frac{\text{Li}(y)}{\varphi(r)} + O_{A,C}\left(\frac{y}{(\log y)^A}\right).$$

The constants, which need not be effective, are independent of y, a, r .

Theorem 2.3 (Multiplicative large sieve). *Let I be a real interval of length H , let a_n be complex numbers supported on $I \cap \mathbb{Z}$, and let $R \geq 1$. Then*

$$\sum_{r \leq R} \frac{r}{\varphi(r)} \sum_{\chi \bmod r}^* \left| \sum_{n \in I \cap \mathbb{Z}} a_n \chi(n) \right|^2 \ll (H + 1 + R^2) \sum_n |a_n|^2,$$

where the asterisk restricts the sum to primitive Dirichlet characters.

We will use absolute character estimates on intervals that may be very short. The next consequence records precisely what is required; it does not require a relative prime asymptotic on every such interval.

Corollary 2.4 (Harmonic character estimates). *Fix $c, C, B, A > 0$. If χ is nonprincipal modulo $r \leq L^B$, then, uniformly for real $|t| \leq L^B$ and intervals $I \subset [\exp(L^c), x^C]$,*

$$\sum_{p \in I} \frac{\chi(p) p^{it}}{p} \ll_{c,C,B,A} L^{-A}.$$

The same conclusion holds for a nonprincipal character induced from a smaller modulus.

Proof. Writing $A_\chi(y) = \sum_{p \leq y} \chi(p)$, sum Theorem 2.2 against χ over reduced residue classes. The main terms cancel, and the loss from the number of classes is at most L^B . Because $\log y \geq L^c$, the permitted moduli are bounded by a fixed power of $\log y$. Thus $A_\chi(y) \ll_K L^B y (\log y)^{-K}$ for every fixed K . Partial summation on $I = (a, b]$ against y^{-1+it} gives boundary terms of size $O_K(L^B (\log a)^{-K})$ and an integral bounded by

$$C_K L^B (1 + |t|) \int_a^b \frac{dy}{y (\log y)^K} \ll_K L^{2B} (\log a)^{1-K}.$$

Choose K sufficiently large. The argument is uniform in both endpoints and also covers an empty interval. An induced nonprincipal character is itself nonprincipal on the modulus on which it is used, so the same argument applies. $\square$

2.2 Divisors, Dirichlet polynomials, and smooth separation

Lemma 2.5 (Fixed divisor moments). *For each fixed nonnegative integer k ,*

$$\sum_{n \leq Y} \frac{\tau(n)^k}{n} \ll_k (\log(2Y))^{C_k}, \quad \sum_{n \leq Y} \tau(n)^k \ll_k Y (\log(2Y))^{C_k}$$

for a constant C_k . Consequently, a convolution of a fixed number of sequences bounded by fixed logarithmic powers has coefficient-square sum $UL^{O(1)}$ on a dyad of size $U \leq x^{O(1)}$. If its coefficients include the reciprocal of the product index, this bound is $L^{O(1)}/U$ instead.

Proof. Positivity and unique factorization bound the harmonic sum by

$$\prod_{p \leq Y} \sum_{a \geq 0} \frac{(a+1)^k}{p^a} = \prod_{p \leq Y} \left(1 + \frac{2^k}{p} + O_k(p^{-2}) \right) \ll_k (\log(2Y))^{C_k},$$

using Equation (2.2). The counting bound follows by multiplying by Y . If there are b convolution factors, their coefficient at n has absolute value at most a fixed logarithmic power times the number of ordered b -factorizations of n . The latter is at most $\tau(n)^{b-1}$: choose the first $b-1$ divisors, which determine the last factor. Apply the counting moment bound with $k = 2(b-1)$. On a dyad the reciprocal index is comparable to $1/U$. $\square$

The next coefficient-independent estimate is a weaker elementary form of the Dirichlet-polynomial mean-value theorem; compare [16, Theorem 2 and Corollary 3].

Lemma 2.6 (Mean squares). *If $P(t) = \sum_{n \leq Y} c_n n^{it}$ and I is an interval of length H , then*

$$\int_I |P(t)|^2 dt \ll (H + Y \log(2Y)) \sum_n |c_n|^2.$$

For a set $\mathcal{T}$ of real points at mutual distance at least one, contained in an interval of length H , one also has

$$\sum_{t \in \mathcal{T}} |P(t)|^2 \ll (H + 1 + Y \log(2Y)) (\log(2Y))^2 \sum_n |c_n|^2. \quad (2.3)$$

The constants are absolute, independently of any factorization used to form the coefficients c_n .

Proof. Integration gives the diagonal $H \sum |c_n|^2$. For $n \neq m$, the absolute value of the exponential integral is at most $2/|\log(n/m)| \ll Y/|n-m|$. Apply $2|c_n c_m| \leq |c_n|^2 + |c_m|^2$ and sum $1/|n-m|$ to obtain the first bound. On the unit interval centered at t , the elementary one-dimensional Sobolev inequality bounds $|P(t)|^2$ by a constant times the integral of $|P|^2 + |P'|^2$. These unit intervals have bounded overlap. The derivative has coefficients $i(\log n)c_n$, so the first bound applied twice gives Equation (2.3). $\square$

We also use two elementary exponential-sum estimates in the following forms; see [15, Corollary 16.6 and Theorem 16.7].

Lemma 2.7 (Derivative tests). *Let f be real-valued on an interval containing H consecutive integers.*

- (i) *If f' is monotone and stays in $[j + \lambda, j + 1 - \lambda]$ for some integer j and $0 < \lambda \leq 1/2$, then $\sum_n e(f(n)) \ll \lambda^{-1}$.*
- (ii) *If f is twice continuously differentiable and $\lambda \leq |f''| \leq C\lambda$ throughout the interval, then $\sum_n e(f(n)) \ll_C H\lambda^{1/2} + \lambda^{-1/2}$.*

Both estimates hold on every subinterval on which the stated hypotheses hold.

Lemma 2.8 (Fourier separation). *Let F be smooth, supported in a fixed bounded box in $\mathbb{R}^b$, where b is fixed. Suppose that for every nonnegative integer j its derivatives of order j are bounded by $C_j L^{C(j+1)}$. Then Fourier inversion separates the variables with integrated absolute coefficient mass $L^{O(1)}$. There is a fixed $C' > 0$, depending on b, C , such that restricting each Fourier frequency to absolute value at most $L^{C'}$ leaves an error $O_A(L^{-A})$ for every fixed A . The exponent C' may be fixed before the desired A .*

Proof. Repeated integration by parts gives a bound for $\widehat{F}(\xi)$ by a derivative norm times an arbitrary fixed negative power of $1 + |\xi|$. Taking more than b derivatives first makes this bound integrable, with a fixed power of L . For the tail, take $j > b$ derivatives and integrate outside $|\xi| \geq L^{C'}$. The resulting bound is at most a constant depending on j times

$$L^{C(j+1)-C'(j-b)}.$$

Fix $C' > C$ sufficiently large for the chosen Fourier convention and box. Increasing j then supplies any prescribed negative power of L . Fourier inversion expresses the integrand as a product of one-variable phases. Applying this in normalized logarithmic coordinates gives the corresponding Mellin separation on fixed dyads. Fixed-dimensional smooth cutoffs localized at logarithmic precision are included in the derivative hypothesis. $\square$

2.3 An elementary weighted sieve

We record a form with explicit coefficient and level bounds. This will be used both for congruence counts and for oscillatory sums over rough integers. The construction is a version of the Brun–Hooley sieve: the product of block upper bounds and its one-block correction are the inequalities of Ford and Halberstam [8, Section 1, Lemma 1]. We prove the form needed here, including its coefficient and level bounds.

Lemma 2.9 (Block sieve). *Consider a finite set of objects with nonnegative weights. For some primes $p \leq z$, let a bad condition be specified. If d is a squarefree product of these primes, suppose the weight of the objects on which all conditions at $p \mid d$ hold is*

$$Xg(d) + r(d),$$

including $d = 1$, where $X \geq 0$, g is multiplicative, and

$$0 \leq g(p) \leq 1 - \eta_0, \quad \sum_{w < p \leq w^2} g(p) \leq C \quad (w > 1) \quad (2.4)$$

for fixed $\eta_0 > 0$ and C . For every sufficiently large even integer h , the weight S of objects avoiding all bad conditions satisfies

$$S = X \prod_{p \leq z} (1 - g(p))(1 + O(e^{-h})) + O\left(\sum_{d \leq z^{4h+2}} |r(d)|\right). \quad (2.5)$$

Only the indicated primes and their squarefree products are used. The implied constants depend on η_0, C , uniformly also when h grows. For the fixed even choice $h = 2$, there is an upper bound by a constant times $X \prod (1 - g(p))$ plus the same remainder sum.

More precisely, the proof supplies polynomials U and V in the indicators of the bad conditions such that

$$V \leq \mathbf{1}_{\text{no bad condition}} \leq U, \quad U \geq 0.$$

Both have coefficients of absolute value at most one, supported on squarefree $d \leq z^{4h+2}$. The nonnegative overcount $U - \mathbf{1}_{\text{no bad condition}}$ is dominated by the pointwise nonnegative polynomial $U - V$, whose absolute coefficients are also at most one and which has the same level bound.

Proof. Partition the indicated primes into blocks

$$\mathcal{B}_j = \{p : z^{2^{-j-1}} < p \leq z^{2^{-j}}\}, \quad j \geq 0.$$

Only finitely many blocks are nonempty. Set $h_j = (j+1)h$, which is even. In block j , write I_j for avoidance of all its bad conditions, U_j for the inclusion–exclusion polynomial through degree h_j , and E_j for the sum of all monomials of degree $h_j + 1$. If exactly t bad conditions hold in the block, then

$$U_j = \sum_{a=0}^{h_j} (-1)^a \binom{t}{a} = \begin{cases} 1, & t = 0, \\ \binom{t-1}{h_j}, & t \geq 1. \end{cases}$$

Here a binomial coefficient is zero when its lower index exceeds its nonnegative upper index. It follows that $U_j \geq I_j \geq 0$ and $U_j - I_j \leq E_j$. Telescoping a product of nonnegative factors gives

$$0 \leq \prod_j U_j - \prod_j I_j \leq \sum_j E_j \prod_{k \neq j} U_k.$$

The upper and lower polynomials are therefore

$$U = \prod_j U_j, \quad V = U - \sum_j E_j \prod_{k \neq j} U_k. \quad (2.6)$$

In U , every block has degree at most h_j . A monomial from correction j has degree exactly $h_j + 1$ in block j and at most h_k elsewhere. These supports are disjoint across j and disjoint from the support of U . Thus the absolute coefficients in U, V are at most one. The same disjointness gives this bound for $U - V$.

The logarithm of a product in U , divided by $\log z$, is at most

$$\sum_{j \geq 0} h(j+1)2^{-j} = 4h.$$

A correction adds at most one more unit to this upper bound. Hence the stated level z^{4h+2} is valid for all the polynomials above.

Now give the bad conditions independent probabilities $g(p)$. Their block avoidance probability $a_j = \mathbb{E}I_j$ obeys

$$a_j = \prod_{p \in \mathcal{B}_j} (1 - g(p)) \geq \exp(-C/\eta_0) =: a_* > 0,$$

because $-\log(1-u) \leq u/\eta_0$ for $0 \leq u \leq 1 - \eta_0$. Also

$$e_j := \mathbb{E}E_j \leq \frac{C^{h_j+1}}{(h_j+1)!}, \quad a_j \leq \mathbb{E}U_j \leq a_j + e_j.$$

For all large even h ,

$$R_h := \sum_j e_j / a_j \ll e^{-h}.$$

Indeed the factorial tails eventually decrease geometrically in their index, and $h_j = (j+1)h$; the same sum is bounded by an absolute constant when $h = 2$. Independence between blocks yields

$$\prod_j a_j \leq \mathbb{E}U \leq \prod_j a_j e^{R_h}, \quad \mathbb{E} \sum_j E_j \prod_{k \neq j} U_k \leq \prod_j a_j R_h e^{R_h}.$$

Consequently the expectations of both bounding polynomials differ from $\prod a_j$ by $O(e^{-h}) \prod a_j$ for large h . For $h = 2$, the upper expectation is at most a constant times $\prod a_j$.

Evaluate the two polynomials in the original counting problem. The main terms of their monomials are exactly their independent expectations multiplied by X . The absolute remainder for either polynomial is at most $\sum_{d \leq z^{4h+2}} |r(d)|$, by the coefficient bound. Squeezing S between them proves the result. $\square$

Corollary 2.10 (A polynomial for rough integers). *For the ordinary bad conditions $p \mid n$ at primes $p \leq z$, let $U(n) = \sum_{d \mid n} \lambda_d$ be the upper polynomial in Equation (2.6), and put $D = z^{4h+2}$. Then*

$$U(n) \geq \mathbf{1}_{P^-(n) > z} \geq 0, \quad |\lambda_d| \leq 1, \quad \lambda_d = 0 \quad \text{unless } d \leq D \text{ is squarefree,}$$

and

$$\sum_{n \in I \cap \mathbb{N}} (U(n) - \mathbf{1}_{P^-(n) > z}) \ll |I|e^{-h} + D \quad (2.7)$$

for every finite interval $I \subset [1, \infty)$ and all sufficiently large even h . The bound is uniform in h, z, I . Moreover,

$$\sum_d \frac{|\lambda_d|}{d} \leq \prod_{p \leq z} (1 + p^{-1}) \ll \log(2z). \quad (2.8)$$

Proof. The pointwise claims and harmonic sum follow from the coefficient and support bounds. For the overcount, use its pointwise upper bound $U - V = \sum_j E_j \prod_{k \neq j} U_k$, which is nonnegative pointwise and has absolute coefficients at most one. In the interval I , the count of multiples of d is $|I|/d + O(1)$. The independent expectation of $U - V$ was bounded in the proof by $O(e^{-h}) \prod_{p \leq z} (1 - 1/p)$; the sum of absolute remainders is $O(D)$. Dropping the avoidance product proves Equation (2.7). Finally, $\prod (1 + 1/p) \leq \prod (1 - 1/p)^{-1} \ll \log(2z)$ by Mertens. $\square$

3 Transference for dilation graphs

We transfer a norm estimate for independent prime labels to an averaged moment of a dilation graph whose labels must divide the integer at their vertex. Averaging a required divisibility condition supplies a factor $1/p$, which explains the reciprocal-prime law in the independent model. A prime used at several vertices pays this cost only once. The main problem is to retain that dependence while using the independent-model norm on most edges. The resulting moment will control pairings in which one endpoint is constant on all mark lists at its integer position. Signed closed walks and repeated prime labels also occur in the divisibility-graph arguments of Helfgott and Radziwiłł [11, Section 1.5] and Pilatte [18, Sections 3 and 5]. The operators and ranges here differ; the memory identity and transference estimate below are proved for the present model.

3.1 Prime groups and the two operators

Definition 3.1 (Prime groups). Fix constants

$$0 < a < b < c < d < 0.47, \quad c_0, C_0, v_-, v_+ > 0, \quad 0 < q < 1, \quad \ell \in \mathbb{N}.$$

For each sufficiently large x , let the finite, pairwise disjoint sets of primes $\mathcal{P}_g$ be indexed by the disjoint sets $\mathcal{S}$ and $\mathcal{B}$, where

$$c_0 T \leq |\mathcal{S}|, |\mathcal{B}| \leq C_0 T.$$

The small and big groups, respectively, satisfy

$$\begin{aligned} \exp(L^a) \leq p \leq \exp(L^b) \quad (p \in \mathcal{P}_g, \quad g \in \mathcal{S}), \\ \exp(L^c) \leq p \leq \exp(L^d) \quad (p \in \mathcal{P}_g, \quad g \in \mathcal{B}), \\ v_- \leq V_g := \sum_{p \in \mathcal{P}_g} \frac{1}{p} \leq v_+, \quad \mu_g(p) := \frac{1}{V_g p}. \end{aligned} \quad (3.1)$$

Write $s = |\mathcal{S}| + |\mathcal{B}|$ and

$$\omega_g(n) = \sum_{p \in \mathcal{P}_g} \mathbf{1}_{p|n}, \quad \omega(n) = \sum_g \omega_g(n) \quad (n \in \mathbb{Z}).$$

Thus divisibility at $n = 0$ has its usual meaning. We call primes in $\bigcup_g \mathcal{P}_g$ group primes.

Let J be a sufficiently large fixed integer and put $M = J + \ell$. An ordered label list has M slots in each group. A pattern ν specifies sets $C_g \subseteq \{1, \dots, J\}$, with

$$C_g = \emptyset \quad (g \in \mathcal{S}), \quad |C_g| \leq m, \quad t_g = \ell + |C_g|.$$

The slots in $\{1, \dots, J\} \setminus C_g$ are shared; their labels are copied from the source list to the target list. The other t_g slots in each endpoint list are unshared. To form D , use the shared labels and, in the C_g slots, independent labels of law μ_g . Thus D has J prime factors from each group, counted with multiplicity. Write $D = D_{\mathcal{S}} D_{\mathcal{B}}$.

For each pattern there is a complex coefficient K_ν , depending only on the ordered big-group source labels, target labels, and the auxiliary free labels used in $D_{\mathcal{B}}$. The finite pattern family may depend on x and J , but

$$\sum_\nu \sup |K_\nu| \leq L^{C_1}, \quad (3.2)$$

where m, C_1 are fixed independently of J . In every operator below, symmetrization means averaging all $M!$ slot permutations independently in each relevant group. This is an orthogonal projection because the measures are invariant under these permutations. Our convention is that a row operator integrates a function at the target and returns a function at the source.

Definition 3.2 (Ideal operator). On the space

$$\mathcal{H}_{\text{id}} = L^2 \left(\prod_{g \in \mathcal{B}} \mathcal{P}_g^M, \bigotimes_{g \in \mathcal{B}} \mu_g^{\otimes M} \right),$$

repetitions of prime values are allowed. For a real ζ , the unsymmetrized ideal row operation for ν retains shared labels, samples all unshared target labels independently with laws μ_g , and samples the auxiliary labels of $D_{\mathcal{B}}$ independently with the same laws. Its multiplier is

$$K_\nu D_{\mathcal{B}}^{i\zeta} e(\Theta D_{\mathcal{B}}).$$

The sum over ν , composed on both sides with big-group symmetrization, is denoted $\mathcal{T}(\Theta)$.

Definition 3.3 (Physical operator). A physical state $(n, \mathbf{p})$ consists of $n \in \mathbb{Z}$ and an ordered list $\mathbf{p} = (p_{g,i})_{g, 1 \leq i \leq M}$ such that, within each group, the labels are distinct and divide n . Give every state at n mass $\prod_g V_g^{-M}$ and use counting measure in n . This defines $\mathcal{H}_{\text{ph}}$.

Fix an integer k with $0 < |k| \leq L^{C_2}$. For pattern ν , sample the auxiliary labels forming D as above and move from n to $n' = n + kD$. Sum over physical target states at n' with the prescribed

shared labels, with coefficient $\prod_g V_g^{-t_g}$ for each target. Forbid every auxiliary free label of D from both endpoint lists. Auxiliary free labels may equal one another. Multiply this row action by

$$K_\nu D^i \zeta q^{(\omega(n)-Ms)/2} q^{(\omega(n')-Ms)/2}. \quad (3.3)$$

The sum over patterns, symmetrized in all groups on both sides, is $\mathcal{A}$.

For clarity, the adjoint moves by $-kD$, interchanges source and target labels in the coefficient, and conjugates the multiplier. Indeed, after both lists and the auxiliary labels have been fixed, the shared product and hence D are unchanged on reversal. The joint measure of the two lists in group g has normalization

$$V_g^{-M-t_g} \quad (3.4)$$

in either direction. The auxiliary-label probability is also unchanged. This proves the assertion before symmetrization, and the symmetrizing projections are self-adjoint.

Lemma 3.4 (Absolute physical bounds). *Replace the coefficients of each row transition of $\mathcal{A}$ by their absolute values before adding patterns or averaging permutations. The resulting operator and its adjoint have row sums at most $L^{C_{\text{abs}}}$, where C_{abs} may depend on m, C_1 and the fixed data in Theorem 3.1, but is independent of J . The same assertion holds if the position is replaced by independent uniform residues at all group primes.*

Proof. If the target has $y \geq M$ divisors from group g , then, after the $M - t_g$ distinct shared labels have been fixed, there are $(y - M + t_g)_{t_g}$ possible ordered unshared target lists. Hence their sum, with their endpoint damping, is at most

$$\sup_{z \geq 0} V_g^{-t_g} (z + t_g)_{t_g} q^{z/2} \leq C. \quad (3.5)$$

Here $t_g \leq \ell + m$, so C is independent of J . The source damping is at most one. Free-label probabilities have total mass one, and restrictions can only decrease an absolute sum. Taking the product over $s = O(T)$ groups and using Equation (3.2) gives the row bound. The reversed normalization in Equation (3.4) gives the identical argument for columns. Neither argument used any property of integer positions beyond divisibility and translation. $\square$

In particular all these operators are bounded on their stated Hilbert spaces, including the physical space with unrestricted integer position.

Theorem 3.5 (Local transference). *For every fixed $E > 0$ there is $E_{\text{id}} > 0$, depending only on E and the fixed data in Theorem 3.1, with the following property. Assume*

$$\sup_{\Theta \in \mathbb{R}} \|\mathcal{T}(\Theta)\| \leq L^{-E_{\text{id}}}. \quad (3.6)$$

Then, for every sufficiently large fixed J , where the lower bound on J may also depend on m, C_1 , put

$$R = \lceil L^{0.52} \rceil, \quad N = 2R.$$

For every fixed $\gamma > 0$ and $I = [X, 2X)$ with $x^\gamma \leq X \leq x^{1/\gamma}$, one has

$$\frac{1}{|I \cap \mathbb{Z}|} \sum_{n \in I \cap \mathbb{Z}} \langle u_n, (\mathcal{A}\mathcal{A}^*)^R u_n \rangle \leq L^{-EN}, \quad (3.7)$$

for all sufficiently large x . Here u_n is one on every physical state at position n and zero elsewhere. The threshold for x may depend on all fixed parameters, including J, C_2, γ , and the bound is uniform subject to these parameters and Equation (3.6). No bound on ζ is needed apart from the assumed ideal estimate.

The vector u_n is the constant function on the mark lists at n ; it is not normalized. Thus the moment in Equation (3.7) sums paths whose final integer position returns to n , with arbitrary initial and final mark lists. Theorem 3.11 will turn precisely this estimate into cancellation against a mark-independent first endpoint. Those are the endpoints supplied in Section 5.

Here is the structure of the proof. First replace the integer root by independent residues and expand each big prime's contribution over intervals containing all its active visits. A memory records its congruence between active uses. The small-prime residues remain physical: the many choices of omitted small marks make transfers between memory and active lists rare. On an edge with no such transfer, Fourier transformation of the residue coordinates leaves the ideal big-label operator. We first allow different lifespans to use the same prime, and then exclude such coincidences by inclusion–exclusion organized by the number of independent equalities. Only the rare-transfer estimate needs the complexity-dependent choice of J ; the ideal accuracy is fixed beforehand.

3.2 Replacing the root by independent residues

Expand the left side of Equation (3.7) into paths with N alternating forward and backward edges. Fix the patterns, all endpoint permutations, the auxiliary labels, and the active lists at visits $0, \dots, N$. These choices fix offsets $h_0 = 0, h_1, \dots, h_N$, with

$$h_{i+1} - h_i = \sigma_i k D_i, \quad \sigma_i \in \{1, -1\}, \quad h_N = 0.$$

Only return to the position is imposed; the last ordered list need not be the first one. Let Λ_i be the set of active labels at visit i . Put $q_0 = q_N = \sqrt{q}$ and $q_i = q$ for $0 < i < N$. The dependence on the initial integer n is precisely the nonnegative factor

$$\prod_{i=0}^N \left(\prod_{p \in \Lambda_i} \mathbf{1}_{n+h_i \equiv 0 \pmod{p}} \right) q_i^{\#\{p \text{ a group prime: } p \notin \Lambda_i, n+h_i \equiv 0 \pmod{p}\}}. \quad (3.8)$$

The label normalization is initially $\prod_g V_g^{-M}$ and at each edge $\prod_g V_g^{-t_g}$, in addition to the free-label probabilities. All remaining factors are independent of n .

Lemma 3.6 (Uniform residue replacement). *For each compatible fixed path and every fixed $A > 0$, the average of Equation (3.8) over $I \cap \mathbb{Z}$ equals its expectation under independent uniform residues at all group primes, multiplied by $1 + O(\exp(-ANT))$. Incompatible active congruences give zero in both models.*

Proof. Let Q_{act} be the product of the distinct active primes over all visits. There are at most $(N+1)Ms$ of them, so

$$\log Q_{\text{act}} = O_J(NTL^d) = o(L). \quad (3.9)$$

Compatibility fixes one congruence modulo Q_{act} . All damping factors belonging to these primes are then deterministic; factor them out in both models. This step permits a relative estimate even when these deterministic factors are very small.

For a remaining prime, collect repeated offsets into distinct residues. Its factor has the form

$$1 - X_p(n), \quad X_p(n) = \sum_r c_{p,r} \mathbf{1}_{n \equiv r \pmod{p}}, \quad 0 \leq c_{p,r} \leq 1,$$

with at most $N+1$ residues and $0 \leq X_p \leq 1$. In the independent model,

$$\lambda := \sum_p \mathbb{E} X_p \leq (N+1) \sum_p \frac{1}{p} = O(NT), \quad \prod_p (1 - \mathbb{E} X_p) \geq \exp(-O(NT)),$$

since $\max_p \mathbb{E}X_p = o(1)$. Bonferroni inequalities for numbers in $[0, 1]$ bracket $\prod_p (1 - X_p)$ by consecutive truncations of its elementary-symmetric expansion. At degree h , the independent expectation of the difference is at most $\lambda^h/h!$. Choose h of order NT , with its fixed constant sufficiently large after A . Stirling's lower bound $h! \geq (h/e)^h$ makes the difference smaller than $\exp(-(A+C)NT)$ for any required fixed C .

Each truncated term specifies one residue at each of at most h additional primes. Its CRT count on I differs from its independent density by $O(|I \cap \mathbb{Z}|^{-1})$, also when the active congruence is imposed. The number of such terms is at most

$$\exp(O(h(L^d + \log N + \log s))) = \exp(o(L)),$$

and their moduli, including Q_{act} , have product $\exp(o(L))$. Here $0.52 + d < 1$. Thus the total counting error is $\exp(-\gamma L + o(L))$. By Equation (3.9), this is negligible relative to $Q_{\text{act}}^{-1} \exp(-O(NT))$. Restoring the factored deterministic damping proves the stated relative estimate. $\square$

This replacement may be summed over paths with absolute coefficients. Indeed retain the full vector of residues as a position, with its Haar probability measure, and translate it by $\sigma_i k D_i$ on an edge. The expected total mass of all initial lists is at most one: in group g it is

$$V_g^{-M} \sum_{\substack{p_1, \dots, p_M \in \mathcal{P}_g \\ \text{distinct}}} \frac{1}{p_1 \cdots p_M} \leq 1. \quad (3.10)$$

By Theorem 3.4, the absolute mass of all length N paths, even without a return condition, is $L^{O(N)}$. Taking A sufficiently large in Theorem 3.6 makes the summed error smaller than $L^{-A'N}$ for any desired fixed A' . We may therefore use the independent residue model. The return condition is now imposed by the exact identity

$$\mathbf{1}_{h_N=0} = \int_0^1 e(\theta h_N) d\theta. \quad (3.11)$$

It suffices to bound the unrestricted signed path expression uniformly in θ . Small-prime residues will remain actual Haar coordinates throughout the argument.

3.3 The exact expansion for one big prime

Put $\eta_i = 1 - q_i$, $P_* := \exp(L^c)$, and, for each big prime,

$$b_p = 1 - \frac{1}{p} \sum_{i=0}^N \eta_i > 0, \quad \tilde{\mu}_g(p) = \frac{1}{V_g p b_p}. \quad (3.12)$$

The ratio $\tilde{\mu}_g(p)/\mu_g(p)$ is $1 + O(N/P_*)$, uniformly in big groups. We extract the common baseline $\prod_p \text{big } b_p \leq 1$ from every path sum.

Suppose first that p is active at some visits. Incompatible active offsets give zero. Otherwise let their common residue be c_p , let a_p, z_p be the first and last active visits, and put $H_i = \mathbf{1}_{h_i \equiv c_p \pmod{p}}$. Its residue expectation is

$$\frac{1}{p} \prod_{i: p \notin \Lambda_i} (1 - \eta_i H_i).$$

The two exact telescoping identities

$$\begin{aligned} \prod_{i < a_p} (1 - \eta_i H_i) &= 1 - \sum_{j < a_p} \eta_j H_j \prod_{j < i < a_p} (1 - \eta_i H_i), \\ \prod_{i > z_p} (1 - \eta_i H_i) &= 1 - \sum_{j > z_p} \eta_j H_j \prod_{z_p < i < j} (1 - \eta_i H_i) \end{aligned} \quad (3.13)$$

follow by subtracting consecutive partial products. Thus, in addition to its active uses, the prime either starts at a_p or has an earlier ghost birth at a hit $j < a_p$, with coefficient $-\eta_j$; it either ends at z_p or has a later ghost termination at a hit $j > z_p$, again with coefficient $-\eta_j$. At strictly internal unmarked hits it receives q_i .

If p is never active, expand its full product and group each term of degree at least two by its earliest and latest indices. This gives

$$b_p + \frac{1}{p} \sum_{j < i} \eta_j \eta_i \mathbf{1}_{h_j \equiv h_i \pmod{p}} \prod_{j < t < i} \left(1 - \eta_t \mathbf{1}_{h_t \equiv h_j \pmod{p}}\right). \quad (3.14)$$

The empty and singleton terms give b_p ; the interior product is the sum over every possible further chosen index. Relative to the baseline, the prime is either absent or has an orphan interval from j to i , $j < i$, of cost $(pb_p)^{-1}(-\eta_j)(-\eta_i)$ and the indicated internal damping.

Consequently each represented big prime has exactly one *lifespan*: an interval containing all its active visits, possibly containing none. Its endpoints and all its active visits have equal offsets modulo p . It pays $(pb_p)^{-1}$ once, the ghost endpoint coefficients if present, and q_i at each strictly internal unmarked hit. No offset outside the lifespan occurs in its weight.

There are two useful consequences of physical compatibility. For large x every group prime exceeds $|k|$. If a prime is active at both ends of an edge, then it divides kD , hence D . It cannot be an auxiliary free label, by the ban, so it must be a prescribed shared label. Conversely, a prime at an unshared entry or exit cannot divide D : it would either duplicate a shared label in that state or violate the free-label ban. Thus the original mark normalization assigns V_g^{-1} to each maximal run of activity, and no additional factor at shared continuation. In particular, a lifespan with r active runs requires total label weight

$$\frac{V_g^{-r}}{pb_p}.$$

The memory construction must therefore charge the reciprocal prime once, at the first appearance, and charge V_g^{-1} once for each active run. Later congruence tests must not introduce further reciprocal-prime costs. These requirements guide the choice of measures and transfer coefficients below; the memory identity will verify them.

3.4 A Hilbert space which remembers lifespans

Let $Y_S = \prod_{p \text{ small}} \mathbb{Z}/p\mathbb{Z}$, with Haar probability measure. A small state at $y \in Y_S$ is an ordered list of M distinct zero coordinates in each small group, with mass $\prod_{g \in S} V_g^{-M}$. Write $\mathcal{H}_S$ for the resulting L^2 space. For a big group set

$$\mathcal{Z}_g = \{(p, r) : p \in \mathcal{P}_g, r \in \mathbb{Z}/p\mathbb{Z}\}, \quad d\lambda_g(p, r) = \tilde{\mu}_g(p) d\text{count}(r).$$

Define its memory space by

$$\mathcal{F}_g = \bigoplus_{j \geq 0} L_{\text{sym}}^2 \left(\mathcal{Z}_g^j, \frac{1}{j!} \lambda_g^{\otimes j} \right), \quad \mathcal{H} = \mathcal{H}_S \otimes L^2 \left(\prod_{g \in \mathcal{B}} \mathcal{P}_g^M, \bigotimes_{g \in \mathcal{B}} \tilde{\mu}_g^{\otimes M} \right) \otimes \bigotimes_{g \in \mathcal{B}} \mathcal{F}_g. \quad (3.15)$$

Here “symmetric” means invariant under permutations of the j particle indices, including when their numerical values coincide. The $j = 0$ summand is $\mathbb{C}$. A pending particle (p, r) records displacement from its required hit, modulo p . The residue measure in λ_g is *counting measure*: requiring $r = 0$ selects one point and incurs no factor $1/p$. The boundary vector $\mathbf{1}_\emptyset$ is one when every memory is empty and zero otherwise; it is independent of all other coordinates.

We next specify row actions. The factorial measure in Equation (3.15) is used for inner products and adjoints, not inserted anew in every row transition. All operations are first understood on finite memory truncations; the absolute bounds below justify the unrestricted sums.

Separate the evolution into an operation G_i at each visit $0 \leq i \leq N$ and an operation $\mathcal{E}_i$ across the following edge for $i < N$. The visit operation handles ghost endpoints and unmarked hits. The edge operation moves residues and transfers particles between active lists and memory: a store keeps a departing active label pending, and a promotion returns a pending particle to an active slot. Thus G_i acts after arrival and promotion at visit i , and before storage and departure.

To make both operations bounded, choose $\rho \in (0, 1)$ with $\rho^2 > \sqrt{q}$. At a strictly internal unmarked hit, split the required damping as

$$q_i = \rho (q_i / \rho^2) \rho.$$

The adjacent edges supply the two factors ρ , and the visit operation supplies the middle factor. The edge factors will control the number of possible promotions, while $q_i / \rho^2 < 1$ damps the visit operation. The ghost endpoint factor is similarly split as $-\eta_i = (-\eta_i / \rho) \rho$; the birth normalization is specified below. The definitions implement these local factorizations, including the initial and terminal visits.

For a group memory $\mathbf{z} = ((p_h, r_h))_{h=1}^j$, write $Z(\mathbf{z}) = \{h : r_h = 0\}$. At visit i the ghost operation G_i is the product over big groups of the following row action, leaving every other coordinate fixed:

$$(G_{i,g}F)(\mathbf{z}) = \sum_{A \subseteq Z(\mathbf{z})} \left(-\frac{\eta_i}{\rho}\right)^{|A|} \left(\frac{q_i}{\rho^2}\right)^{|Z(\mathbf{z})|-|A|} \sum_{b \geq 0} \frac{1}{b!} \left(-\frac{\eta_i V_g}{\rho}\right)^b \cdot \int_{\mathcal{P}_g^b} F(\mathbf{z} \setminus A, (p'_1, 0), \dots, (p'_b, 0)) \prod_{h=1}^b d\tilde{\mu}_g(p'_h). \quad (3.16)$$

Thus selected old zeros terminate, continuing zeros are damped, and a new unordered batch is born at zero. Termination precedes birth; an orphan cannot be born and terminate at the same visit.

The edge operation $\mathcal{E}_i$, $0 \leq i < N$, sums the given patterns in orientation σ_i , with independent source and target symmetrizations in all active lists. Between these permutations its row action is as follows.

- (i) Form D from shared source labels and the independent free draws. Retain the ban of free labels from both endpoint active lists. Multiply by the oriented coefficient $K_\nu D^{i\zeta}$, or its conjugate with roles interchanged, and by $e(\theta \sigma_i k D)$.
- (ii) Multiply by ρ for every old pending zero, before storing any active particle. In each big group select any subset of its t_g unshared source slots to store, appending their particles at residue zero; drop the other unshared source particles. Shared particles remain active.
- (iii) Translate y and all pending residues, including the newly stored ones, by $\Delta = \sigma_i k D$. Choose any subset of the t_g unshared big target slots and an injection of those ordered slots into distinct *old* pending particle indices of that group whose translated residue is zero. Promote the selected particles into those slots, remove them from memory, and multiply by V_g^{-1} per promotion. Newly stored particles cannot be promoted across this edge. Require $p \nmid D$ for every stored or promoted particle. Fill the other unshared target slots by independent fresh integrations against $\tilde{\mu}_g$.
- (iv) Multiply by ρ for every pending zero remaining after promotions. In the small groups use the physical target sum, with normalization $V_g^{-\ell}$, and the two small endpoint factors $q^{(\omega_g - M)/2}$. These counts concern small states at y and $y + \Delta$.

Selections of promotion slots and injections are summed without a factorial divisor. All choices of source or target slots refer to their canonical order between the sampled endpoint permutations. No big endpoint damping is included apart from the pending-particle factors specified above. Auxiliary free draws of D are not particle births.

For the moment impose, in the complete path sum, the rule that all prime values assigned at *different births* are distinct. Births comprise initial active slots, fresh active target slots, and ghost births. This is a global rule, even if two births occur far apart. Apart from it, place no distinctness restriction on big active lists or memories.

Lemma 3.7 (Exact memory identity). *With the distinct-birth rule, the independent-residue path expression with the phase in Equation (3.11) equals*

$$\left(\prod_{p \text{ big}} b_p \right) \langle \mathbf{1}_\emptyset, G_0 \mathcal{E}_0 G_1 \cdots \mathcal{E}_{N-1} G_N \mathbf{1}_\emptyset \rangle_{\text{distinct}}. \quad (3.17)$$

The subscript means that the indicator is inserted in the expanded path integral, not that each local operator separately enforces it.

Proof. A particle born active begins with its required hit at that visit. When it leaves an active run, storing it at zero and then translating records subsequent displacement from that hit. Promotion checks displacement zero. A later store resets the anchor at a congruent offset, so it changes no congruence requirement. A ghost birth sets the anchor at an unmarked hit, and a ghost termination also checks zero. Shared continuation automatically respects the congruence.

The ρ factors cancel locally. At an internal visit a continuing pending zero pays $\rho(q_i/\rho^2)\rho = q_i$. A ghost birth pays $(-\eta_i/\rho)\rho = -\eta_i$, and a ghost termination pays $\rho(-\eta_i/\rho) = -\eta_i$. An active particle pays none of these factors: promotion occurs before output damping and storage after input damping. At visit zero memory starts empty, and at visit N it must end empty, so there is no missing boundary factor. The values $q_0 = q_N = \sqrt{q}$ are therefore treated exactly.

For a prime with r active runs, a first active birth costs $1/(V_g p b_p)$ and each of its $r - 1$ later promotions costs $1/V_g$. A first ghost birth costs $1/(p b_p)$ and its r promotions cost V_g^{-r} . Both give

$$\frac{V_g^{-r}}{p b_p}.$$

An orphan has $r = 0$. These are precisely the baseline-relative costs in Equations (3.13) and (3.14), including all original mark normalizations.

Conversely, fix an original compatible path and a chosen lifespan term for every represented big prime. Its active visits determine exactly when it is shared, stored, promoted, or dropped; its ghost endpoints determine its birth and termination. Between these uses it must remain pending. The exclusions $p \nmid D$ follow from unshared entries and exits as proved above, and immediate promotion of a newly stored particle is impossible for the same reason. A numerical prime occupies at most one active slot or pending particle, because it has only one birth. Distinct ghost births in an unordered batch are counted once by its factorial divisor. Thus these constructions are mutually inverse and preserve all coefficients. This proves Equation (3.17). $\square$

Figure 2 illustrates an admissible lifespan with two runs of activity and ghost endpoints.

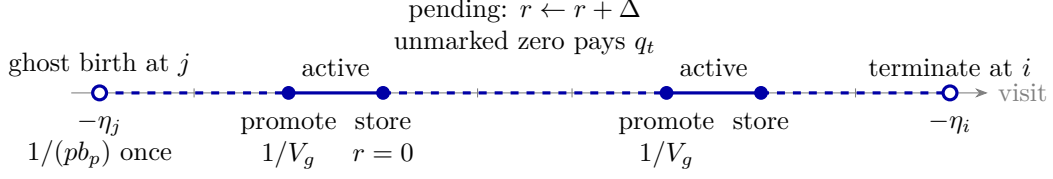


Figure 2: One possible lifespan. Solid segments denote shared activity; dashed segments denote memory. Every marked use and ghost endpoint has the same offset modulo p . The reciprocal prime factor is charged only at birth; $\eta_t = 1 - q_t$ at visit t .

3.5 Absolute bounds, including the adjoint measures

The memory identity is exact with the global distinct-birth rule. We now allow different births to have equal prime values, so that each operation acts locally on $\mathcal{H}$. We will restore the global rule after estimating this enlarged evolution. The absolute bounds proved here justify the memory truncation and remain available when equality constraints later modify individual operations. Their column bounds require the factorial adjoint measures explicitly.

For a positive row kernel K , if a positive function w satisfies $Kw \leq Rw$ and $K^*w \leq Cw$, then

$$\|K\|_{2 \rightarrow 2} \leq \sqrt{RC}. \quad (3.18)$$

Indeed, Cauchy's inequality in each row bounds $|Kf|^2$ by $(Kw)K(|f|^2/w)$; integration and the column inequality prove Equation (3.18). We apply this to absolute kernels, with

$$w(\text{state}) = v^{j_{\text{tot}}}, \quad j_{\text{tot}} = \sum_{g \in \mathcal{B}} j_g,$$

where $v > 1$ is a fixed constant chosen below.

Here is a direct verification of the measures in the ghost adjoint. In one group abbreviate

$$A_i = \frac{q_i}{\rho^2}, \quad D_i = -\frac{\eta_i}{\rho}, \quad C_i = V_g D_i, \quad \nu = \tilde{\mu}_g,$$

and let $z(\mathbf{u})$ count zero residues in a memory list. For test functions f, h , the joint integral from Equation (3.16) is

$$\begin{aligned} \langle f, G_{i,g} h \rangle &= \sum_{k,t,b \geq 0} \frac{1}{k!t!b!} \int A_i^{z(\mathbf{u})} D_i^t C_i^b \overline{f(\mathbf{u}, \boldsymbol{\alpha}^0)} h(\mathbf{u}, \boldsymbol{\beta}^0) \\ &\quad \cdot d\lambda_g^{\otimes k}(\mathbf{u}) d\nu^{\otimes t}(\boldsymbol{\alpha}) d\nu^{\otimes b}(\boldsymbol{\beta}). \end{aligned} \quad (3.19)$$

Here $\boldsymbol{\alpha}^0$ means that all its residues are zero. The identity follows by selecting t old zero indices from a list of size $k + t$:

$$\frac{1}{(k+t)!} \binom{k+t}{t} = \frac{1}{k!t!}.$$

Swapping $(t, \boldsymbol{\alpha})$ with $(b, \boldsymbol{\beta})$ and conjugating shows that the adjoint has the same row formula with birth coefficient D_i and termination coefficient C_i . In particular, reversal moves the factor V_g from births to terminations; it does not create any factor p or $1/p$.

For the absolute ghost kernel the weighted row ratio is exactly

$$\exp(|C_i|v\nu(\mathcal{P}_g)) \left(A_i + \frac{|D_i|}{v} \right)^{z(\mathbf{z})}, \quad (3.20)$$

and its column ratio has C_i, D_i interchanged. Choose v large enough that, for both possible q_i ,

$$\frac{q_i}{\rho^2} + \frac{2 \max(1, v_+) \eta_i}{\rho v} < 1. \quad (3.21)$$

This is possible since $q_i \leq \sqrt{q} < \rho^2$. The same choice works when every original ghost birth receives an extra factor two. Since $\nu(\mathcal{P}_g) = 1 + o(1)$, the exponential factor in Equation (3.20) is bounded by a fixed constant per group. Thus for some fixed C_v ,

$$|G_i|w \leq L^{C_v} w, \quad |G_i|^* w \leq L^{C_v} w, \quad \|G_i\| \leq L^{C_v}. \quad (3.22)$$

The exponent C_v is independent of m, C_1, J . All these assertions include the version with doubled ghost births.

We verify the edge adjoint just as explicitly. Fix one big group, a pattern, endpoint permutations, free labels, and the chosen transfer subsets. Suppose a target slots are promotions and b source slots are stores, so $a, b \leq t_g$. Let the source memory have size j , and write u for its $j - a$ surviving particles. Let $p_1, \dots, p_a$ be the promoted prime labels and $s_1, \dots, s_b$ the stored labels. For step Δ , the endpoint memories are

$$X = u \sqcup ((p_h, -\Delta \bmod p_h))_{h \leq a}, \quad Y = T_\Delta u \sqcup ((s_h, \Delta \bmod s_h))_{h \leq b}, \quad (3.23)$$

where T_Δ translates every residue. The joint measure on these variables is

$$\frac{d\lambda_g^{\otimes(j-a)}(u)}{(j-a)!} \prod_{h \leq a} d\tilde{\mu}_g(p_h) \prod_{h \leq b} d\tilde{\mu}_g(s_h), \quad (3.24)$$

along with one $\tilde{\mu}_g$ integration for every shared active label, dropped source label, and fresh target label. The transfer coefficient is V_g^{-a} , and the damping is $\rho^{z(X)+z(Y)}$. The factorial accounting is

$$\frac{(j)_a}{j!} = \frac{1}{(j-a)!} = \frac{(j-a+b)_b}{(j-a+b)!}. \quad (3.25)$$

The first equality selects old memory indices into the a specified ordered target slots. The second selects b indices of the output memory into the original source slots on reversal. Each required residue in Equation (3.23) selects exactly one point of counting measure. Translation preserves that measure.

It follows that the reversed row operation retrieves the original stores and stores the original promotions; the coefficient V_g^{-a} remains unchanged. If it is expressed using the forward convention that charges V_g^{-b} for its reversed promotions, its additional factor is V_g^{b-a} . This is bounded uniformly for $a, b \leq \ell + m$. Fixing shared and free D -labels fixes Δ independently of which memory indices are retrieved, as required for this change of variables. In small groups Haar translation and the joint normalization $V_g^{-M-\ell}$ are invariant on reversal.

The identities above continue to hold with numerical coincidences. For example, a memory multiset with multiplicities n_z has mass $\prod_z \lambda_g(z)^{n_z} / n_z!$. Subset deletions and ordered retrievals retain their index multiplicities. No passage from indices to distinct numerical values was made in Equation (3.19) or Equation (3.25).

Lemma 3.8 (Absolute memory bounds). *There is C_a , allowed to depend on m, C_1 but independent of J , such that every absolute edge has weighted row and column bounds L^{C_a} . These bounds and Equation (3.22) persist under arbitrary multipliers of modulus at most one on individual local transition choices, and under memory-size projections.*

Proof. For a row, fix a pattern, free labels, permutations, and transfer-slot subsets of sizes $a_g, b_g \leq t_g$. If Z' old pending particles have zero translated residue in a group, their ordered promotions and the remaining output damping cost at most

$$(Z')_{a_g} \rho^{Z' - a_g} \leq \sup_{z \geq a_g} (z)_{a_g} \rho^{z - a_g} < \infty. \quad (3.26)$$

Damping of stored particles can be discarded for this upper bound. The Schur weight contributes $v^{b_g - a_g}$, promotion coefficients contribute $V_g^{-a_g}$, and fresh integrations have mass $\tilde{\mu}_g(\mathcal{P}_g)^{t_g - a_g} \leq 2^{t_g}$ for large x . There are at most 4^{t_g} choices of transfer subsets. Each cost is bounded by a constant per group, independent of J . Small transitions satisfy Equation (3.5). Multiplying over groups, averaging permutations and free labels, and summing Equation (3.2) proves the row bound.

For the column, use Equations (3.23) and (3.25) in reverse. Original stores are now ordered retrievals. Their count is controlled by the original input ρ factors after reversed translation, in exactly the form in Equation (3.26). The remaining factor $V_g^{b_g - a_g}$ is bounded per group. This proves the column bound. Restrictions and local multipliers of modulus at most one are dominated by these absolute kernels. $\square$

We now restrict the total memory size to

$$B = \lceil L^2 \rceil \quad (3.27)$$

before and after every operation, writing the projections implicitly. At most $(\ell + m)|\mathcal{B}|N$ particles can be supplied by stores. Thus a path violating this restriction has at least $B - O((\ell + m)sN)$ ghost births. Multiply every ghost birth by two in the absolute path sum and use the preceding bounds. The total omitted mass, even with the distinct-birth indicator, is at most

$$2^{-B + O((\ell + m)sN)} L^{O(N)} = \exp(-\Omega(L^2)) = o(L^{-AN}) \quad (3.28)$$

for every fixed A . The empty-memory boundary vector has bounded norm: the small part has mass at most one by Equation (3.10), and the big part has mass

$$\prod_{g \in \mathcal{B}} \tilde{\mu}_g(\mathcal{P}_g)^M = 1 + O_J(sN/P_*) = 1 + o(1).$$

These estimates also prove absolute summability before truncation, for instance by first restricting all batch sizes and then applying monotone convergence to absolute kernels.

3.6 Rare transfers and the role of the small groups

It remains to obtain a small signed edge norm and then restore the omitted distinct-birth rule. Decompose each edge as $\mathcal{E}_i = \mathcal{E}_{i,\text{clean}} + \mathcal{E}_{i,\text{dirty}}$: a term is *dirty* if it has at least one store or promotion, and *clean* otherwise. Dirty terms connect memory to the active lists; we suppress them by averaging the omitted small labels. Clean terms only translate memory residues and refresh unshared active labels; the next subsection will reduce their norm to the ideal estimate. Define

$$\mathcal{D} = \binom{M}{\ell}^{|\mathcal{S}|}. \quad (3.29)$$

Lemma 3.9 (Rare transfer). *With the memory truncation in Equation (3.27), the aggregate of all dirty terms of any edge has norm at most*

$$2L^{C_a} \left(\frac{B}{\mathcal{D}} \right)^{1/2}, \quad (3.30)$$

after enlarging C_a independently of J if necessary.

Proof. First consider the positive sum of terms with a promotion. Fix its source, pattern, big-group permutations and shared choices, and free D -labels. For some one of its at most B old pending particles (p, r) , promotion requires

$$r + \sigma_i k D_{\mathcal{B}} D_{\mathcal{S}} \equiv 0 \pmod{p}, \quad p \nmid D. \quad (3.31)$$

In each small group the source symmetrization omits an independent uniformly chosen ℓ -subset of its M distinct labels. If F is the full product of all small source labels and U the product of the omitted labels, then $D_{\mathcal{S}} = F/U$. Unique factorization and disjointness of groups show that the $\mathcal{D}$ choices give distinct positive integers U . Moreover,

$$U \leq \exp(\ell |\mathcal{S}| L^b) < \exp(L^c) \leq p$$

for sufficiently large x . They are therefore distinct modulo p . The quantities $F, k, D_{\mathcal{B}}$ are invertible modulo every p eligible in Equation (3.31). Hence each pending particle allows at most one omission choice. At most B of the $\mathcal{D}$ equally likely choices allow any promotion.

Conditional on every omission choice, the remaining weighted row costs are uniformly bounded by the proof of Theorem 3.8. Thus the promotion part improves its row bound to $L^{C_a} B / \mathcal{D}$, while its column bound remains L^{C_a} . Schur's test gives the first half of Equation (3.30). Terms with stores and no promotion have the same improvement in their column bound by reversing the edge: a stored particle becomes a retrieval from output memory. The exclusion $p \nmid D$ is retained under reversal, even when numerical coincidences have been allowed. Add these two bounds. $\square$

The coefficient restriction to big labels is used here: conditioning on all such labels does not bias the small omission choices. In particular,

$$2L^{C_a} \sqrt{B/\mathcal{D}} \ll L^{C_a+1-(c_0/2)\log(J/\ell)}. \quad (3.32)$$

An arbitrarily strong fixed saving is obtained by increasing J , without changing the ideal norm target.

3.7 Clean edges and Fourier reduction

Lemma 3.10 (Clean norm). *There is C_s , depending only on the fixed data in Theorem 3.1, such that the clean part of every edge satisfies*

$$\|\mathcal{E}_{i,\text{clean}}\| \leq 2L^{-E_{\text{id}}+C_s} \quad (3.33)$$

for all sufficiently large x , under Equation (3.6). In particular C_s is independent of m, C_1, J .

Proof. A clean edge leaves all memory prime lists and sizes unchanged, translates their residues, drops every unshared big source slot, and fills every unshared big target slot freshly. Its input and output ρ factors and its memory projections are contractions. They commute with active-list symmetrizations. Absorb these factors and the symmetrizations into the two test vectors w_1, w_2 . These vectors remain symmetric in the big active lists.

Fix an ordered tuple $\mathbf{d}$ of J shared labels in each small group and let d' denote their joint product. Define a map to unrestricted small residue functions by

$$(P_{\mathbf{d}} w)(y) = \left(\prod_{g \in \mathcal{S}} V_g^{-J/2-\ell} \right) q^{\frac{1}{2} \sum_{g \in \mathcal{S}} (\omega_g(y) - M)} \sum_{\mathbf{R}: (\mathbf{d}, \mathbf{R}) \text{ a small state at } y} w(y, (\mathbf{d}, \mathbf{R})). \quad (3.34)$$

All other coordinates are left untouched. The map is zero when the displayed list is not a small state, so the damping is used only where $\omega_g(y) \geq M$. The two endpoint factors in Equation (3.34) give $V_g^{-J-2\ell} = V_g^{-M-\ell}$, exactly the physical joint normalization.

By Cauchy's inequality in $\mathbf{R}$, followed by summation over $\mathbf{d}$, one has

$$\sum_{\mathbf{d}} \|P_{\mathbf{d}} w\|^2 \leq L^{C_s} \|w\|^2. \quad (3.35)$$

To see the independence of J , the multiplier relative to the original small state measure in group g is bounded by

$$\sup_{z \geq M} q^{z-M} \frac{(z - M + \ell)_{\ell}}{V_g^{\ell}},$$

a constant depending only on q, ℓ, v_- . Taking products over $O(T)$ groups proves Equation (3.35).

The clean bilinear form is now exactly

$$\sum_{\mathbf{d}} \langle P_{\mathbf{d}} w_1, \mathcal{T}_{\mathbf{d}}' P_{\mathbf{d}} w_2 \rangle. \quad (3.36)$$

The operator between these maps uses the big-list transitions and translates all unrestricted small and memory residues by $\sigma_i k d' D_{\mathcal{B}}$. Fix the memory sizes and their prime lists. Fourier transformation on the finite abelian group consisting of the small residue torus and the pending residue coordinates diagonalizes these translations. Counting measure on pending residues and Haar measure on the small torus each have the usual unitary finite Fourier transform. One can first use ordered memory representatives; restriction to symmetric functions preserves the bound. Big active-list symmetry is unaffected.

At any fixed Fourier frequency, translation and the closure factor contribute

$$e(\sigma_i \Theta D_{\mathcal{B}})$$

for a real Θ depending on $\mathbf{d}$, the memory prime list, the frequency, and θ , but independent of the variable big active labels. The scalar factor $(d')^{\sigma_i i \zeta}$ has modulus one. Consequently the remaining operator, on symmetric big active tests, is $\mathcal{T}(\Theta)$ or its adjoint, with two changes only: the measures of active labels and fresh target integrations are $\tilde{\mu}_g$ in place of μ_g , and the free/active collision ban remains in force.

Both changes are negligible in operator norm. Across the $M|\mathcal{B}|$ active coordinates, the product density changes by

$$1 + O_J(sN/P_*).$$

The corresponding square-root density identification is unitary between the two L^2 spaces and commutes with symmetrization. Fresh integration densities obey the same bound. The ideal absolute row and column sums are at most L^{C_1} before these changes, since all its unrestricted fresh laws are probabilities. Schur's test therefore bounds the density perturbation by $O_J(L^{C_1} sN/P_*)$.

For the ban, in either row direction condition on the fixed active state. Each free auxiliary draw has maximal atom $O(1/P_*)$; its chance to match a fixed active label is bounded by that quantity. A match to a fresh opposite unshared label has the same bound by independence. There are $O_{J,m}(s)$ possible comparisons. Reversal has the same estimate. Thus removing the banned transitions costs $O_{J,m}(L^{C_1} s/P_*)$ in norm. These estimates are $o(L^{-A})$ for every fixed A . They do not require exclusion of coincidences with pending labels: such labels are fixed in the Fourier decomposition and are already accounted for by translation.

It follows uniformly in $\mathbf{d}$ that the between-map norm is at most $2L^{-E_{\text{id}}}$ for large x . Apply Cauchy's inequality to Equation (3.36) and then Equation (3.35); this proves Equation (3.33). $\square$

Choose, in this order,

$$G > E + C_v + 3, \quad E_{\text{id}} > G + C_s + 2, \quad (3.37)$$

and finally choose the fixed integer J so large that Equation (3.32) is smaller than $\frac{1}{2}L^{-G}$ for large x . For example it suffices that

$$\frac{c_0}{2} \log \binom{J + \ell}{\ell} > C_a + G + 3.$$

The clean estimate supplies the other half. Hence the whole signed truncated edge, with birth distinctness still omitted, satisfies

$$\|\mathcal{E}_i\| \leq L^{-G}. \quad (3.38)$$

Only this final choice of J needs to depend on m, C_1 . This establishes the required quantifier order, but a further argument is necessary to impose distinct births without losing the signed contraction on every edge.

3.8 Restoring distinct births by equality rank

A single absolute collision estimate cannot finish the proof: one reciprocal-prime saving P_*^{-1} does not absorb a length- N absolute path cost $L^{O(N)}$. Instead, expand the global distinctness condition by equalities between birth values. Many independent equalities yield many point-mass savings. A small number of independent equalities can be imposed by phases at their birth operations, preserving the signed contraction on all the other edges.

In the truncated evolution allocate potential birth addresses as follows: initial big slots; every possible big target slot at every edge, in its canonical order before final symmetrization; and indices $1, \dots, B$ in each group's ordered ghost batch at each visit. A target slot is performed as an address precisely when it is a fresh birth rather than shared or promoted. A ghost address is performed precisely when its batch is at least that large. These are conditions on the choices made in their single local operation. The number A_* of potential addresses satisfies

$$A_* \leq M|\mathcal{B}|(N + 1) + (N + 1)|\mathcal{B}|B = L^{O(1)}. \quad (3.39)$$

The implied exponent can be taken independent of J once x is sufficiently large for fixed J .

We use the following elementary form of distinctness inclusion-exclusion. Sum over collections π of disjoint blocks of potential addresses, every block having size at least two. Let $\mathcal{C}_\pi$ require that all addresses in those blocks be performed and that the prime values within each block be equal. Set

$$\mu(\pi) = \prod_{B' \in \pi} (-1)^{|B'| - 1} (|B'| - 1)!, \quad r(\pi) = \sum_{B' \in \pi} (|B'| - 1). \quad (3.40)$$

For a fixed realized path,

$$\mathbf{1}_{\text{distinct performed birth values}} = \sum_{\pi} \mu(\pi) \mathbf{1}_{\mathcal{C}_\pi}. \quad (3.41)$$

To verify this, interpret a block of size h as a cycle on its h addresses: there are $(h - 1)!$ such cycles, each with sign $(-1)^{h-1}$. The right side is the sum of signs of permutations of the performed addresses which preserve their numerical prime values. It factors over equal-value classes. The sign sum is one for a singleton class and zero for any class of size at least two, by pairing permutations with their product by a fixed transposition.

A rank- r collection involves at most $2r$ addresses. More precisely its total absolute coefficient, summed over all collections of rank r , obeys

$$\sum_{r(\pi)=r} |\mu(\pi)| \leq A_*^{2r} \leq L^{C'r}. \quad (3.42)$$

For instance the unsigned cycle generating polynomial on A_* addresses is $\prod_{j=0}^{A_*-1} (1 + jt)$, where the power of t is the sum of cycle lengths minus the number of cycles. Its coefficient of t^r is at most $(\sum_j j)^r / r! \leq A_*^{2r}$.

Large rank: chronological fresh integrations. Fix π . Order birth addresses chronologically, using canonical slot order inside an edge or batch, and call the first address in each block its pivot. Every other address is a dependent birth. When such an address is performed, its fresh prime integration is constrained to the already determined pivot value. Since

$$\sup_{g \in \mathcal{B}, p \in \mathcal{P}_g} \tilde{\mu}_g(p) \ll P_*^{-1}, \quad (3.43)$$

this improves the weighted absolute row bound by $O(P_*^{-1})$ per dependent birth.

Here the chronological assertion uses genuinely fresh integrations. At an edge choose stores, promotions, and the step first, and then fill fresh target slots in canonical order. The step, promotion count, and memory weight do not depend on these new prime values; the coefficient is bounded by its supremum and exclusions may be discarded. In the proof of [Theorem 3.8](#), replace the total fresh mass in each constrained slot by its single-point bound. For a ghost operation keep its ordered batch integral and factorial divisor; constraining indicated fresh coordinates replaces their mass factors by the same single-point bound. The birth coefficient V_g and the memory weight only give fixed constants per constrained coordinate. Initial active slots satisfy the same product estimate. Pivots and dependents within one operation are handled in their prescribed integration order.

For completeness, these conditional row bounds can be iterated although a pivot may have died before its dependent birth. Retain all pivot values in the conditioning. Bound the remaining terminal sum by backward induction using the uniform weighted row bounds for every possible conditioning, beginning with $\mathbf{1}_\emptyset \leq w$. Whenever a dependent birth is reached, the preceding point-mass improvement is uniform in its retained pivot value. At the initial boundary $w = 1$, and the remaining integrated boundary mass is bounded. Thus the total absolute contribution for this fixed collection is at most

$$L^{C_h(N+1)} (C/P_*)^{r(\pi)}, \quad (3.44)$$

for fixed C_h, C ; both may depend on m, C_1 .

Choose a sufficiently large fixed C'' , after these constants and C' , and set

$$r_0 = \left\lceil \frac{C''NT}{\log P_*} \right\rceil. \quad (3.45)$$

By [Equation \(3.42\)](#) and [Equation \(3.44\)](#), the sum over $r \geq r_0$ is bounded by

$$L^{C_h(N+1)} \sum_{r \geq r_0} (CL^{C'}/P_*)^r = o(L^{-EN}).$$

Indeed $\log P_* = L^c \gg T$, and increasing C'' beats the fixed exponent $C_h + E$.

Small rank: phases only at the affected births. For a block with pivot β_0 and other addresses $\beta_1, \dots, \beta_{h-1}$, equality of the integer prime values has the exact representation

$$\mathbf{1}_{p_{\beta_j}=p_{\beta_0} \ (1 \leq j < h)} = \int_{[0,1]^{h-1}} e \left(\sum_{j=1}^{h-1} \tau_j (p_{\beta_j} - p_{\beta_0}) \right) d\boldsymbol{\tau}, \quad (3.46)$$

on paths performing all these addresses. For fixed Fourier parameters, collect the factors by their birth operation. Each affected operation is multiplied, on its individual local choices, by

$$\mathbf{1}_{\text{designated local births are performed}} e \left(\sum_{\beta \text{ local}} c_{\beta} p_{\beta} \right),$$

of modulus at most one. Initial-slot factors merely modify the initial boundary vector by such a multiplier. An affected edge can lose its signed norm saving, but [Theorem 3.8](#) bounds its new norm by L^{C_a} . Unaffected edges retain their complete signed pattern sum and both symmetrizations, so [Equation \(3.38\)](#) still applies.

The ordered indices in a ghost batch do not require ordered-memory functions. For a batch of size b , insert its local multiplier $\phi_b(p_1, \dots, p_b)$ inside the existing $1/b!$ product integral in [Equation \(3.16\)](#). The rest of that integral is symmetric in its fresh variables, so its value is unchanged on replacing ϕ_b by

$$\frac{1}{b!} \sum_{\sigma \in S_b} \phi_b(p_{\sigma(1)}, \dots, p_{\sigma(b)}).$$

This average has modulus at most one, also on numerical diagonals. The resulting operator acts on symmetric memories, and its reversed multiplier is the conjugate on the corresponding deleted batch in [Equation \(3.19\)](#). Its absolute row and column bounds remain [Equation \(3.22\)](#). In particular no phase has to follow a particle through later storage, propagation, or promotion: equality was encoded entirely at its two birth addresses.

At most $2r$ edges are affected by a rank- r collection. The boundary norms are bounded, and all $N + 1$ ghost operations cost at most L^{C_v} . Therefore its Fourier-integrated contribution is at most

$$O \left(L^{(-G+C_v)N+C_v+2r(G+C_a)} \right). \quad (3.47)$$

Since

$$\frac{r_0}{N} = O(T/L^c) + O(N^{-1}) = o(1),$$

the sum over $r < r_0$, with the coefficient bound [Equation \(3.42\)](#), is

$$L^{(-G+C_v+o(1))N} = o(L^{-EN})$$

by [Equation \(3.37\)](#).

Thus the high-rank terms are controlled by their fresh-label costs, while the low-rank terms retain signed contraction on $N - o(N)$ edges. Together they bound the expansion with distinct births, which is the expression required by [Theorem 3.7](#).

Completion of Theorem 3.5. Apply [Equation \(3.41\)](#) in the truncated memory identity. The large- and small-rank estimates bound it by $o(L^{-EN})$, uniformly in θ . Restore the tail [Equation \(3.28\)](#); multiply by the baseline, which is at most one; and integrate [Equation \(3.11\)](#). Finally restore the uniform residue-replacement error, choosing its precision after the absolute path bound. The result is $o(L^{-EN})$, and therefore [Equation \(3.7\)](#) for sufficiently large x . The choice of E_{id} in [Equation \(3.37\)](#) depends only on E, C_v, C_s , and the dependence of these constants was stated above. This completes the proof with the asserted quantifiers. $\square$

3.9 The endpoint pairing consequence

We now convert the root moment into the pairing estimate used in [Section 5](#). Constancy on a mark fiber is needed only at the first endpoint; the second endpoint can be any vector with the stated norm.

Corollary 3.11. *Under the hypotheses and choices of [Theorem 3.5](#), let $f \in \mathcal{H}_{\text{ph}}$ be supported on positions $I = [X, 2X)$ and independent of the chosen marks, with value f_n at position n . Suppose*

$$\sup_{n \in I} |f_n| \leq \exp(O(\sqrt{L})), \quad \|f\|, \|g\| \leq X^{1/2} L^{C_3}.$$

Then

$$|\langle f, \mathcal{A}g \rangle| \ll XL^{-E+2C_3}. \quad (3.48)$$

Only the first endpoint must be independent of its mark list.

Proof. Let $H \geq 1$ bound every displacement $|kD|$. Since D has J s factors, all at most $\exp(L^d)$,

$$\log H = O_J(TL^d). \quad (3.49)$$

Partition I into consecutive intervals I_j of length $\lceil H \rceil$, except possibly the last. Put $f_j = \mathbf{1}_{I_j} f$ and restrict g to the H -neighborhood of I_j to obtain g_j . These neighborhoods have bounded overlap, and

$$\langle f, \mathcal{A}g \rangle = \sum_j \langle f_j, \mathcal{A}g_j \rangle, \quad \sum_j \|g_j\|^2 \ll \|g\|^2.$$

For $B_0 = \mathcal{A}\mathcal{A}^* \geq 0$, spectral Hölder gives

$$|\langle f_j, \mathcal{A}g_j \rangle| \leq \|g_j\| \|f_j\|^{1-1/R} \langle f_j, B_0^R f_j \rangle^{1/(2R)}.$$

The finite matrix $(\langle u_n, B_0^R u_m \rangle)_{n,m \in I_j \cap \mathbb{Z}}$ is positive semidefinite. Its largest eigenvalue is at most its trace, so mark independence implies

$$\langle f_j, B_0^R f_j \rangle \leq \left(\sum_{n \in I_j \cap \mathbb{Z}} |f_n|^2 \right) d_j, \quad d_j := \sum_{n \in I_j \cap \mathbb{Z}} \langle u_n, B_0^R u_n \rangle.$$

There is no assertion here that the u_n have equal norm, or that ordered lists return to their starting values. By [Equation \(3.49\)](#),

$$\left(O(H) \sup |f_n|^2 \right)^{1/(2R)} = \exp \left(O_J \left(\frac{TL^d + \sqrt{L}}{L^{0.52}} \right) \right) = O(1).$$

Sum the resulting block bounds by Hölder with exponents 2, $2R/(R-1)$, and $2R$. This gives

$$|\langle f, \mathcal{A}g \rangle| \ll \|g\| \|f\|^{1-1/R} \left(\sum_j d_j \right)^{1/(2R)}.$$

The theorem bounds the last sum by $|I \cap \mathbb{Z}| L^{-2RE} \ll XL^{-2RE}$. Substitution of the two endpoint norm bounds proves [Equation \(3.48\)](#); the extra factor $L^{-C_3/R}$ is bounded for fixed C_3 . $\square$

4 Small ideal kernels at every frequency

We use the groups and probability measures of [Section 3](#). In this section each big group $\mathcal{P}_g$, $g \in \mathcal{B}$, is the set of primes in an interval contained in $[\exp(L^c), \exp(L^d)]$. In particular, intersecting a group with a further interval again gives an interval of primes. This additional assumption allows us to use [Theorem 2.2](#) with one label varying and all other labels fixed.

Theorem 4.1 (Construction of the residual ideal operator). *Fix $E_{\text{id}} > 0$ and a fixed frequency exponent $C_4 \geq 0$. There are fixed integers m, J_0 and a fixed constant C_1 , depending only on these parameters, ℓ , and the prime-group bounds, with the following property. For every fixed $J \geq J_0$, there is a family consisting of the raw pattern*

$$C_g = \emptyset \quad \text{for every } g, \quad K_0 = 1,$$

and signed comparison patterns satisfying

$$C_g = \emptyset \quad (g \in \mathcal{S}), \quad |C_g| \leq m \quad (g \in \mathcal{B}), \quad \sum_{\nu} \sup |K_{\nu}| \leq L^{C_1},$$

such that the symmetrized ideal operator of [Section 3](#) satisfies

$$\sup_{\Theta \in \mathbb{R}, |\zeta| \leq L^{C_4}} \|\mathcal{T}(\Theta)\|_{2 \rightarrow 2} \leq L^{-E_{\text{id}}} \quad (4.1)$$

for sufficiently large x . The coefficients and the family are independent of Θ and of the individual value of ζ . Their defining parameters are independent of J ; J only supplies the available shared slots. The threshold for x may depend on J .

More precisely, designate m of the first J slots in each big group as probes, and split the big groups into two blocks. Each comparison frees nonempty sets A and B of designated probes in the respective blocks, and its coefficient has the form

$$-(-1)^{|A|+|B|} \mathcal{K}_A(D_A, X_A) \mathcal{K}_B(D_B, X_B). \quad (4.2)$$

Here D_A, D_B are the products of the free labels used to form D , X_A is the product of the target labels in A , and X_B the product of the input labels in B . These coefficients use no shared labels and none of the last ℓ labels. Each kernel is a finite sum of products of matched log-cell indicators and pairs of Dirichlet characters of conductor bounded by a fixed power of L .

The construction separates an approximation from a cancellation. We first build kernels that replace selected shared products by independent products in a bilinear pairing, for each prescribed fixed error exponent. The tests may depend on the entire ordered label tuples, as they will when we condition on the labels outside the freed slots.

The cancellation then decomposes the input test in the first-block probes and the target test in the second-block probes. Each decomposition is orthogonal: its pieces are mean zero in a specified set of these probe coordinates and independent of the others. Freeing A averages the source coordinates in A , and freeing B averages the target coordinates in B ; the asymmetric coefficient above leaves these coordinates absent from the multiplier. Thus only components independent of the freed coordinates survive. Alternating subset signs cancel every pair for which the input component is independent of at least one first-block probe and the target component is independent of at least one second-block probe. Every remaining pair contains a component that is mean zero in every probe of one block. Such components have small norms by permutation symmetry and the averaging of the last ℓ coordinates. We prove this cancellation after the comparison estimate, so that its use on conditional tuple tests is explicit.

All label tuples in the next three subsections have their product probability measure, with repetitions allowed.

4.1 Products of labels and an elementary bilinear bound

Let Λ be a nonempty set of slots, using $k_g \leq m$ slots from group g , and let P_Λ denote the product of its labels. Disjointness of the prime groups and unique factorization give, on the support of this product,

$$\rho_\Lambda(n) := \mathbb{P}(P_\Lambda = n) = \frac{1}{n} \prod_g \frac{k_g!}{V_g^{k_g} \prod_{p \in \mathcal{P}_g} a_p!} \leq \frac{L^\kappa}{n}, \quad (4.3)$$

$$\kappa = C_0(\log(m!) + m \log \max(1, v_-^{-1})),$$

where a_p is the multiplicity of p in n . The probability is zero if the prescribed group multiplicities do not hold. Indeed, each factor apart from $1/n$ is at most $m! \max(1, v_-^{-1})^m$, and there are at most $C_0 T$ big groups. In particular, this exponent does not depend on J .

This estimate applies to tests on the entire ordered label tuple. If f is any such L^2 test, supported where $c_1 U \leq P_\Lambda \leq c_2 U$ for fixed $c_1, c_2 > 0$, put $a_n = \mathbb{E}[f \mathbf{1}_{P_\Lambda=n}]$. Cauchy–Schwarz on each fiber gives

$$\sum_n |a_n|^2 \leq \sum_n \rho_\Lambda(n) \mathbb{E}[|f|^2 \mathbf{1}_{P_\Lambda=n}] \leq \frac{L^\kappa}{c_1 U} \|f\|_2^2. \quad (4.4)$$

No assumption that f is a function of the product has been made.

Lemma 4.2 (Integer bilinear estimate). *Let $U, V \geq 1$. Suppose a_n, b_t are supported in integer intervals of lengths at most CU, CV , respectively, where C is fixed. If*

$$\alpha = u/r + \beta, \quad (u, r) = 1, \quad r \geq 1, \quad |\beta| \leq r^{-2},$$

then

$$\left| \sum_{n,t} a_n b_t e(\alpha n t) \right| \ll_C \|a\|_{\ell^2} \|b\|_{\ell^2} [U + (1 + V/r) \{U + r \log(2r)\}]^{1/2}. \quad (4.5)$$

The containing intervals need not begin at 1, and the coefficients may vanish on arbitrary subsets of them.

Proof. Extend the coefficients by zero to their containing intervals. Cauchy–Schwarz in n , followed by expansion of the square and the geometric-sum bound on the full interval of n , gives

$$\begin{aligned} \left| \sum_{n,t} a_n b_t e(\alpha n t) \right|^2 &\ll_C \|a\|_{\ell^2}^2 \left(U \sum_t |b_t|^2 + \sum_{1 \leq |h| \leq CV} \min\{U, \|\alpha h\|_{\mathbb{R}/\mathbb{Z}}^{-1}\} \sum_t |b_t b_{t+h}| \right) \\ &\ll_C \|a\|_{\ell^2}^2 \|b\|_{\ell^2}^2 \left(U + \sum_{1 \leq h \leq CV} \min\{U, \|\alpha h\|_{\mathbb{R}/\mathbb{Z}}^{-1}\} \right). \end{aligned}$$

At a zero denominator the minimum is interpreted as U . For $r \geq 2$, split the h -range into $O_C(1 + V/r)$ consecutive blocks, each of diameter at most $r/2$. If $h \neq h'$ belong to a block, their difference is nonzero and has magnitude less than r , so

$$\|\alpha(h - h')\|_{\mathbb{R}/\mathbb{Z}} \geq \|u(h - h')/r\|_{\mathbb{R}/\mathbb{Z}} - \frac{|h - h'|}{r^2} \geq \frac{1}{2r}.$$

Thus the phases in a block are $1/(2r)$ -separated on the circle. Ordering them by distance from zero bounds their contribution by

$$O_C \left(U + \sum_{1 \leq j \leq 2r} \frac{r}{j} \right) = O_C(U + r \log(2r)).$$

For $r = 1$, the trivial bound $O_C(UV)$ for the whole sum gives the asserted estimate. This proves the lemma. $\square$

For two independent selected products of sizes U, V , combining [Equations \(4.4\) and \(4.5\)](#) shows that the probability-space bilinear form with phase $e(\alpha P_A P_B)$ has norm at most

$$CL^\kappa \left(\frac{1}{V} + \frac{1}{r} + \frac{\log(2r)}{U} + \frac{r \log(2r)}{UV} \right)^{1/2}. \quad (4.6)$$

The same assertion holds with the phase multiplied by $(P_A P_B)^{i\zeta}$ for any real ζ , since this factor separates between the two tests and preserves their norms.

4.2 Log cells and conditional character operators

For a fixed nonempty slot set Λ , let D, X be independent products of label tuples of this type. Given $Q \geq 1$, $0 < h \leq 1$, and $0 < \xi \leq 1$, put

$$I_j = [jh, (j+1)h), \quad \nu_{\Lambda j} = \mathbb{P}(\log D \in I_j), \quad \mathcal{C}_Q = \{\chi : \chi \text{ primitive, } \text{cond } \chi \leq Q\}.$$

The conductor-1 character is included. Define

$$\mathcal{K}_\Lambda(D, X) = \sum_{j: \nu_{\Lambda j} \geq \xi} \frac{\mathbf{1}_{\log D \in I_j} \mathbf{1}_{\log X \in I_j}}{\nu_{\Lambda j}} \sum_{\chi \in \mathcal{C}_Q} \overline{\chi(D)} \chi(X). \quad (4.7)$$

We use $\mathcal{K}_A, \mathcal{K}_B$ for the corresponding slot sets. All powers of L chosen below are fixed before J .

There are at most Q^2 characters in $\mathcal{C}_Q$. The number N_Λ of cells of positive mass satisfies

$$N_\Lambda \ll_m 1 + TL^d/h, \quad \sup |\mathcal{K}_\Lambda| \leq Q^2 \xi^{-1}, \quad \sup_X \mathbb{E}_D |\mathcal{K}_\Lambda(D, X)| \leq Q^2. \quad (4.8)$$

The last inequality follows because, for fixed retained X , the mass $\nu_{\Lambda j}$ of its cell cancels the denominator. The kernel is zero for discarded X and has support only where $|\log D - \log X| < h$.

We record explicitly the character estimate under a cell restriction. If Q is a fixed power of L and χ, χ' are distinct members of $\mathcal{C}_Q$, then, for every fixed $D_* > 0$,

$$\left| \mathbb{E} \left[\mathbf{1}_{\log D \in I_j} \overline{\chi(D)} \chi'(D) \right] \right| \ll_{D_*} L^{-D_*}, \quad (4.9)$$

uniformly in j and in the nonempty slot set with at most m slots per group. To see this, hold all but one label fixed. The cell and the selected prime group restrict the remaining prime to an interval, possibly empty. The character $\overline{\chi} \chi'$ on the common modulus is nonprincipal: otherwise the uniqueness of primitive induction would imply $\chi = \chi'$. Its modulus is at most Q^2 . For every fixed K , [Theorem 2.2](#), summed against the character over reduced residue classes and applied with a larger accuracy exponent, gives

$$A_\rho(t) := \sum_{p \leq t} \rho(p) \ll_K t(\log t)^{-K}$$

for these nonprincipal characters, uniformly for $t \geq y := \exp(L^c)$. The allowed modulus is a fixed power of $\log y$, as required there. Partial summation on any subinterval of $[y, \exp(L^d)]$ bounds its reciprocal-prime sum by

$$O_K \left((\log y)^{-K} + \int_y^\infty \frac{dt}{t(\log t)^K} \right) = O_K((\log y)^{1-K}).$$

Divide by $V_g \geq v_-$ and choose K sufficiently large in terms of D_* . Averaging the other labels proves Equation (4.9); its precision is independent of the width or location of the interval.

For a retained cell of mass ν , the integral operator defined by Equation (4.7) is the character frame operator on the conditional probability space:

$$f \mapsto \sum_{\chi \in \mathcal{C}_Q} \chi \langle \chi, f \rangle_{L^2(\mathbb{P}(\cdot|I_j))}. \quad (4.10)$$

Its nonzero eigenvalues are those of the character Gram matrix. Every diagonal entry is 1, since all selected primes exceed Q^2 for large x . By Equation (4.9), every off-diagonal entry is $O_{D_*}(\xi^{-1}L^{-D_*})$. The Gram matrix therefore has norm at most

$$1 + O_{D_*}(Q^2\xi^{-1}L^{-D_*}) \leq 2 \quad (4.11)$$

once the character precision has been chosen sufficiently large. Passing from the conditional to the original measure multiplies both squared norms by ν , so the operator norm is unchanged. Different cells are orthogonal blocks, and discarded cells have zero operator. Consequently kernel integration on the full label space has norm at most 2. The adjoint and transpose have the same bound, so the integration may be moved onto either test in a bilinear form. In particular neither a cell-count factor nor a factor ξ^{-1} is lost in this operator norm.

4.3 Uniform comparison for arbitrary tuple tests

Lemma 4.3 (Comparison kernel). *Fix m , $A_0 > 0$, and $C_4 \geq 0$. There are fixed powers Q, h^{-1}, ξ^{-1} of L , independent of J , such that the kernels in Equation (4.7) have the following property. Let A, B be nonempty sets of slots, using at most m slots per big group. Take independent label tuples for D_A, X_A, D_B, X_B with the prescribed slot laws. If f and g are arbitrary L^2 tests on the tuples making up X_B and X_A , respectively, then for every $\alpha \in \mathbb{R}$ and $|\zeta| \leq L^{C_4}$,*

$$\begin{aligned} & \left| \mathbb{E} \overline{f(X_B)} g(X_A) [(X_A X_B)^{i\zeta} e(\alpha X_A X_B) \right. \\ & \quad \left. - \mathcal{K}_A(D_A, X_A) \mathcal{K}_B(D_B, X_B) (D_A D_B)^{i\zeta} e(\alpha D_A D_B)] \right| \leq L^{-A_0} \|f\|_2 \|g\|_2. \end{aligned} \quad (4.12)$$

Here notation such as $f(X_B)$ denotes a tuple test, not an assumption of dependence only on the product. The kernels are independent of α and ζ .

Proof. Split the tests according to dyads $X_A \in [U, 2U)$ and $X_B \in [V, 2V)$. Each selected product lies between $\exp(L^c)$ and $\exp(mC_0TL^d)$; thus for sufficiently large x there are at most L^{d+1} dyads for each product. We prove a bound L^{-B_0} on each pair of dyads, with $B_0 > A_0 + 2(d+1) + 10$. Summing the bounds proves the lemma, even using the crude bound of the original test norm for every dyadic restriction.

Let $Z = UV$ and introduce a further fixed power S of L . Dirichlet approximation with maximum denominator $\lfloor Z/S \rfloor$ gives a reduced fraction satisfying

$$\alpha = u/r + \beta, \quad 1 \leq r \leq Z/S, \quad |\beta| \leq r^{-2}, \quad |\beta|Z \leq 2S/r. \quad (4.13)$$

Here $Z/S \rightarrow \infty$ and the factor 2 accounts for the integer part. This approximation is used only in the proof: the kernel does not depend on the choice of the rational approximation.

Denominators larger than Q . Suppose $r > Q$. Apply Equation (4.6) to the raw term. For the comparison term, move the two separate kernel integrations onto the two tests. Their norms increase by at most 2 each by Equation (4.11). The resulting tests on D_A, D_B are supported in $[U/2, 4U]$ and $[V/2, 4V]$ for large x , by the log localization. The same bilinear bound therefore applies with an absolute constant change. In both uses we collapse arbitrary tuple tests by Equation (4.4) before applying the integer lemma.

The two scales exceed every fixed power of L , and $\log(2r) \leq L^{d+1}$ for large x . Since $r > Q$ and $r \leq UV/S$, the bound for either term, apart from an absolute constant and the two test norms, is

$$L^\kappa \left(o(L^{-A}) + Q^{-1} + L^{d+1} S^{-1} \right)^{1/2} \quad (4.14)$$

for every fixed A . Choosing Q, S sufficiently large gives the required dyadic precision. This part is uniform in all real ζ , because the Mellin factor separates.

Denominators at most Q . Suppose $r \leq Q$. All products under consideration are units modulo r . Fourier expansion on the finite group of units gives

$$e(uv/r) = \sum_{\psi \bmod r} c_\psi \psi(v), \quad \sum_{\psi} |c_\psi| \leq \sqrt{\varphi(r)} \leq \sqrt{Q}. \quad (4.15)$$

Indeed, in normalized counting measure the left side has squared norm 1, so Parseval gives $\sum |c_\psi|^2 = 1$ and Cauchy–Schwarz proves the stated bound. The unique primitive character inducing ψ has conductor at most r and is included in $\mathcal{C}_Q$.

First remove discarded cells. Their total probability for either product is at most

$$\mathbb{P}(\mathcal{E}) \leq (N_A + N_B)\xi, \quad \mathcal{E} = \{X_A \text{ or } X_B \text{ lies in a discarded cell}\}. \quad (4.16)$$

The comparison term is zero on this event. The raw term there has absolute value at most

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\mathcal{E}} |f(X_B)g(X_A)|] &\leq \mathbb{P}(\mathcal{E})^{1/2} (\mathbb{E}|f(X_B)g(X_A)|^2)^{1/2} \\ &= \mathbb{P}(\mathcal{E})^{1/2} \|f\|_2 \|g\|_2. \end{aligned}$$

The last equality uses independence of the two tuple spaces. This argument also applies to tests concentrated in a discarded cell; it does not assert that restriction to a small event is small as an L^2 operator.

On retained cells, write the remaining scalar multiplier in the log variable as

$$M(v) = \exp(i\zeta v) e(\beta e^v).$$

On the enlarged product range its derivative has magnitude $O(L^{C_4} + S)$, by Equation (4.13). Since both matched log products differ by less than h , we may replace $M(\log(D_A D_B))$ by $M(\log(X_A X_B))$ while leaving the rational phase unchanged. For each fixed retained pair (X_A, X_B) , Equation (4.8) bounds the error after integration in D_A, D_B by

$$O(Q^4 h(S + L^{C_4})). \quad (4.17)$$

This is a pointwise error. Pairing it with the tests costs at most the same bound times $\|f\|_2 \|g\|_2$, since their spaces are independent.

For a character ψ in Equation (4.15), the Gram calculation also gives the following pointwise reproduction formula for retained X :

$$\mathbb{E}_D \mathcal{K}_\Lambda(D, X) \psi(D) = \psi(X) + O_{D_*}(Q^2 \xi^{-1} L^{-D_*}). \quad (4.18)$$

In fact, the summand corresponding to its inducing primitive character is exactly $\psi(X)$: the diagonal conditional expectation is 1. Each other summand is bounded by Equation (4.9) divided by the retained-cell mass. There are at most Q^2 such summands. Choose D_* so that the displayed error is at most 1. Applying the formula on both sides and then Equation (4.15) reproduces $e(uX_A X_B/r)$ with pointwise error

$$O_{D_*}(Q^{5/2} \xi^{-1} L^{-D_*}). \quad (4.19)$$

Multiplication by the fixed scalar $M(\log(X_A X_B))$ has modulus one and does not change this error. Together, Equations (4.16), (4.17) and (4.19) establish the desired comparison on a pair of dyads.

A noncircular choice of parameters. For completeness the following sufficient inequalities fix all the precisions just used. Write

$$Q = L^{q_*}, \quad S = L^{s_*}, \quad h = L^{-h_*}, \quad \xi = L^{-x_*}.$$

After m, A_0, C_4 have been fixed, choose successively

$$\begin{aligned} B_0 &> A_0 + 2(d+1) + 10, \\ q_*, s_* &> 2(B_0 + \kappa + 10) + d + 1, \\ h_* &> 4q_* + \max(s_*, C_4) + B_0 + 10, \\ x_* &> 2B_0 + d + 1 + h_* + 10, \\ D_* &> B_0 + x_* + \frac{5}{2}q_* + 10. \end{aligned} \quad (4.20)$$

Indeed $N_A, N_B \leq L^{d+1+h_*}$ for large x . The successive lines control, respectively, dyadic summation, Equation (4.14), Equation (4.17), the square root of Equation (4.16), and both Equations (4.11) and (4.19). Fixed implied constants are absorbed by the margins in these inequalities. Finally use Theorem 2.2 at the precision needed for this D_* . None of these choices involves J . $\square$

4.4 Symmetric probes and exact cancellation

Proof of Theorem 4.1. Split $\mathcal{B}$ into blocks $\mathcal{B}_1, \mathcal{B}_2$ of sizes s_1, s_2 differing by at most one. For sufficiently large x each size is at least $c_0 T/3$. In each group designate the first m of the first J slots as probes; their sets in the two blocks are $\mathcal{I}_1, \mathcal{I}_2$. Let $P_i = |\mathcal{I}_i| = ms_i$ and $P = P_1 + P_2$. For every pair of nonempty subsets $A \subseteq \mathcal{I}_1, B \subseteq \mathcal{I}_2$, free exactly $A \cup B$ and use the coefficient in Equation (4.2), with the kernels just constructed. The other comparison target slots, including the last ℓ , have exactly the independent laws in the ideal operator's definition.

It suffices to bound pairings against symmetric vectors F, H , since the operator has the orthogonal symmetrizing projection on both sides. In every pairing, average out the last ℓ input and target slots in each group. No multiplier uses these coordinates. Denote the resulting functions on the first J coordinates by F', H' . Both operations are contractions. The raw pairing is

$$R(F', H') = \mathbb{E} \left[\overline{F'} H' D_{\mathcal{B}}^{i\zeta} e(\Theta D_{\mathcal{B}}) \right], \quad (4.21)$$

where input and target use the same first J labels. Its bilinear norm is at most 1.

We use the orthogonal product-space decomposition of Efron and Stein [5, Section 2, Decomposition Lemma], and then count the components retained by the symmetry and probe constraints. For a coordinate i , write E_i for averaging its label. Decompose F' by the probes in the first block, and H' by those in the second:

$$\begin{aligned} F' &= \sum_{S \subseteq \mathcal{I}_1} F_S, & F_S &= \prod_{i \in S} (1 - E_i) \prod_{i \in \mathcal{I}_1 \setminus S} E_i F', \\ H' &= \sum_{T' \subseteq \mathcal{I}_2} H_{T'}, & H_{T'} &= \prod_{i \in T'} (1 - E_i) \prod_{i \in \mathcal{I}_2 \setminus T'} E_i H'. \end{aligned}$$

These are orthogonal decompositions. The indicated components are separately mean zero in each indicated probe and independent of the other probes of their block. Put $F_{\text{full}} = F_{\mathcal{I}_1}$ and $H_{\text{full}} = H_{\mathcal{I}_2}$.

We need the precise effect of having first averaged out the last ℓ slots. In one group, decompose the original symmetric vector on all $M = J + \ell$ coordinates by the same coordinate averaging projections. Among its components on subsets of cardinality j , permutation symmetry makes their squared norms equal. Requiring all m probes to occur in the subset and all last ℓ slots to be absent retains exactly the fraction

$$\frac{\binom{M-m-\ell}{j-m}}{\binom{M}{j}} = \frac{(j)_m (M-j)_\ell}{(M)_{m+\ell}}. \quad (4.22)$$

The fraction is zero when a required cardinality is impossible. If $M \geq (m+\ell)(m+\ell-1)$, then

$$(M)_{m+\ell} = M^{m+\ell} \prod_{i=0}^{m+\ell-1} (1 - i/M) \geq \frac{1}{2} M^{m+\ell},$$

using $\prod(1 - a_i) \geq 1 - \sum a_i$ for $0 \leq a_i \leq 1$. Consequently Equation (4.22) is at most

$$2(j/M)^m (1 - j/M)^\ell \leq 2 \left(\frac{m}{m+\ell} \right)^m \left(\frac{\ell}{m+\ell} \right)^\ell \leq 2\ell^\ell m^{-\ell} =: \theta_m. \quad (4.23)$$

This bound tensorizes even when the vectors are not products across groups. Indeed, first resolve the orthogonal decomposition by the tuple of component cardinalities in the groups of the block. Independent within-group permutations give equal squared norms for all tuples of subsets with those cardinalities. The fraction surviving the stated inclusions and exclusions is the product of Equation (4.22) over those groups. Sum the resulting inequality over the cardinality tuples. Averaging the last ℓ coordinates in other groups is a further contraction. We obtain

$$\|F_{\text{full}}\|_2 \leq \theta_m^{s_1/2} \|F\|_2, \quad \|H_{\text{full}}\|_2 \leq \theta_m^{s_2/2} \|H\|_2. \quad (4.24)$$

Choose m so large that $\theta_m < 1$ and $-(c_0/6) \log \theta_m > E_{\text{id}} + 3$. The right sides are then at most $L^{-E_{\text{id}}-3} \|F\|_2$ and $L^{-E_{\text{id}}-3} \|H\|_2$. This is the only use of large m , and it is possible because $\ell \geq 1$.

Now fix a component pair $(F_S, H_{T'})$ and let

$$U = \mathcal{I}_1 \setminus S, \quad V = \mathcal{I}_2 \setminus T'$$

be its missing probes. A comparison term with free sets A, B vanishes unless $A \subseteq U$ and $B \subseteq V$. For if $i \in A \cap S$, the input label in coordinate i is unshared, is absent from the multiplier, and is absent from the target. Integrating it annihilates the mean-zero input component. The same reasoning applies to a target label in $B \cap T'$. This uses the deliberate asymmetry in Equation (4.2): its A labels come from the target, and its B labels from the input.

Suppose the inclusions hold. Condition on all shared first- J labels outside $A \cup B$, and write C for their product. The input component is independent of the A labels, and the target component is independent of the B labels. Thus in the raw term the remaining tests are an arbitrary tuple test on X_B and one on X_A , respectively, on independent product spaces. In the comparison term, the unused input A labels and target B labels integrate out, leaving exactly the same two tests and independent free products D_A, D_B . The fixed shared product contributes $C^{i\zeta}$, of modulus one, and changes the additive frequency to $\alpha = \Theta C$. [Theorem 4.3](#) therefore matches the comparison term before its sign to $R(F_S, H_{T'})$ with error at most

$$L^{-A_0} \|F_S\|_2 \|H_{T'}\|_2. \quad (4.25)$$

To justify this bound after conditioning, apply the lemma to the two conditional tests, average its error over the shared labels, and use Cauchy–Schwarz. The averages of their squared conditional norms are precisely $\|F_S\|_2^2$ and $\|H_{T'}\|_2^2$. Uniformity in every real α is essential at this step.

When U, V are nonempty, the signs give exact cancellation:

$$\sum_{\substack{\emptyset \neq A \subseteq U \\ \emptyset \neq B \subseteq V}} (-1)^{|A|+|B|} = \left(\sum_{\emptyset \neq A \subseteq U} (-1)^{|A|} \right) \left(\sum_{\emptyset \neq B \subseteq V} (-1)^{|B|} \right) = (-1)(-1) = 1. \quad (4.26)$$

The initial minus sign in the comparison coefficient therefore cancels the raw pairing for this component pair, up to the errors just estimated. If either missing set is empty, no comparison survives. All these exceptional raw component pairs together are exactly

$$R(F_{\text{full}}, H') + R(F' - F_{\text{full}}, H_{\text{full}}).$$

We sum them before taking absolute values. By [Equations \(4.21\) and \(4.24\)](#), their total is at most $2L^{-E_{\text{id}}-3} \|F\|_2 \|H\|_2$. In particular there is no subset-count loss in this bound.

For the approximation errors, there are at most 3^P admissible triples consisting of a component pair and its free subsets: each probe is either indicated, missing but not freed, or freed. Since $P \leq mC_0T$, choose

$$A_0 > E_{\text{id}} + mC_0 \log 3 + 10.$$

Then the sum of [Equation \(4.25\)](#), even bounding each component norm by the original norm, is at most $L^{-E_{\text{id}}-10} \|F\|_2 \|H\|_2$. Together with the exceptional contribution this proves [Equation \(4.1\)](#) for sufficiently large x .

Finally, the number of comparisons is at most $2^P \leq L^{mC_0 \log 2}$, and [Equation \(4.8\)](#) bounds each coefficient by $Q^4 \xi^{-2}$. For example, after making the choices in [Equation \(4.20\)](#), any fixed

$$C_1 > mC_0 \log 2 + 4q_* + 2x_* + 1$$

bounds the total coefficient cost, including the raw pattern. Expanding the two cell sums and two character sums also has only a fixed power of L terms. Both free products contain a prime, so $D_A, D_B \geq \exp(L^c)$. The coefficient uses only the four products specified in [Equation \(4.2\)](#); all assertions about labels and subsequent separation follow directly from [Equation \(4.7\)](#).

The order of choices is now explicit: choose m from [Equation \(4.24\)](#), then A_0 to pay for 3^P , then the dyadic and kernel precisions in [Equation \(4.20\)](#), and hence C_1 . Only afterward require $J \geq m$ and $J + \ell \geq (m + \ell)(m + \ell - 1)$. These requirements define J_0 . A still larger fixed J can therefore be chosen to meet [Theorem 3.5](#) without changing any kernel parameter or the exponent C_1 . $\square$

5 Lifted shifted correlations

We retain the prime groups and their notation from [Sections 3 and 4](#). Thus the small groups have primes between $\exp(L^a)$ and $\exp(L^b)$, the big groups are intervals of primes between $\exp(L^c)$ and $\exp(L^d)$, and $0 < a < b < c < d < 0.47$. All groups are disjoint, their harmonic masses V_g lie in $[v_-, v_+]$, and each kind has between fixed positive multiples of T groups. The parameters $q \in (0, 1)$ and $\ell \geq 1$ are fixed. Write $\mathcal{P} = \bigcup_g \mathcal{P}_g$ and let n_* be the part of n supported on primes outside $\mathcal{P}$.

5.1 Endpoint hypotheses and the correlation theorem

For a vector $\mathbf{t} = (t_g)_g$ of nonnegative integers, define

$$W_{\mathbf{t}}(n) = q^{\omega(n) - \sum_g t_g} \prod_g \frac{(\omega_g(n))_{t_g}}{V_g^{t_g}}. \quad (5.1)$$

The falling factorial counts ordered lists of distinct group primes dividing n . More generally, if $\mathbf{b}$ is a function on these lists, put

$$W_{\mathbf{t}}^{\mathbf{b}}(n) = q^{\omega(n) - \sum_g t_g} \prod_g V_g^{-t_g} \sum_{\substack{(p_{g,1}, \dots, p_{g,t_g}) \text{ distinct in each group} \\ p_{g,j} \in \mathcal{P}_g, p_{g,j} | n}} \mathbf{b}((p_{g,j})_{g,j}).$$

An empty admissible-list set gives value zero. Write W_{ℓ} when all $t_g = \ell$. For each fixed $t_{\max}$,

$$\sup_{\max t_g \leq t_{\max}} |W_{\mathbf{t}}^{\mathbf{b}}(n)| \leq \|\mathbf{b}\|_{\infty} L^{C(t_{\max})}. \quad (5.2)$$

Indeed, $q^{y-t}(y)_t/V_g^t$ has a bounded supremum over integers $y \geq t$, uniformly for $t \leq t_{\max}$, and there are $O(T)$ groups.

An endpoint is a function invariant under multiplication by $\mathcal{P}$ -integers of the form

$$F_0(n) = \sum_{m p r = n_*} \alpha_m \chi_0(m) m^{i\sigma_0} \mathbf{1}_{p \in I_p, P^-(p) > W} p^{i\sigma_1} c_r. \quad (5.3)$$

Here $|\alpha_m|, |c_r| \leq L^C$, $\alpha_m = 0$ unless $P^-(m) > W$, $I_p \subset [x^\tau, x^\eta]$ is an interval, $0 < \tau < \eta < 1/4$ are fixed, χ_0 is a Dirichlet character of modulus at most L^C , and $|\sigma_0|, |\sigma_1| \leq L^C$. The variable p ranges over integers. The sequences may depend on x , and all the parameters and sequences can differ at the two endpoints. Since $W = \exp(\sqrt{L})$ exceeds every group prime, m and p are $\mathcal{P}$ -free for sufficiently large x . We impose the following condition on the α of at least the first endpoint:

$$\left| \sum_{m \in I} \frac{\alpha_m \chi(m) m^{iu}}{m} \right| \leq L^{-B} \quad \begin{cases} I \subset [1, x^2] \text{ an interval,} \\ \chi \text{ a Dirichlet character of modulus at most } L^B, \\ |u| \leq L^B. \end{cases} \quad (5.4)$$

This includes principal and imprimitive characters. An existing range restriction on α is part of the sequence in this condition.

Theorem 5.1 (Lifted shift cancellation). *Fix all the group and endpoint bounds above. Fix $D_0, C > 0$ and a fixed compact interval $[c_\Psi, C_\Psi] \subset (0, \infty)$. Suppose $0 < |k| \leq L^C$ is integral, $xL^{-C} \leq Z \leq xL^C$, $|a_1|, |a_2| \leq L^C$, and Ψ is supported in that interval with $\|\Psi^{(j)}\|_{\infty} \ll_j L^{C(j+1)}$ for every $j \geq 0$. There exists a fixed B , depending only on these bounds and D_0 , such that [Equation \(5.4\)](#) implies*

$$\sum_{\substack{z > 0 \\ z+k > 0}} \frac{\Psi(z/Z)}{z} z^{ia_1} (z+k)^{ia_2} \overline{F_0(z)} G_0(z+k) W_{\ell}(z) W_{\ell}(z+k) \ll L^{-D_0}. \quad (5.5)$$

The bound is uniform over the indicated sequences, intervals, characters, frequencies, and cutoffs. It also holds when the discrepancy-bearing endpoint or its coefficients have been conjugated.

Conjugation in the last assertion is harmless: conjugate Equation (5.4) and replace (χ, u) by $(\overline{\chi}, -u)$. Products with additional characters and bounded power twists are covered by increasing B . We give the proof in several stages.

5.2 The physical lift and removal of shared labels

We first show that the raw correlation together with its signed comparison correlations is small. The harmonic factor $1/n$ will turn sums over shared prime divisors into draws with laws μ_g when the shared product is removed.

Take the raw pattern and signed comparison patterns of Theorem 4.1. Write m_{probe} for the fixed probe bound m in that theorem; the letter m in Equation (5.3) denotes an integer factor. As in the physical construction, $M = J + \ell$, the edge product is D , and the positions are $n, n' = n + kD$. Insert into the physical pairing the root weight $\Psi(n/(ZD))/n$, the edge phase $D^{-i(a_1+a_2)}$, and the mark-independent vectors

$$f(n, \mathbf{p}) = n^{-ia_1} F_0(n) q^{(\omega(n) - Ms)/2}, \quad g(n', \mathbf{p}') = (n')^{ia_2} G_0(n') q^{(\omega(n') - Ms)/2}. \quad (5.6)$$

Our pairing is conjugate-linear in its first entry, so its integrand contains $n^{ia_1} \overline{F_0(n)}$ at the source and $(n')^{ia_2} G_0(n')$ at the target. Every factor involving D is inside the slot symmetrizations. On the support, $n \asymp ZD = x^{1+o(1)}$ and $n'/n = 1 + O(L^C/Z)$. A partition into at most $O(L)$ dyads $n \asymp X$ therefore suffices, and one can restrict the other endpoint to an interval of length $O(X)$.

We check carefully that the exponent in the endpoint norm bound is independent of J and of the probe count. On a physical state $\omega(n) \geq Ms$, so the damping in Equation (5.6) is at most one. There are at most $\tau(n_*)^2$ ordered factorizations in Equation (5.3), whence $|F_0(n)| \leq L^{2C} \tau(n_*)^2$. Fix an ordered physical list S' of M distinct labels per group, with product $P(S')$. For $P(S') \mid n$, one has $n_* \mid n/P(S')$ and hence

$$\sum_{\substack{n \asymp X \\ P(S') \mid n}} \tau(n_*)^4 \leq \sum_{v \asymp X/P(S')} \tau(v)^4 \ll \frac{X}{P(S')} L^{C_5}.$$

Here $P(S') \leq \exp(O_J(TL^d)) = x^{o(1)}$, so the interval in v is still of positive power length. Moreover

$$\sum_{S'} \frac{1}{P(S')} \prod_g V_g^{-M} \leq \prod_g \left(\sum_{p \in \mathcal{P}_g} \frac{1}{p V_g} \right)^M = 1.$$

Thus both endpoint vectors have norm at most $X^{1/2} L^{C_6}$ in the physical state measure, with C_6 independent of J and the probe count. Also $|F_0(n)| \leq \exp(O(\sqrt{L}))$: an integer of size $x^{O(1)}$ has $O(\sqrt{L})$ prime factors above W , with multiplicity, so it has at most $2^{O(\sqrt{L})}$ choices for each of the rough divisors m, p . The vectors are independent of the selected marks.

For clarity, let $\Phi(u) = \Psi(e^u)$. Logarithmic Fourier inversion gives

$$\Psi(n/(ZD)) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\Phi}(v) (n/Z)^{iv} D^{-iv} dv.$$

The derivative hypotheses imply an $L^{O(1)}$ integral norm, and, for a fixed sufficiently large C_7 , tails $|v| > L^{C_7}$ have arbitrarily large negative log powers. This cutoff exponent can be fixed from the

derivative bounds: increasing the number of integrations by parts increases the saving without increasing C_7 . On the dyad multiply the first vector by X/n and $(n/Z)^{-iv}$, and divide the pairing by X . The conjugation in the pairing then supplies the factor $(n/Z)^{iv}$ above. The required ideal frequency is $\zeta = -a_1 - a_2 - v$. Consequently [Theorem 3.11](#) and [Theorem 4.1](#) bound the entire residual pairing by L^{-D_1} , for any prescribed D_1 , after choosing the transfer target first, then the ideal family, then J . The norm exponent just proved makes this ordering possible. The discarded Fourier tails obey the same conclusion by [Theorem 3.4](#); their accuracy may be chosen after the absolute comparison costs are known. All these steps apply to the sum of the patterns, with their signs.

We next calculate the individual pattern after this lift. In group g write c_g for its free-slot count and $s_{R,g} = J - c_g$ for its shared-slot count; thus $t_g = \ell + c_g$. Let D_R, D_C be the products of the shared and free labels, so $D = D_R D_C$. This partitions labels by their role on the edge; the earlier factorization $D = D_S D_B$ partitions them by prime-group size. Set $n = D_R w$, $n' = D_R w'$, so $w' = w + k D_C$. The physical state and target-list normalizations in this group are

$$V_g^{-M-t_g} = V_g^{-s_{R,g}-2t_g}, \quad M + t_g = s_{R,g} + 2t_g.$$

The factor $1/D_R$ from $1/n$ therefore changes each shared-label sum into a draw of mass $1/(pV_g)$, and leaves $V_g^{-t_g}$ at each endpoint for its unshared lists. Away from shared overlaps,

$$\omega_g(D_R w) - M = \omega_g(w) - t_g.$$

The damping already in the graph and that in [Equation \(5.6\)](#) give this full power of q at each endpoint. Invariance gives $F_0(D_R w) = F_0(w)$, and the phases become

$$(D_R w)^{ia_1} (D_R w')^{ia_2} (D_R D_C)^{-i(a_1+a_2)} = (w/D_C)^{ia_1} (w'/D_C)^{ia_2}.$$

The coefficient K_ν of a comparison uses the free labels and designated unshared big labels, and uses no shared labels. Thus its dependence is preserved when these shared harmonic draws are summed.

Here are error bounds justifying the exclusions in this calculation. For fixed w, w', D_C and any earlier shared draws, only $O_J(L)$ values are forbidden to a new shared draw: endpoint prime divisors, free labels, and previous shared labels. Each atom has mass at most $v_-^{-1} \exp(-L^a)$. The $O_J(T)$ draws have total excluded probability $L^{O_J(1)} \exp(-L^a)$. For distinct shared values the exact identity is

$$\omega(D_R w) - Ms = \omega(w) - \sum_g t_g - \#\{p : p \mid D_R, p \mid w\}.$$

In bounding the actual overlap terms by the unrestricted marked sum, the damping loss is at most $q^{-2Ms} = L^{O_J(1)}$. Unshared divisors still divide w, w' and their count cannot increase. By [Equation \(5.2\)](#), endpoint Cauchy–Schwarz and divisor moments, their remaining absolute harmonic average is $L^{O(1)}$. Shared exclusions therefore cost $L^{O_J(1)} \exp(-L^a)$.

Once shared overlaps are excluded, the remaining ban excludes free labels from the unshared endpoint mark lists. Every excluded configuration therefore has a free prime p' dividing an endpoint. Since $p' \mid D_C$, it follows that both $p' \mid w$ and $p' \mid w'$; it suffices to bound all such multiples. On $w \asymp Y_0 = Z D_C$, put $w = p'v$ and $w' = p'(v + k D_C/p')$. Invariance and Cauchy–Schwarz give

$$\sum_{\substack{w \asymp Y_0 \\ p' \mid w}} \frac{|F_0(w) G_0(w + k D_C)|}{w} \ll \frac{L^{O(1)}}{Y_0} \left(\sum_{v \asymp Y_0/p'} \tau(v)^4 \right)^{1/2} \left(\sum_{v \asymp Y_0/p'} \tau(v + k D_C/p')^4 \right)^{1/2} \ll \frac{L^{O(1)}}{p'}.$$

Both shifted ranges stay in positive intervals of size $O(Y_0/p')$, and $Y_0/p' = x^{1+o(1)}$. The mark weights, coefficients, and total pattern costs are fixed log powers. Summing over the free slots therefore makes this error negligible too. This is an averaged bound over multiples of p' , not a restriction on the arbitrary coefficients.

We have obtained, up to errors smaller than every fixed negative log power, the following expression for each pattern:

$$\mathbb{E}_{\text{free}} \sum_{\substack{w>0 \\ w'=w+kD_C>0}} \frac{\Psi(w/(ZD_C))}{w} (w/D_C)^{ia_1} (w'/D_C)^{ia_2} \overline{F_0(w)} G_0(w') W_{\mathbf{t}}(w) W_{\mathbf{t}}(w') \mathbb{E}_{\text{marks}} K_{\nu}. \quad (5.7)$$

The outer expectation uses independent free-label laws μ_g . Conditional on these labels and the positions, the inner expectation uses independent uniform ordered unshared lists at each endpoint; its term is zero if either list set is empty. The raw pattern has $D_C = 1$ and $t_g = \ell$, so it is exactly the sum in Equation (5.5). It remains to bound all the comparison terms.

5.3 Comparison shifts and their major arcs

Expand a comparison coefficient from Theorem 4.3 by its two logarithmic cells and two characters. Its factors on the free products D_A, D_B separate, and the remaining factors are bounded weights on specified big unshared marks at the two endpoints. Cell normalizations, the number of terms, and the integral costs are fixed log powers, independent of J . Let H' be the product of the lower size endpoints of the two cells. Then $D_A D_B \asymp H'$ and

$$H' \geq \exp(L^c), \quad H' \leq \exp(O(m_{\text{probe}} T L^d)), \quad Y = ZH' = x^{1+o(1)}. \quad (5.8)$$

Insert smooth size cutoffs at $w, w' \asymp Y$, separate $\Psi(w/(ZD_A D_B))$ logarithmically, and include Y/w in the first cutoff. Apart from $1/Y$ and fixed log-power costs, every resulting term is

$$\mathbb{E}_{D_A, D_B} b_1(D_A) b_2(D_B) \sum_w \overline{\mathcal{U}(w)} \mathcal{V}(w + kD_A D_B). \quad (5.9)$$

Here $|b_1|, |b_2| \leq 1$ after extracting constants; each is supported on its cell. Each endpoint sequence is its invariant endpoint times a smooth cutoff at size Y , a power w^{iv} with $|v| \leq L^{O(1)}$, and a weight $W_{\mathbf{t}}^{\mathbf{b}}(w)$. The additional mark function uses only big labels. In particular

$$\sum_w |\mathcal{U}(w)|^2 + \sum_w |\mathcal{V}(w)|^2 \ll Y L^{C_8}. \quad (5.10)$$

All frequency and cutoff bounds here are fixed before B .

The endpoint sequences are now independent of the free products. Fourier inversion isolates the weighted shift average as a multiplier: bilinear cancellation will control it away from small rational frequencies, and endpoint Fourier energy will control the remaining contribution. Use $\hat{F}(\theta) = \sum_w F(w) e(\theta w)$ on $\mathbb{R}/\mathbb{Z}$. The multiplier in Equation (5.9) is

$$M(\theta) = \mathbb{E} b_1(D_A) b_2(D_B) e(-k\theta D_A D_B).$$

We claim that, for any fixed desired saving D_2 , it is $O(L^{-D_2})$ outside

$$\mathfrak{M} = \bigcup_{1 \leq r \leq L^{C_{\text{arc}}}} \bigcup_{u \bmod r} \{\theta : |\theta - u/r|_{\mathbb{R}/\mathbb{Z}} \leq L^{C_{\text{arc}}}/H'\},$$

if C_{arc} is large enough. To verify the coefficient hypothesis of [Theorem 4.2](#), collapse a free product of size U_0 to integer weights $\rho(n)$. Each group contributes at most m_{probe} slots. Unique factorization bounds the multiplicity at n by $\prod_g (m_{\text{probe}}!)$, and the V_g normalizations cost another fixed constant per slot. Hence $\rho(n) \leq L^{O(1)}/n$ and $\sum_n \rho(n) = 1$; for bounded tests,

$$\sum_{n \asymp U_0} |b(n)\rho(n)|^2 \ll L^{O(1)}/U_0.$$

Both free products are nonempty and each contains a big prime, so both sizes exceed every fixed power of L .

Put $Q_D = \lfloor H'/L^{C'} \rfloor$. Dirichlet approximation to $k\theta$ gives a reduced u'/r' with $r' \leq Q_D$ and $|k\theta - u'/r'| \leq 1/(r'Q_D) \leq (r')^{-2}$, where $1/(r'Q_D) \ll L^{C'}/(r'H')$. If $r' \leq L^{C'}$, division by k places θ on one of the stated arcs after increasing C_{arc} ; reduction can only decrease the denominator. Otherwise $L^{C'} < r' \ll H'/L^{C'}$. The normalized bilinear bound of [Theorem 4.2](#) then saves any prescribed log power by increasing C' . This proves the claim. Parseval and [Equation \(5.10\)](#) control the complementary contribution to [Equation \(5.9\)](#), including its $1/Y$.

Put $H = H'/L^{C_{\text{arc}}}$. On each arc it is enough to prove, for any fixed $A > 0$,

$$\int_{|\beta| \leq 1/H} |\widehat{\mathcal{U}}(u/r + \beta)|^2 d\beta \ll_A Y L^{-A}. \quad (5.11)$$

Indeed $|M| \leq 1$, and Cauchy–Schwarz with [Equation \(5.10\)](#) gives $L^{-A/2+C_s/2}$ per arc after division by Y . There are at most $L^{2C_{\text{arc}}+O(1)}$ arcs. We establish [Equation \(5.11\)](#) with A chosen after all these costs. Its proof uses three features of the first endpoint: one small prime can be extracted from its ordered marks, every factorization contains the long rough-integer factor in [Equation \(5.3\)](#), and the m coefficients satisfy [Equation \(5.4\)](#). After conversion to Mellin frequencies, these supply cancellation in different ranges.

5.4 One small mark and the rational character expansion

Choose any small group g_s and a designated slot among its $t_{g_s} = \ell \geq 1$ ordered marks. The endpoint mark weight is independent of the small labels. Whenever no prime of $\mathcal{P}_{g_s}$ has its square dividing w , removal of the prime in this slot gives the exact identity

$$W_{\mathbf{t}}^{\mathbf{b}}(w) = \sum_{\substack{p_s \in \mathcal{P}_{g_s} \\ p_s | w}} \frac{1}{V_{g_s}} W_{\mathbf{t}-\mathbf{e}_{g_s}}^{\mathbf{b}}(w/p_s). \quad (5.12)$$

Removing the designated slot is a bijection of the ordered lists; both ω_{g_s} and t_{g_s} decrease by one. There is therefore no extra factorial or damping factor. Both sides, even on the exceptional integers, are $L^{O(1)}$ by [Equation \(5.2\)](#) and $\omega_{g_s}(w) = O(L)$. Since $w_* \mid w/p_s^2$ on a square multiple, the squared norm of the error after multiplication by F_0 is at most

$$L^{O(1)} \sum_{p_s \in \mathcal{P}_{g_s}} \sum_{\substack{w \asymp Y \\ p_s^2 | w}} \tau(w_*)^4 \ll Y L^{O(1)} \sum_{p_s \in \mathcal{P}_{g_s}} p_s^{-2} \ll Y L^{O(1)} e^{-L^a}.$$

Every Y/p_s^2 is $x^{1+o(1)}$. Parseval makes this error negligible in [Equation \(5.11\)](#).

After [Equation \(5.12\)](#), combine the remaining $\mathcal{P}$ -part with the residual factor of [Equation \(5.3\)](#). The full integer factorization is $w = mp_s pr$, with residual coefficient

$$c_{r_*} W_{\mathbf{t}-\mathbf{e}_{g_s}}^{\mathbf{b}}(r).$$

This identity uses that m, p have no group factors. The residual coefficient is a function of r alone and is bounded by $L^{O(1)}$.

All of m, p_s, p are units modulo the arc denominator $r_0 \leq L^{C_{\text{arc}}}$. Separate residual factors by $d_0 = (r, r_0)$ and put $q_0 = r_0/d_0$. On units modulo q_0 the exact finite character expansion is

$$e(uv/q_0) = \sum_{\chi \bmod q_0} c_{u,q_0}(\chi) \chi(v), \quad c_{u,q_0}(\chi) = \frac{1}{\varphi(q_0)} \sum_{v \bmod q_0}^* e(uv/q_0) \overline{\chi(v)}.$$

Each coefficient has modulus at most one. Use this with $v = mp_s p(r/d_0)$; the condition $(r, r_0) = d_0$ guarantees that v is a unit modulo q_0 . The factors $\chi(r/d_0)$ and the gcd restriction are absorbed into the residual coefficient. This includes every character modulo the actual quotient, so principal and imprimitive components are retained. The number of divisors and characters is a fixed log power, and products with χ_0 have polylogarithmic modulus.

It now suffices to prove the local energy bound at zero for sequences

$$a_w = \Psi_1(w/Y) w^{iv_1} \sum_{mp_s pr=w} \alpha_m \chi_m(m) m^{i\sigma_0} b_s(p_s) \mathbf{1}_{p \in I_p, P^-(p) > W} \chi_p(p) p^{i\sigma_1} d_r, \quad (5.13)$$

where $p_s \in \mathcal{P}_{g_s}$, $|b_s| \ll 1$, $|d_r| \leq L^{O(1)}$, the characters and frequencies have fixed log-power bounds, and Ψ_1 has the same kind of smooth size support as above. Fixed-order divisor moments give $\sum_w |a_w|^2 \ll Y L^{O(1)}$.

5.5 From local Fourier energy to a Mellin integral

We record the scale conversion explicitly. Choose a smooth nonnegative K of integral one, supported in a sufficiently small fixed neighborhood of zero that $|\widehat{K}(\xi)| \geq 1/2$ for $|\xi| \leq 1$, using the Fourier convention with $e(-\xi u)$. Plancherel on $\mathbb{R}$ gives

$$\int_{|\beta| \leq 1/H} |\widehat{a}(\beta)|^2 d\beta \ll H^{-2} \int_{\mathbb{R}} \left| \sum_w a_w K((w-v)/H) \right|^2 dv. \quad (5.14)$$

Only $v \asymp Y$ contributes. Keep the integral restricted to this range when replacing the kernel by $K(v \log(w/v)/H)$; no estimate for this logarithmic kernel near $v = 0$ is used. On the union of the two kernels' supports within this range, $|w-v| \ll H$ and their arguments differ by $O(H/Y)$. Therefore, by Cauchy–Schwarz over $O(H)$ integers and then integrating the centers allowed for each w , the squared norm of the error in Equation (5.14) is

$$\ll (H/Y)^2 \sum_w |a_w|^2. \quad (5.15)$$

This is smaller than $Y L^{-A}$ for every fixed A , since $H = x^{o(1)}$ and $Y = x^{1+o(1)}$.

Set $h = H/Y$ and $P(t) = \sum_w a_w w^{it}$. With $v = Y e^u$, logarithmic Fourier inversion gives

$$\sum_w a_w K(v \log(w/v)/H) = \frac{h}{2\pi} \int_{\mathbb{R}} e^{-u} \widehat{K}(h e^{-u} t / (2\pi)) v^{-it} P(t) dt.$$

Insert a smooth compactly supported function $\rho(u)$ equal to one on the required range. The Fourier transform in u of $\rho(u) e^{-u} \widehat{K}(h e^{-u} t / (2\pi))$, denoted $A(s, t)$, satisfies, for each fixed j ,

$$|A(s, t)| \ll_j (1 + |s|)^{-2} (1 + |ht|)^{-j}.$$

This follows by two integrations by parts in u , since $\widehat{K}$ is Schwartz and u stays in a fixed compact interval. Fourier-expand in u , apply Minkowski's integral inequality in s , and apply Plancherel in u for each s . As $dv \ll Y du$, the normalization is $H^{-2}Yh^2 = Y^{-1}$. After replacing j by a larger integer, we obtain

$$\int_{|\beta| \leq 1/H} |\widehat{a}(\beta)|^2 d\beta \ll_j Y^{-1} \int_{\mathbb{R}} (1 + |ht|)^{-j} |P(t)|^2 dt + (H/Y)^2 \sum_w |a_w|^2. \quad (5.16)$$

By [Theorem 2.6](#), for any dyadic time scale $T' \geq 1$,

$$Y^{-2} \int_{T' \leq |t| \leq 2T'} |P(t)|^2 dt \ll (1 + T'/Y) L^{C_9}.$$

Consequently the portion $|t| > L/h$ in [Equation \(5.16\)](#) is $O(YL^{-A})$ after choosing j large enough. For the remaining portion split the four factor variables in [Equation \(5.13\)](#) into dyads. There are $O(L^4)$ boxes and only boxes whose joint size is comparable to Y contribute. Write

$$\frac{P(t)}{Y} = \sum_w \frac{1}{w} ((w/Y)\Psi_1(w/Y)) w^{i(t+v_1)} \sum_{mp_s pr=w} (\text{the four coefficients}).$$

Fourier-separate the parenthesized cutoff in logarithmic coordinates. Its integral cost is a fixed log power; truncate at a fixed log-power frequency with arbitrary saving. The tails are bounded by the preceding mean values. The frequency shifts enlarge $|t| \leq L/h$ only to $|t| \leq 2L/h$, because $1/h$ exceeds every fixed log power. We have reduced the assertion to

$$\int_{|t| \leq 2L/h} |P_s(t)P_l(t)\mathcal{R}(t)|^2 dt \ll L^{-A'} \quad (5.17)$$

for any prescribed fixed A' , on each retained box, where

$$\begin{aligned} P_s(t) &= \sum_{p_s \asymp P} b_s(p_s) p_s^{-1+it}, \\ P_l(t) &= \sum_{\substack{p \asymp P', \ p \in I_p \\ P^-(p) > W}} \chi_p(p) p^{-1+i\sigma_1+it}, \\ \mathcal{R}(t) &= \left(\sum_{m \asymp M_0} \alpha_m \chi_m(m) m^{-1+i\sigma_0+it} \right) \left(\sum_{r \asymp R_0} d_r r^{-1+it} \right). \end{aligned}$$

Here and below dyads may be intersected with their existing support. We have

$$e^{L^a}/2 \leq P \leq e^{L^b}, \quad x^\tau/2 \leq P' \leq x^\eta, \quad PM_0P'R_0 \asymp Y. \quad (5.18)$$

For any subproduct of these four polynomials, collapse its coefficients to $\sum_{n \asymp U} c_n n^{it}$. A fixed number of convolution factors and [Theorem 2.5](#) give

$$\sum_n |c_n|^2 \ll L^{C_{10}}/U. \quad (5.19)$$

This uses a fixed divisor moment: the number of marked groups has already been absorbed by [Equation \(5.2\)](#), not by a growing divisor order. Individually all these polynomials are also $L^{O(1)}$ in absolute value by their harmonic sums.

The remaining integral has three regimes. Most times are controlled by the small-prime polynomial P_s : where $|P_s(t)|$ is a sufficiently small negative power of L , the ordinary mean square of $P_l\mathcal{R}$ suffices. We will show that the exceptional times occupy only $Y^{o(1)}$ unit intervals. On these intervals at large times, P_l is small by oscillation over its long rough-integer range; a sparse mean square for $\mathcal{R}$ makes that saving sufficient. For $|t|$ bounded by a fixed power of L , the m factor in $\mathcal{R}$ is small by the original discrepancy hypothesis. We prove the exceptional-time and long-factor estimates first, then choose the discrepancy precision in the completion of the argument.

5.6 Exceptional times and a sparse mean square

The split into ordinary and exceptional times, followed by high prime-polynomial moments and a sparse mean-square estimate, is in the spirit of the method of Matomäki and Radziwiłł; compare [13, Section 2.1 and Section 4, Lemmas 8–9]. The estimates needed here are proved below; we do not invoke their short-interval theorem.

Fix a large constant A_s and let $\mathcal{E} = \{t : |t| \leq 2L/h, |P_s(t)| > L^{-A_s}\}$. Outside $\mathcal{E}$, use the indicated gain and the mean square of $P_l\mathcal{R}$, of joint size Y/P . Its length exceeds the time range, since

$$\frac{Y/P}{2L/h} = \frac{H}{2LP} \rightarrow \infty$$

by $b < c$ and Equations (5.8) and (5.18). Thus this portion of Equation (5.17) is $O(L^{-2A_s+C_{11}})$.

The factorial coefficient estimate in the next proof is the standard prime-polynomial moment argument; compare [20, Lemma 3 and its proof]. We keep its dependence on the growing power explicit.

Lemma 5.2 (Number of exceptional unit intervals). *The set $\mathcal{E}$ meets at most*

$$\exp\left(O(L^b + L^{1-a} \log L)\right) = Y^{o(1)}$$

intervals $[j, j+1]$, uniformly in the bounded coefficients b_s .

Proof. Choose a point of $\mathcal{E}$ in each occupied interval, and split the interval indices into three residue classes to obtain separated points. Put $j_0 = \lfloor \log Y / \log(2P) \rfloor$. The polynomial $P_s(t)^{j_0}$ has indices at most Y . Its coefficient-square sum is at most

$$j_0! \left(\sum_{p_s \lesssim P} |b_s(p_s)|^2 p_s^{-2} \right)^{j_0} \leq j_0! (C/P)^{j_0}.$$

Indeed an integer has at most $j_0!$ ordered representations as a product of j_0 primes; Cauchy–Schwarz on each fiber proves the first inequality even with repeated primes.

For a polynomial $Q(t) = \sum_{n \leq Y} c_n n^{it}$ and unit-separated points in $[-O(Y), O(Y)]$, the unit-interval Sobolev inequality and bounded overlap imply

$$\sum_j |Q(t_j)|^2 \ll \int_{-O(Y)}^{O(Y)} (|Q(t)|^2 + |Q'(t)|^2) dt \ll Y(\log(2Y))^{O(1)} \sum_n |c_n|^2.$$

In the last step use Theorem 2.6; differentiation multiplies a coefficient by $i \log n$, so the constants here are independent of j_0 . The range $2L/h$ is $o(Y)$, as required. If $N_{\mathcal{E}}$ is the number of occupied intervals, it follows that

$$N_{\mathcal{E}} \ll Y L^{O(1)} j_0! (C L^{2A_s} / P)^{j_0}.$$

Now $j_0 = O(L^{1-a})$ and $\log Y - j_0 \log P = O(L^b + j_0)$ by the definition with $2P$. Taking logarithms and using $\log(j_0!) \leq j_0 \log j_0$ proves the asserted estimate. $\square$

Lemma 5.3 (Sparse residual mean square). *For the occupied unit intervals I_j in Theorem 5.2,*

$$\sum_j \sup_{t \in I_j} |\mathcal{R}(t)|^2 \ll L^{C_{12}}.$$

Proof. The collapsed residual size is

$$U \asymp Y/(PP') \geq Y^{1-\eta-o(1)} > Y^{0.6}$$

for large x , and its coefficient-square sum is $L^{O(1)}/U$ by Equation (5.19). On a containing integer interval $n \asymp U$, the Gram kernel satisfies, uniformly for $|z| \ll Y$,

$$\left| \sum_{n \asymp U} n^{iz} \right| \ll \frac{U}{1+|z|} + Y^{1/2}. \quad (5.20)$$

For $|z| < 1$ this is trivial. For $1 \leq |z| \leq cU$ with small fixed c , the phase $z \log n/(2\pi)$ has monotone first derivative of magnitude comparable to $|z|/U$ and less than $1/2$; Theorem 2.7 gives $O(U/|z|)$. For $cU < |z| \ll Y$ the second derivative test gives $O(|z|^{1/2} + U|z|^{-1/2}) = O(Y^{1/2})$.

Choose a maximum point of $|\mathcal{R}|$ in each closed I_j and again split the indices into three residue classes. In each class these points are at least unit separated. For $R = Y^{o(1)}$ such points, the absolute row sums of their Gram matrix are at most

$$C(U \log(2Y) + RY^{1/2}) \ll U \log(2Y).$$

The first term follows by grouping distances into unit intervals; the second is absorbed by $U > Y^{0.6}$. Schur's bound on this Gram matrix, applied to the map $(c_n) \mapsto (\sum_n c_n n^{it_j})_j$, now gives

$$\sum_j |\mathcal{R}(t_j)|^2 \ll U \log(2Y) \sum_n |c_n|^2 \ll L^{O(1)}.$$

Summing the three classes proves the statement, including the suprema over the intervals. $\square$

5.7 Cancellation of the long rough-integer factor

The following elementary estimate is the reason for requiring a rough integer factor of positive power length in every endpoint.

Lemma 5.4 (Long rough-integer polynomial). *Fix $\tau > 0$, $\eta < 1/4$, $C > 0$, and $A > 0$. Let $x^\tau/2 \leq P' \leq x^\eta$, let $J' \subset [P', 2P']$ be an interval, and let χ have modulus at most L^C . If $|\sigma| \leq L^C$, then, for a sufficiently large fixed B_0 , uniformly whenever $L^{B_0} \leq |t|$ and $|t + \sigma| \leq x^2$,*

$$\left| \sum_{\substack{n \in J' \\ P^-(n) > W}} \chi(n) n^{-1+i(t+\sigma)} \right| \ll L^{-A}.$$

The constant B_0 can be chosen after C, A, τ, η .

Proof. Set $h_s = 2\lceil T^2 \rceil$ and $D = W^{4h_s+2} = \exp(O(\sqrt{L}T^2)) = x^{o(1)}$. By Theorem 2.10, there is a polynomial

$$U(n) = \sum_{d|n} \lambda_d \geq \mathbf{1}_{P^-(n) > W},$$

whose coefficients have modulus at most one, are supported on squarefree $d \leq D$ with primes at most W , and satisfy

$$\sum_{n \in [P', 2P']} (U(n) - \mathbf{1}_{P^-(n) > W}) \ll P' e^{-h_s} + D.$$

The summand is nonnegative, so restriction to any J' preserves this upper bound. Replacing roughness by U in the normalized sum therefore costs at most $O(e^{-h_s} + D/P')$, smaller than every fixed negative power of L . Also

$$\sum_d \frac{|\lambda_d|}{d} \leq \prod_{p \leq W} (1 + 1/p) \ll \log W = \sqrt{L}. \quad (5.21)$$

Write q' for the character modulus. Terms with $(d, q') > 1$ vanish. For the other d , split $n = dv$ by $v = q'y + a$ modulo q' . The character is constant on a class. Its y variable ranges over an interval at size $N = P'/(dq') \geq x^{\tau/2}$ for large x . Put $u = t + \sigma$. By taking $B_0 > C + 1$ we ensure $|u| \geq L^{B_0}/2$. We claim, on every subinterval of the allowed y range,

$$\left| \sum_y \exp(iu \log(q'y + a)) \right| \ll N(|u|^{-1} + N^{-\delta}) \quad (5.22)$$

for a fixed $\delta > 0$ depending only on τ .

If $|u| \leq cN$, the monotone first derivative test gives $O(N/|u|)$, since the derivative of $u \log(q'y + a)/(2\pi)$ stays away from nonzero integers. Otherwise put $\varepsilon_0 = 1/10$. The overlapping ranges

$$N^{k-2+\varepsilon_0} \leq |u| \leq N^{k-\varepsilon_0}, \quad k \geq 2,$$

cover $cN < |u| \leq x^2$ using $2 \leq k \leq K = \lceil 4/\tau \rceil + 3$. The k th derivative of this phase has constant sign and magnitude comparable to $|u|N^{-k}$. For $k = 2$, the second derivative test directly saves a positive power. For $k > 2$, apply the van der Corput differencing inequality $k - 2$ times with positive integer shifts at most $H_0 = \lfloor N^{\varepsilon_0/(2k)} \rfloor$. After shifts $h_1, \dots, h_{k-2}$, the second derivative is an iterated integral of the k th derivative, and hence has constant sign and magnitude comparable to

$$|u|N^{-k}h_1 \cdots h_{k-2}, \quad \text{between } c_k N^{-2+\varepsilon_0} \text{ and } C_k N^{-\varepsilon_0/2}.$$

On an interval of any length $m \leq CN$, the second derivative bound is

$$O\left(m\sqrt{\lambda} + \lambda^{-1/2}\right) \ll N^{1-\varepsilon_0/4}.$$

To justify the estimate uniformly for short subintervals, extend each sequence by zero in a containing interval of length $O(N)$. The differencing inequality for its normalized sum has the form $B_j^2 \ll H_0^{-1} + B_{j+1}$, where B_{j+1} bounds the average of the absolute normalized correlations. Their supports are intersections of translates of the original interval, and so are again intervals. All shifts are $o(N)$, preserving the derivative bounds. Iterating from the final exponent $\varepsilon_0/4$ proves a bound $O(N^{-\delta_k})$ for the original normalized sum, with $\delta_k = \varepsilon_0/(k2^{k-1})$. Taking $\delta = \varepsilon_0/(K2^{K-1})$ proves Equation (5.22) in every case. The differencing inequality itself follows by averaging H_0 translates and applying Cauchy–Schwarz; its zero-shift term is precisely the displayed $O(H_0^{-1})$.

Partial summation of $1/[d(q'y + a)]$ costs $O(1/P')$ times the uniform unweighted bound. Summing the at most q' classes gives

$$\left| \sum_{\substack{n \in J' \\ d|n}} \chi(n) n^{-1+iu} \right| \ll \frac{1}{d} (|u|^{-1} + x^{-\tau\delta/2}).$$

Finally Equation (5.21) bounds the total by

$$O(e^{-h_s} + D/P') + O\left(\sqrt{L}(L^{-B_0} + x^{-\tau\delta/2})\right).$$

Choosing $B_0 > A + 2$ in addition to its previous constraint proves the lemma. $\square$

5.8 Completion and order of parameters

We finish Equation (5.17). Outside $\mathcal{E}$ its integral is $O(L^{-2A_s+C_{11}})$, so choose A_s sufficiently large. On the exceptional intervals with $|t| \geq L^{B_0}$, Theorem 5.4 gives an arbitrarily strong uniform bound for P_t . Its range condition holds since $2L/h = o(Y) = x^{1+o(1)} < x^2/2$; the fixed frequency shift does not change this. The polynomial P_s is bounded, and Theorem 5.3 gives

$$\int_{\mathcal{E} \cap \{|t| \geq L^{B_0}\}} |P_s(t)P_t(t)\mathcal{R}(t)|^2 dt \ll \left(\sup_{|t| \geq L^{B_0}} |P_t(t)|^2 \right) L^{O(1)}.$$

Here the supremum is restricted to the actual time range, and each occupied interval has length one. Choose the saving in the long-polynomial lemma after the exponent in the sparse bound.

For all remaining $|t| \leq L^{B_0}$ use Equation (5.4) on the m polynomial in $\mathcal{R}$. The retained dyads satisfy $PM_0P'R_0 \asymp Y$, so $m \ll Y = x^{1+o(1)} < x^2$ for large x , also after the smooth cutoff has been separated. Its character modulus is a fixed log power, and its frequency is $t + \sigma_0$. A single sufficiently large B therefore bounds this polynomial by L^{-B} throughout the range. Every other factor has absolute value $L^{O(1)}$, and the length of this time interval is $2L^{B_0}$. The resulting integral is $L^{-2B+B_0+O(1)}$, giving the required accuracy. This proves Equation (5.17), then Equation (5.11) by Equation (5.16), and then every comparison bound. The residual pairing and the raw identity in Equation (5.7) prove Equation (5.5).

For completeness, the choices occur in the following order. First fix the endpoint bounds, group geometry, cutoff derivative bounds, and desired saving D_0 . The uniform endpoint norm bound fixes the physical transfer target and its required ideal target; the fixed Mellin cutoff fixes the ideal frequency range. Next choose the probe count and comparison parameters of Theorem 4.1, then choose the fixed J required by transference. All pattern, cell, and absolute norm costs are now determined. Choose the separation accuracies, arc exponent, local energy target, and Mellin tail precision. Next choose A_s , the accuracy of the long rough polynomial and its B_0 . Finally choose one B in Equation (5.4) exceeding all the modulus, frequency, and low-time saving requirements, and let x tend to infinity. The residual L^2 bounds used before the choice of J depend only on the endpoint coefficient bounds and a fixed divisor moment. Increasing later Fourier tail accuracies uses more integrations by parts, with no enlargement of the already fixed ideal frequency range. Thus none of these choices is circular.

6 A Type II estimate

We apply the shifted-correlation estimate to factorizations $mn = 2u + 1$, where u carries a weight supported on many geometric prime bands. The coefficient of m will satisfy the discrepancy hypothesis of Theorem 5.1; the coefficient of n may be arbitrary within its stated bound. The task is to retain that discrepancy while converting multiplicative factorizations into additive correlations.

We split $u = eh$ with e slightly smaller than the scale of m , using a smooth allocation of the band primes. Cauchy's inequality in (e, n) then has a small enough outside factor to control its diagonal. Off the diagonal, the two factorizations give cross-products whose difference is a small nonzero integer. These are the shifted endpoints. All group factors and the compulsory rough integer are assigned to h , so both endpoints retain the form required by Theorem 5.1.

6.1 The weight and the uniform statement

Use the prime groups and the weight W_ℓ of [Section 5](#). In particular, the groups are disjoint, their harmonic masses belong to a fixed interval $[v_-, v_+] \subset (0, \infty)$, and there are between fixed positive multiples of T small groups and big groups. For fixed $0 < a < b < c < d < 0.47$, the small primes lie in $[\exp(L^a), \exp(L^b)]$ and the big primes in $[\exp(L^c), \exp(L^d)]$. The big groups have the interval structure required there. The damping $q \in (0, 1)$ and the integer $\ell \geq 1$ are fixed. Write u_* for the factor of u supported outside all these groups and $u_P = u/u_*$. Thus

$$W_\ell(u) = \prod_g q^{\omega_g(u) - \ell} \frac{(\omega_g(u))_\ell}{V_g^\ell}.$$

All constants describing these groups are among the fixed data below.

Fix $0 < c_1 < c_2$, $s' > 0$, and $\lambda > 1$, with $c_2 s' < 1/4$. For $0 \leq j \leq j_x$ put

$$s_j = s' L \lambda^{-j}, \quad \mathcal{Q}_j = \{p \text{ prime} : c_1 s_j \leq \log p \leq c_2 s_j\}.$$

Assume that these intervals are pairwise disjoint and disjoint from the prime groups above, and that $c_- T \leq s_{j_x} \leq c_+ T$ for fixed $0 < c_- \leq c_+$. Choose a fixed integer r_0 such that

$$(r_0 - 1)c_1/\lambda \geq c_2 + 1. \quad (6.1)$$

There are r_0 labelled slots in every band. Each normally takes a prime in $\mathcal{Q}_j$, with repetitions allowed. In one specified slot of a fixed band j_{**} replace the prime by an integer in

$$\begin{aligned} I_{**} &= [x^{\tau_*}, x^{\eta_*}], & P^-(p) &> W, \\ \tau_* &= c_1 s' \lambda^{-j_{**}}, & \eta_* &= c_2 s' \lambda^{-j_{**}}. \end{aligned} \quad (6.2)$$

The notation p for this slot will never assert that it is prime. For sufficiently large x , this slot is free of group primes, since $\exp(L^d) < W$.

Define

$$\begin{aligned} a_Q(v) &= \left(\prod_{j=0}^{j_x} \frac{1}{r_0!} \right) \# \{\text{admissible labelled slot lists with product } v\}, \\ A(u) &= a_Q(u_*) W_\ell(u). \end{aligned} \quad (6.3)$$

The exceptional integer can be fixed as a divisor of v . Thereafter the prime multiplicities in each disjoint band determine the list up to at most $r_0!$ permutations. Consequently

$$0 \leq a_Q(v) \leq \tau(v), \quad 0 \leq A(u) \leq L^{C'_A} \tau(u_*), \quad \sum_v \frac{a_Q(v)}{v} \leq L^{C'_A} \quad (6.4)$$

for a fixed exponent C'_A . Here the bound for W_ℓ follows by maximizing $q^{t-\ell}(t)_\ell/V_g^\ell$ in each group. For the last bound, factor the harmonic sum slot by slot: every prime slot has bounded harmonic mass, the exceptional slot has mass $O(L)$, and there are $O(T)$ slots. The exponent in [Equation \(6.4\)](#) depends only on the fixed data.

Theorem 6.1 (Type II). *Fix $D_* > 0$, $b_* > 0$, and $C > 0$, and require $\eta_* < b_*/4$ in [Equation \(6.2\)](#). Let U, V be dyadic scales satisfying*

$$UV \asymp x, \quad x^{b_*} \leq U, V \leq x^{1-b_*}.$$

Let Ψ be a fixed smooth function with compact support in a fixed compact subinterval of $(0, \infty)$. Suppose that $|\alpha_m|, |\beta_n| \leq L^C$, and that $\alpha_m = 0$ unless $P^-(m) > W$. There is a fixed exponent B , depending only on these data and the fixed group and band constants, such that the discrepancy hypothesis in Equation (5.4) on α implies

$$\sum_{\substack{m \asymp U, n \asymp V, u \geq 1 \\ mn = 2u+1}} A(u) \Psi(u/x) \alpha_m \beta_n \ll x L^{-D_*}. \quad (6.5)$$

Here the sequence to which the discrepancy hypothesis is applied includes the actual restriction $m \asymp U$ and any additional interval restriction on m . Explicitly, for that restricted sequence one requires

$$\left| \sum_{m \in I} \frac{\alpha_m \chi(m) m^{it}}{m} \right| \leq L^{-B} \quad \left(I \subset [1, x^2], \quad \text{mod}(\chi) \leq L^B, \quad |t| \leq L^B \right).$$

There is no discrepancy hypothesis on β . The constants are uniform over the sequences, scales, and interval restrictions satisfying these conditions.

6.2 Removing a large group-prime factor

Proof of Theorem 6.1. We first discard $u_P > x^{b_*/4}$ with an error smaller than every fixed negative power of L times x . Set $\theta = L^{-d}$. For one group write $a_p = (p^{1-\theta} - 1)^{-1}$. Since $p^\theta \leq e$ and the group primes tend to infinity, $a_p \leq 2e/p$. Its tilted marked harmonic sum is

$$\begin{aligned} & \sum_{v: \text{ all prime factors of } v \text{ in } \mathcal{P}_g} \frac{q^{\omega_g(v)-\ell} (\omega_g(v))_\ell}{V_g^\ell v^{1-\theta}} \\ &= \frac{1}{V_g^\ell} \prod_{p \in \mathcal{P}_g} (1 + qa_p) \sum_{\substack{p_1, \dots, p_\ell \in \mathcal{P}_g \\ \text{distinct, ordered}}} \prod_{i=1}^{\ell} \frac{a_{p_i}}{1 + qa_{p_i}} \leq C_0. \end{aligned}$$

The last constant is fixed because $v_- \leq V_g \leq v_+$. Taking the product over $O(T)$ groups gives

$$\sum_{\substack{v > x^{b_*/4} \\ v_* = 1}} \frac{W_\ell(v)}{v} \leq \exp(-\tfrac{1}{4} b_* L^{1-d}) L^{C_1}. \quad (6.6)$$

On the support of the sum in Equation (6.5), $u \asymp x$. The number of divisors m of $2u+1$ all of whose prime factors exceed W is at most $2^{\log(2u+1)/\log W} = \exp(O(\sqrt{L}))$. Using Equations (6.4) and (6.6), multiplying harmonic mass by $O(x)$, and using $1-d > 1/2$, the total discarded contribution is at most

$$x L^{C_2} \exp(-\tfrac{1}{4} b_* L^{1-d} + O(\sqrt{L})).$$

Call the remaining slot tuples good. This estimate is independent of the splitting precision introduced next.

6.3 A smooth partition into two factors

Put $E = UL^{-K_0}$, where the fixed integer K_0 will be chosen after the splitting costs have been bounded. Our target is a weighted partition into factorizations $u = eh$ with

$$EL^{-C_s} \leq e \leq E, \quad (6.7)$$

where C_s depends only on the band constants. The reason for placing e below U is that Cauchy's inequality will contribute an outside factor $EV \asymp xL^{-K_0}$. We will use this logarithmic saving to control the diagonal; it is therefore essential that the splitting costs be independent of K_0 .

Allocate all of u_P and the exceptional integer slot to h . This preserves the group weight and the long rough factor at each eventual shifted endpoint. Process the remaining prime slots in increasing order of their band index, and in a fixed order within each band. Write $v_i = \log p_i$, and let $s_{j(i)}$ be the scale of slot i . If $\delta_i \in \{0, 1\}$ records whether it is allocated to e , put

$$R_i = \log E - \sum_{t < i} \delta_t v_t.$$

Fix $\gamma \in C^\infty(\mathbb{R})$ with $0 \leq \gamma \leq 1$, $\gamma(t) = 0$ for $t \leq 0$, and $\gamma(t) = 1$ for $t \geq 1$. At slot i take it into e with probability $\gamma((R_i - v_i)/s_{j(i)})$; otherwise put it into h .

We verify the range in Equation (6.7) by tracking the unfilled logarithmic size. At every stage the induction target is

$$0 \leq R_i \leq \sum_{t \geq i} v_t + (c_2 + 1)s_{j_x}.$$

The terminal allowance accounts for a skip in the final band. On a good tuple the factors assigned to h before the procedure starts have product at most $x^{b_*/2}$. The sum of the available prime logarithms is therefore at least $(1 - b_*/2)L + O(1)$, whereas $0 < \log E \leq (1 - b_*)L - K_0T$ for large x . This proves the initial bound, even without the terminal allowance. Taking a slot of positive probability requires $R_i > v_i$ and subtracts v_i from both the residual and the available logarithm, so it preserves the bound and nonnegativity. Skipping a slot of positive probability requires

$$R_i < v_i + s_{j(i)} \leq (c_2 + 1)s_{j(i)}. \quad (6.8)$$

If a later band exists, its unprocessed prime slots have total logarithm at least $(r_0 - 1)c_1 s_{j(i)+1} \geq (c_2 + 1)s_{j(i)}$ by Equation (6.1). The subtraction of one slot covers the possible exceptional integer in that later band. Hence, after such a skip, the residual fits inside the remaining available logarithm. In the final band, a skip leaves at most $(c_2 + 1)s_{j_x}$, and subsequent operations cannot increase this residual. This proves the induction target. After processing the last slot it gives

$$0 \leq \log E - \log e \leq (c_2 + 1)s_{j_x}.$$

Since $s_{j_x} \leq c_+T$, this proves Equation (6.7) with $C_s = (c_2 + 1)c_+$. The geometric series for the remaining band scales also bounds $R_i/s_{j(i)}$ by a fixed constant. Both bounds are independent of K_0 .

Choose a fixed smooth compactly supported function ρ with values in $[0, 1]$, equal to one throughout the resulting possible range of $(R_i - v_i)/s_{j(i)}$. Replace the two transition functions by

$$f_1 = \rho\gamma, \quad f_0 = \rho(1 - \gamma).$$

This changes no branch on a good tuple. For an arbitrary tuple the sum of the two transitions is $\rho \leq 1$, so its total branch mass is at most one. Insert the resulting partition, with Equation (6.7), into Equation (6.3). Keep every original factorial normalization. The branches sum to one on good tuples, and extending back to all tuples costs at most the error in Equation (6.6). Every resulting h contains the exceptional rough integer.

Here are quantitative details of the smooth separation, including its independence from K_0 . Let $N = O(T)$ be the number of processed slots, and use the Fourier convention

$$f(t) = (2\pi)^{-1} \int \hat{f}(\xi) e^{i\xi t} d\xi.$$

Choose a fixed $C_F \geq 1$ bounding the Fourier L^1 norms of both f_0, f_1 , with the normalization $(2\pi)^{-1}$ included. Summing the Fourier total variations over the 2^N patterns gives at most $(2C_F)^N = L^{O(1)}$. Truncating each $|\xi_i|$ at L has total error bounded, for every fixed M , by

$$2^N N C_M C_F^{N-1} L^{-M}.$$

Thus any prescribed power saving is available. In a fixed pattern the product of Fourier exponentials is a scalar of modulus one times $\prod_t p_t^{i\sigma_t}$, where

$$\sigma_t = -\frac{\xi_t}{s_{j(t)}} - \delta_t \sum_{i>t} \frac{\xi_i}{s_{j(i)}}. \quad (6.9)$$

Indeed the scalar is $\exp(i \log E \sum_i \xi_i / s_{j(i)})$. Since $\sum_i 1/s_{j(i)} \ll 1/T$, the frequencies in Equation (6.9) are $O(L/T)$. All the bounds in this paragraph are independent of K_0 .

Separate $\Psi(eh/x)$ by Fourier inversion in $\log(eh/x)$ as well. Its Fourier L^1 norm is fixed, and the tail outside frequency L has arbitrary power saving; its two factors are powers of e and h . These errors can all be summed before Cauchy's inequality. Indeed, the absolute sum over the original factorizations is bounded by

$$L^{O(1)} \sum_{u \asymp x} \tau(u) \tau(2u+1) \ll x L^{O(1)} \quad (6.10)$$

by Cauchy's inequality and Theorem 2.5. The same estimate applies to each uniform transition or cutoff error, using the total Fourier variations of the other factors. Thus the original sum is, up to $O(xL^{-D^*-1})$, an integral and a sum of total variation at most L^{C_f} of expressions

$$B_0 = \sum_{\substack{EL^{-C_s} \leq e \leq E \\ n \asymp V}} a_e(e) \beta_n \sum_{\substack{m \asymp U, h \geq 1 \\ mn=2eh+1}} \alpha_m a_h(h). \quad (6.11)$$

The exponent C_f and every coefficient exponent in the next display are independent of K_0 :

$$\begin{aligned} |a_e(e)| &\leq L^{C_a}, & a_h(h) &= h^{i\varrho} W_\ell(h) H(h_*), \\ H(t) &= \sum_{pr=t} \mathbf{1}_{p \in I_{**}, P^-(p) > W} p^{i\sigma} c_r, & |c_r| &\leq L^{C_a}. \end{aligned} \quad (6.12)$$

In particular $|a_h(h)| \leq L^{O(1)} \tau(h_*)$. To verify these statements, only group-free prime slots go into e , so $(eh)_* = eh_*$ and $W_\ell(eh) = W_\ell(h)$. For a fixed pattern, the number of ordered pure-prime lists at a fixed product is at most $(r_0!)^{j_x+1} = L^{O(1)}$. This proves both the a_e bound and the bound for the residual coefficient c_r . Fixing the exceptional integer as a divisor gives the bound for a_h . Individual slot twists have modulus one; the common power of h from the cutoff is $h^{i\varrho}$. The frequencies ϱ, σ have fixed power bounds in L .

6.4 Cauchy's inequality and the diagonal

By Cauchy's inequality in (e, n) , Equation (6.11) satisfies

$$|B_0|^2 \leq EV L^{C_3} \mathcal{N}, \quad \mathcal{N} = \sum_{e,n} \eta_e(e/E) \eta_n(n/V) \left| \sum_{\substack{m \asymp U, h \geq 1 \\ mn=2eh+1}} \alpha_m a_h(h) \right|^2. \quad (6.13)$$

Here η_e, η_n are nonnegative smooth majorants, bounded by a fixed constant, equal to one on the original ranges. They vanish unless

$$\frac{1}{2}EL^{-C_s} \leq e \leq 2E, \quad n \asymp V.$$

Their derivatives of order j in their displayed arguments are $O_j(L^{C_4(j+1)})$, with C_3, C_4 independent of K_0 . For instance the lower edge of η_e is obtained by rescaling a fixed smooth cutoff by L^{C_s} .

On the diagonal $m = m'$ the two equations force $h = h'$. For fixed m, n , the allowable e divide $(mn - 1)/2$, and

$$\sum_{e|(mn-1)/2} \tau((mn-1)/(2e))^2 \leq \tau(mn-1)^3.$$

Collecting by $v = mn$ and using [Theorem 2.5](#), we obtain

$$\mathcal{N}_{\text{diag}} \ll L^{O(1)} \sum_{v \asymp x} \tau(v) \tau(v-1)^3 \ll xL^{C_5}. \quad (6.14)$$

The exponents C_3, C_5 are independent of K_0 . Since $EV \asymp xL^{-K_0}$, its contribution to $|B_0|^2$ is $O(x^2L^{-K_0+C_3+C_5})$. We can therefore fix K_0 large enough that its square root, even multiplied by L^{C_f} , is $O(xL^{-D_*-1})$. All subsequent precisions may depend on this fixed K_0 .

6.5 The exact determinant parametrization

Consider an off-diagonal term of [Equation \(6.13\)](#), with

$$mn = 2eh + 1, \quad m'n = 2eh' + 1, \quad m \neq m'. \quad (6.15)$$

The shared variables e, n force the two m values to be congruent modulo $2e$: the first equation makes n a unit modulo $2e$, and subtracting the equations gives $2e \mid m' - m$. Write $m' - m = 2ek$, with $k \neq 0$. The resulting identities are

$$m' - m = 2ek, \quad h' - h = kn, \quad z = m'h, \quad z + k = mh'. \quad (6.16)$$

The last identity follows from $mh' - m'h = k(mn - 2eh) = k$. The enlarged e range gives

$$0 < |k| \ll L^{K_0+C_s}. \quad (6.17)$$

Conversely, fix such a k , positive integers m, m', h, h' with $P^-(m), P^-(m') > W$, and representations $m'h = z, mh' = z + k$. Suppose

$$m' \equiv m \pmod{2|k|}, \quad e = \frac{m' - m}{2k} > 0.$$

Then e is an integer and

$$m(h' - h) = k(2eh + 1). \quad (6.18)$$

Every prime factor of m exceeds W , whereas $|k|$ is bounded by a fixed power of L . Hence $(m, k) = 1$ for sufficiently large x . [Equation \(6.18\)](#) implies $m \mid 2eh + 1$. Set $n = (2eh + 1)/m$, a positive integer. The same identity gives $h' - h = kn$, and substituting $m' = m + 2ek$ recovers the second equation of [Equation \(6.15\)](#). This proves a bijection, with all variables recovered by the displayed formulas. In particular, no additional divisibility condition is being dropped. The coprimality $(m, k) = 1$ is necessary to this converse.

[Figure 3](#) displays the cross-pairing that produces the two shifted endpoints.

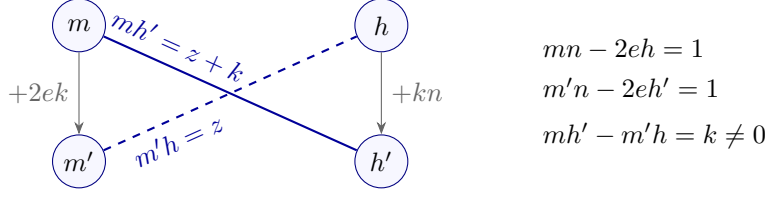


Figure 3: Forward determinant identities for factorizations with common e, n . Cross-paired factors give $z = m'h$ and $z + k = mh'$; the vertical arrows record signed increments, so either sign of k is allowed. The converse reconstruction, including its congruence, positivity, and coprimality conditions, is proved in the text.

Both m and m' are units modulo $q_k = 2|k|$, so the congruence is imposed exactly by

$$\mathbf{1}_{m' \equiv m \pmod{q_k}} = \frac{1}{\varphi(q_k)} \sum_{\chi \bmod q_k} \chi(m) \overline{\chi(m')}. \quad (6.19)$$

Thus $\mathcal{N}_{\text{off}}$ is a sum over k, z and these characters of terms having coefficient

$$\alpha_m \overline{\alpha_{m'}} \chi(m) \overline{\chi(m')} a_h(h) \overline{a_h(h')}, \quad m'h = z, \quad mh' = z + k,$$

with the enlarged smooth (e, n) cutoffs. Their arguments are now the exact functions

$$e = \frac{m' - m}{2k}, \quad n = \frac{(m' - m)z}{kmm'} + \frac{1}{m}. \quad (6.20)$$

The cutoffs are extended smoothly by zero to nonpositive arguments, so they impose the positivity and size conditions as well.

6.6 Smooth separation and the shifted endpoints

On the support just obtained, $m, m' \asymp U$, $n \asymp V$, and $h \asymp x/e$. Hence $z = m'h$ lies in $[xL^{-C_6}, xL^{C_6}]$ for a fixed C_6 now allowed to depend on K_0 . Insert a smooth dyadic partition in z with $O(T)$ terms, and denote one scale by $Z = xL^{O(1)}$. Keep its smooth cutoff $\psi_Z(z/Z)$ outside the ensuing Fourier expansion.

For completeness, the pulled-back cutoffs have the following uniform regularity. Put $t_1 = \log(m/U)$, $t_2 = \log(m'/U)$, $t_3 = \log(z/Z)$. Multiply by fixed smooth localizations in these three variables equal to one on the ranges in use. From Equation (6.20), every derivative of e/E in the t_i is $O_j(U/(|k|E))$, and every derivative of n/V is

$$O_j(Z/(|k|UV) + (UV)^{-1}).$$

Both are bounded by fixed powers of L ; moreover, the normalized derivatives of η_e, η_n have the bounds already given. Repeated use of the chain rule therefore bounds derivatives of the localized product cutoff by $O_j(L^{C_7(j+1)})$. Its support in (t_1, t_2, t_3) is a fixed compact box. Fourier inversion, as in Theorem 2.8, consequently represents this cutoff with total variation $L^{O(1)}$ as a superposition of

$$m^{i\xi_1} (m')^{i\xi_2} z^{i\xi_3}. \quad (6.21)$$

Here is an explicit tail estimate. Enlarge a fixed exponent C_8 so that, with $P = L^{C_8}$, the localized cutoff F satisfies $\|\partial^\alpha F\|_\infty \ll_\alpha P^{|\alpha|}$; its zeroth derivative is bounded independently of L . Integration

by parts then gives

$$|\widehat{F}(\xi)| \ll_j (1 + |\xi|/P)^{-j}, \quad \int_{\mathbb{R}^3} |\widehat{F}(\xi)| d\xi \ll P^3, \quad \int_{|\xi| > PL} |\widehat{F}(\xi)| d\xi \ll_j P^3 L^{3-j}.$$

Thus truncation at PL has uniform error $O(L^{-A})$ for every fixed A , by choosing $j > A + 3C_8 + 3$. These later exponents may depend on K_0 .

There is sufficient absolute control to sum these errors. If the coefficients in Equation (6.12) and the two m sequences are replaced by their absolute values, each resulting endpoint convolution is at most $L^{O(1)}\tau(z_*)^2$. Its product with $W_\ell(z)$ has squared sum $O(ZL^{O(1)})$ on $z \asymp Z$. Cauchy's inequality, also at $z + k$, therefore bounds the absolute correlation by $ZL^{O(1)}$. This uses only Theorem 2.5. It remains true after dropping the congruence; hence it controls the errors just described, after summing the polynomial number of k values, dyads, and character terms.

We spell out the endpoint identification to track exactly which sequence satisfies discrepancy. Use the notation of Equation (6.12) and fix a character χ and frequencies in Equation (6.21). Since m, m' are free of group primes,

$$z_* = m'h_*, \quad (z + k)_* = mh'_*, \quad W_\ell(z) = W_\ell(h), \quad W_\ell(z + k) = W_\ell(h').$$

Define the invariant endpoint functions

$$F_0(v) = \sum_{mpr=v_*} \alpha_m \chi(m) m^{i(\varrho - \xi_2)} \mathbf{1}_{p \in I_{**}, P^-(p) > Wp^{-i\sigma} \overline{c_r}}, \quad (6.22)$$

$$G_0(v) = \sum_{mpr=v_*} \alpha_m \chi(m) m^{i(\xi_1 + \varrho)} \mathbf{1}_{p \in I_{**}, P^-(p) > Wp^{-i\sigma} \overline{c_r}}. \quad (6.23)$$

Direct substitution shows that the separated summand is

$$\psi_Z(z/Z) z^{i(\varrho + \xi_3)} (z + k)^{-i\varrho} \overline{F_0(z)} G_0(z + k) W_\ell(z) W_\ell(z + k),$$

up to scalar phases of modulus one and the Fourier coefficient. For example, at $z = m'h$ the factor $h^{i\varrho}$ becomes $z^{i\varrho}(m')^{-i\varrho}$, which explains the first endpoint's power and the conjugation of c_r . In particular, the sequence inside F_0 is the original α , with its actual interval restriction.

Both Equations (6.22) and (6.23) have exactly the form in Equation (5.3). The character modulus $2|k|$, all power frequencies, and the coefficient bounds are fixed powers of L . The interval I_{**} lies in $[x^\tau, x^\eta]$ for fixed $0 < \tau < \eta < 1/4$, by Equation (6.2) and the hypothesis $\eta_* < b_*/4$. The residual c_r has a fixed power bound, and no roughness condition on r is needed. The original α is supported on W -rough integers and satisfies Equation (5.4); the explicit character and power factors are precisely the factors permitted in Equation (5.3). Thus the discrepancy assumption has not been silently transferred to a new coefficient sequence.

Finally set $\tilde{\psi}_Z(t) = t\psi_Z(t)$. The last unweighted correlation equals Z times

$$\sum_{z > 0, z+k > 0} \frac{\tilde{\psi}_Z(z/Z)}{z} z^{i(\varrho + \xi_3)} (z + k)^{-i\varrho} \overline{F_0(z)} G_0(z + k) W_\ell(z) W_\ell(z + k).$$

This is Equation (5.5), including its harmonic normalization. Given any fixed D_0 , Theorem 5.1 bounds it by L^{-D_0} once the discrepancy exponent B is sufficiently large. Consequently each separated unweighted correlation is $O(ZL^{-D_0})$. Choose D_0 after K_0 , the k range, all separation costs, and the desired saving in Equation (6.13). The total off-diagonal contribution is then $O(xL^{-A})$ for any prescribed fixed A . Choose B after this application of Theorem 5.1. Together with Equation (6.14), this gives $|B_0| \ll xL^{-D_* - C_f - 1}$. Summing its pre-Cauchy expansion and the previously estimated tails proves Equation (6.5). $\square$

Remark 6.2 (Order of the precisions). The band constants, j_{**} , b_* , coefficient bound C , and target D_* are fixed first. The transition functions, their total variation, and the coefficient and diagonal exponents are independent of K_0 ; only then is K_0 chosen. The smooth separation after Cauchy's inequality and the accuracy required from [Theorem 5.1](#) are fixed next, and the required discrepancy exponent B is fixed last. Accordingly, when α is the difference between a test and its rough-integer proxy, the precision of the cells defining that proxy may be chosen after B , provided the proxy's coefficient bound is already uniform in that cell precision. This is the order used in [Section 7](#).

7 Extracting primes with smooth predecessors

We prove [Theorem 1.2](#). Throughout this section $0 < \delta < 1/4$ is fixed. The task is to detect primes among $N_u := 2u + 1$ for a positive weight on integers u all of whose prime factors are small. The distribution estimate of [Theorem 6.1](#) will replace tests on a factor of N_u by simpler tests on rough integers. We specify the order of all remaining parameter choices at the end of the section.

The extraction separates the composites according to their least prime factor. A preliminary sieve retains those N_u with no prime factor below x^{b_1} , where $b_1 > 0$ is fixed and small. For a fixed small $\kappa > 0$ and least prime factors up to $x^{1/2-\kappa}$, Type II replaces the roughness condition on the complementary factor by a local density on ordinary integers. Integrating these densities accounts for the composite part of the sieve mass. The remaining composites have two prime factors close to $x^{1/2}$; a separate upper-bound sieve makes their contribution small with κ . Its constant must be independent of κ , so we will establish that uniformity before making the final choices.

7.1 The candidate and its mass

Fix a nonnegative smooth function Ψ supported in $[1, 2]$ that is bounded below by a positive constant on a closed interval of positive length contained in $(1, 2)$. In the marked weight of [Section 5](#) take $q = 1/2$ and $\ell = 1$. Set

$$c_1 = 1, \quad c_2 = \frac{6}{5}, \quad c_3 = \frac{3}{2}, \quad c_4 = \frac{9}{5}, \quad s' = \frac{1}{r_0(c_1 + c_2)}, \quad s_j = s' 2^{-j} L,$$

where the fixed integer r_0 is sufficiently large that

$$c_2 s' < \delta, \quad \frac{(r_0 - 1)c_1}{2} \geq c_2 + 1. \quad (7.1)$$

Let j_x be the largest integer with $c_1 s_{j_x} \geq 20T$, and let

$$Q_j = \{p \text{ prime} : c_1 s_j \leq \log p \leq c_2 s_j\} \quad (0 \leq j \leq j_x).$$

At each scale for which $[c_3 s_j, c_4 s_j]$ lies wholly in $[L^{1/10}, L^{1/5}]$ or $[L^{3/10}, L^{2/5}]$, take the primes in that interval of logarithms as a small or big P-group, respectively. The P-groups and the Q-bands are pairwise disjoint: the four constants lie in the displayed order, and $c_4 < 2c_1$. The prime number theorem and partial summation give fixed upper and positive lower bounds for each of their reciprocal prime masses. Both numbers of P-groups are comparable to T , and their exponent ranges and endpoint ratios meet the hypotheses of the preceding sections.

There are r_0 ordered slots in each Q-band. Except for one designated slot in a fixed band j_{**} , a slot contains a prime of its band. The designated slot instead contains any integer v satisfying

$$e^{c_1 s_{j_{**}}} \leq v \leq e^{c_2 s_{j_{**}}}, \quad P^-(v) > W.$$

We always require $j_{**} \geq j_0$, where j_0 is a sufficiently large fixed lower bound chosen below using only the candidate geometry. In particular $j_{**} \geq 1$. Define

$$a_Q(v) = \prod_{j=0}^{j_x} \frac{1}{r_0!} \# \{\text{admissible ordered Q-lists with product } v\}, \quad A(u) = a_Q(u_*) W_1(u).$$

The P-free part u_* and the marked weight W_1 are those of [Section 5](#). The rough slot has no P-prime divisor for large x , since all P primes are less than W .

The candidate must have substantial ordinary mass near x , despite the restrictions on all its factors. Reciprocal weights make the slot choices independent. The band geometry leaves room for one prime in the largest band to adjust their product to size x ; under this harmonic sampling, that prime supplies a local probability of order $1/L$. The next proposition records the resulting mass and the pointwise bound needed to convert that mass into a count of distinct primes.

Proposition 7.1 (Candidate mass). *Put*

$$X_A = \sum_u A(u) \Psi(u/x), \quad H_Q = \sum_v \frac{a_Q(v)}{v}, \quad H_P = \sum_{r:r_*=1} \frac{W_1(r)}{r}.$$

*There are constants C, C_A depending only on the early candidate choices such that, for sufficiently large x after fixing j_{**} ,*

$$0 \leq A(u) \leq L^C \tau(u_*), \quad A(u) \leq \exp(C\sqrt{L}) \quad (u \asymp x), \quad (7.2)$$

$$H_P \sim \exp\left(q \sum_g V_g\right), \quad X_A \asymp \frac{x}{L} H_Q H_P \gg x L^{-C_A}. \quad (7.3)$$

*The comparison constants in (7.3) can be chosen independently of $j_{**} \geq j_0$. On the support of $A(u) \Psi(u/x)$ every prime divisor of u lies in $[L^{20}, x^\delta]$.*

Proof. After fixing the value of the rough slot as a divisor of v , the remaining prime multiplicities determine their band assignments. The normalized number of ordered lists in each band is at most one. This gives $a_Q(v) \leq \tau(v)$. For $v \ll x$, any possible rough-slot divisor divides the part of v supported on primes exceeding W . That part has at most $O(L/\log W) = O(\sqrt{L})$ prime factors counted with multiplicity, and hence at most $\exp(O(\sqrt{L}))$ divisors. Together with the pointwise bound $W_1(u) \leq L^{O(1)}$ this proves (7.2). The claimed lower and upper bounds on prime factors follow from the definitions, (7.1), and $\exp(L^{2/5}) < x^{c_2 s'}$ for large x .

Harmonic masses and the P-part. The Q harmonic sum factorizes into the reciprocal masses of all its slots, divided by $r_0!$ in each band. Each pure-prime slot has mass between two fixed positive constants. The rough-slot mass is at most $O(L)$ and is bounded below by a fixed positive constant for sufficiently large x after fixing j_{**} ; for the latter assertion one may restrict that slot to primes of $Q_{j_{**}}$, which then all exceed W . Since there are $O(T)$ slots,

$$L^{-C} \leq H_Q \leq L^C. \quad (7.4)$$

This exponent and these eventual bounds do not depend on j_{**} .

For one P-group write $F_g(q) = \prod_{p \in \mathcal{P}_g} (1 + q/(p-1))$. Summing over all powers of its prime divisors shows that its factor in H_P is

$$\frac{F'_g(q)}{V_g} = F_g(q) \frac{1}{V_g} \sum_{p \in \mathcal{P}_g} \frac{1}{p-1+q}. \quad (7.5)$$

As $\sum_{g,p \in \mathcal{P}_g} p^{-2} = o(1)$ and V_g is bounded above and below, multiplying (7.5) proves the asymptotic for H_P . It also proves the useful exact inequality

$$H_P \geq \prod_{p \text{ a P prime}} \left(1 + \frac{q}{p-1}\right), \quad (7.6)$$

since $p-1+q \leq p$.

The harmonic P-measure assigns negligible mass to $r > x^\sigma$ for any fixed $\sigma > 0$. Indeed set $\xi = L^{-2/5}$ and replace p^{-1} by $p^{-1+\xi}$ in the preceding Euler calculation. Since $p^\xi \leq e$ for every P prime, the tilted factor of each group is bounded by a fixed constant. Consequently

$$\sum_{r > x^\sigma, r_* = 1} \frac{W_1(r)}{r} \leq \exp(-\sigma L^{3/5}) L^{O(1)}. \quad (7.7)$$

Localizing the product near x . Normalize the independent P choice and all the Q-slot harmonic choices by $H_P H_Q$, and denote their product by U . Then

$$X_A = H_P H_Q \mathbb{E}\{U \Psi(U/x)\}.$$

For the upper comparison we need probability $O(1/L)$ throughout $x \leq U \leq 2x$; for the lower comparison we need probability $\gg 1/L$ where U/x lies in the positivity interval of Ψ . Both are intervals of bounded length for $\log U$. The P-tail estimate ensures that the P-part does not move the product out of reach of the largest Q-band. Condition on everything except one pure-prime slot in Q_0 . The prime number theorem implies that its logarithm falls in any interval of bounded length with probability $O(1/L)$, uniformly in the location of that interval. Thus $\mathbb{P}(x \leq U \leq 2x) = O(1/L)$, giving the upper mass bound.

Here is a uniform lower bound. The sum of the midpoints of all the infinite Q-bands, counting their r_0 slots, is exactly

$$\sum_{j \geq 0} r_0 \frac{c_1 + c_2}{2} s' 2^{-j} L = L.$$

Let $\Delta = (c_2 - c_1)s'L$, the logarithmic width of the free slot. Choose a fixed prefix of bands so long that the sum of all later band widths and midpoints is less than $\Delta/24$. Within that prefix, restrict all slots except the free slot to fixed small relative neighborhoods of their midpoints, so that their total deviation is at most $\Delta/24$. This event has probability bounded below by a positive constant: it involves only a fixed number of pure-prime slots. Choose j_0 larger than the prefix length so that the exceptional slot is always in the unrestricted tail. Its full range obeys the same deterministic tail bound. Missing bands after j_x also obey that bound. Finally restrict the P choice to $\log r \leq \Delta/12$, at negligible loss by (7.7). On this event, a target interval for $\log U$ coming from the positivity interval of Ψ gives an interval of fixed positive length for the free slot, lying a distance at least $\Delta/4$ from its range endpoints for large x . The prime number theorem gives conditional probability $\gg 1/L$. Multiplying by $U \asymp x$ proves the lower comparison in (7.3). This argument uses no property of the tail distribution beyond its range, so its constants are uniform in the later choice of j_{**} . The final lower bound follows from (7.4) and the Euler calculation. $\square$

7.2 Congruence distribution and the preliminary sieve

For odd squarefree d define

$$r(d) = \sum_{d|N_u} A(u) \Psi(u/x) - \frac{X_A}{\varphi(d)}.$$

Proposition 7.2 (Type I distribution). *For every fixed $\vartheta < 1/2$ and $D > 0$,*

$$\sum_{\substack{d \leq x^\vartheta \\ d \text{ odd and squarefree}}} |r(d)| \ll xL^{-D} + X_AL^{-18}. \quad (7.8)$$

Proof. The congruence $2u + 1 \equiv 0 \pmod{d}$ specifies a unit class for u . Use character orthogonality on that class. Replacing the principal coprime mass by X_A has total cost at most X_AL^{-18} , because for each supported u ,

$$\sum_{\substack{d \leq x^\vartheta \\ (d,u) > 1}} \frac{\mu^2(d)}{\varphi(d)} \leq \sum_{p|u} \frac{1}{p-1} \sum_{d' \leq x} \frac{\mu^2(d')}{\varphi(d')} \ll L^{-18}.$$

Here every $p \mid u$ is at least L^{20} , there are at most $O(L/\log L)$ such primes, and the last sum is $O(L)$ by its Euler product and Mertens' theorem.

Extract one fixed Q_0 slot, as in the proof of [Theorem 7.1](#), to write

$$A(u) = \sum_{\substack{pr=u \\ p \in Q_0}} b(r), \quad 0 \leq b(r) \leq L^{O(1)}\tau(r).$$

Split p, r into dyads of sizes P, U with $PU \asymp x$. There is a fixed $a > 0$ such that $x^a \ll P, U \ll x^{1-a}$ in every contributing dyad. A nonprincipal character modulo squarefree d is induced by a primitive nonprincipal character modulo $f > 1$, where $d = hf$ and $(h, f) = 1$. It acts as that primitive character together with the restrictions $(p, h) = (r, h) = 1$, and $\varphi(d) = \varphi(h)\varphi(f)$.

Fix h . On conductors $f \asymp R > L^{C'}$, separate $\Psi(pr/x)$ by Mellin inversion, which has bounded integrated absolute cost. For each resulting twist, [Theorem 2.3](#), Cauchy–Schwarz, and [Theorem 2.5](#) give

$$\begin{aligned} & \sum_{f \asymp R} \frac{1}{\varphi(f)} \sum_{\chi \bmod f}^* \left| \sum_{p \asymp P} a_p \chi(p) \right| \left| \sum_{r \asymp U} b_r \chi(r) \right| \\ & \ll \frac{L^{O(1)}}{R} \sqrt{(P + R^2)(U + R^2)PU} \\ & \ll xL^{O(1)} \left(R^{-1} + P^{-1/2} + U^{-1/2} + R/\sqrt{x} \right). \end{aligned}$$

The coefficients include coprimality to h and unit-modulus Mellin twists, so their squared sums are $O(P)$ and $UL^{O(1)}$. Discarding $(h, f) = 1$ and squarefreeness only enlarges this positive bound. Summing dyads and $1/\varphi(h)$ costs a fixed power of L . Since $R \leq x^\vartheta$, a sufficiently large C' gives any prescribed logarithmic saving.

For $1 < f \leq 2L^{C'}$, fix r and apply [Theorem 2.2](#) to the p -sum with the original smooth cutoff $\Psi(pr/x)$, using partial summation. The nonprincipal character sum is $O_A(PL^{-A})$ for arbitrary fixed A . Omitting primes dividing h costs only $O(1)$ terms, since such primes in this range are at least x^a and $h \leq x^\vartheta$. Their total cost after summing r , moduli and dyads is $x^{1-a}L^{O(1)}$. The remaining costs are fixed powers of L , absorbed by choosing A large. This proves [\(7.8\)](#). $\square$

Define

$$V(y) = \prod_{p \leq y} (1 - 1/p), \quad V_1(y) = \prod_{3 \leq p \leq y} (1 - 1/(p-1)), \quad \mathfrak{S} = 2 \prod_{p \geq 3} \left(1 - \frac{1}{(p-1)^2} \right) > 0.$$

Writing γ_E for Euler's constant, Mertens' theorem gives

$$V(y) \sim \frac{e^{-\gamma_E}}{\log y}, \quad V_1(y) \sim \mathfrak{S}V(y). \quad (7.9)$$

We will choose a small fixed $0 < \kappa < 1/50$ and then a sufficiently small fixed $b_1 > 0$. Set $b_* = b_1/3$ for [Theorem 6.1](#), and eventually require

$$j_{**} \geq j_0, \quad c_2 s' 2^{-j_{**}} < b_*/4. \quad (7.10)$$

Apply [Theorem 2.9](#) to odd prime divisors of N_u up to x^{b_1} , with density $g(p) = 1/(p-1)$ and the remainders from [Theorem 7.2](#). Take

$$h = 2 \left\lfloor \frac{1}{40b_1} \right\rfloor, \quad \vartheta = \frac{1}{2} - \frac{\kappa}{2}.$$

For sufficiently small b_1 , h is large enough for that lemma and $b_1(4h+2) < \vartheta$. Choose $D > C_A + 10$ in (7.8). It follows that

$$\sum_{P^-(N_u) > x^{b_1}} A(u) \Psi(u/x) \geq \frac{\mathfrak{S}X_A}{L} \left\{ \frac{e^{-\gamma_E}}{b_1} (1 - O(e^{-c/b_1})) + o(1) \right\}. \quad (7.11)$$

The constants in the exponential error are uniform for small b_1 and $0 < \kappa < 1/50$.

7.3 Proxies with character and Mellin discrepancy

We now construct the coefficients that Type II will compare with the factor tests. The proxy has exactly the same sum as the test on each short logarithmic cell, but is constant among the W -rough integers in that cell. This count matching handles the principal characters; prime estimates and the elementary sieve will give the nonprincipal cancellation independently of Type II. The cell width may therefore be chosen after Type II specifies its required discrepancy.

At fixed b_1, κ , partition $(b_1, 1/2 - \kappa]$ into finitely many exponent bins $(\gamma, \gamma']$. Their mesh will be chosen later. On the factor ranges used below consider the tests

$$I_\gamma(m) = \mathbf{1}_{P^-(m) > x^\gamma}, \quad I_{\text{pr}}(m) = \mathbf{1}_{m \text{ prime}}.$$

Every such range lies between two fixed positive powers of x ; in particular we can work throughout the ambient interval $[x^{b_1/4}, x^{1-b_1/4}]$. Put $\Delta_L = L^{-S}$, with S a fixed precision to be chosen last. Partition the logarithmic axis into intervals of width Δ_L . For a full cell $C_Y = (Y, e^{\Delta_L} Y]$ meeting this ambient interval, set

$$c_I(C_Y) = \frac{\sum_{m \in C_Y} I(m)}{\#\{m \in C_Y : P^-(m) > W\}}, \quad B_I(m) = c_I(C_Y) \mathbf{1}_{P^-(m) > W} \quad (m \in C_Y). \quad (7.12)$$

The counts defining c_I are full-cell counts even when the factor range cuts a cell. Range restrictions are imposed afterwards.

Lemma 7.3 (Proxy discrepancy). *The denominators in (7.12) are positive for large x . Moreover $0 \leq B_I \leq 1$. Given any fixed discrepancy exponent B , taking $S > 2B + 3$ makes*

$$\alpha_m = (I(m) - B_I(m)) \mathbf{1}_{m \in K}$$

satisfy (5.4), uniformly for every interval K inside a dyad in the stated factor ranges, for each of the finitely many tests above. Thus the coefficient bound required by [Theorem 6.1](#) is independent of S .

Proof. Use [Theorem 2.9](#) on ordinary integers in a cell with $z = W$, $g(p) = 1/p$, and $h = 2\lceil T^2 \rceil$. Each divisibility remainder is $O(1)$, and the sieve level is

$$D_W = W^{4h+2} = \exp(O(\sqrt{L}T^2)) = x^{o(1)}.$$

Since $Y \gg x^{b_1/4}$, uniformly for these cells,

$$\#\{m \in C_Y : P^-(m) > W\} = (e^{\Delta_L} - 1)YV(W)(1 + O(e^{-T^2})) + O(D_W) \sim \Delta_L YV(W). \quad (7.13)$$

Both actual tests select subsets of these rough integers for large x : for the prime test all integers in the cell exceed W , and for the other tests $x^\gamma > W$. Hence $0 \leq c_I \leq 1$ exactly.

Every prime divisor of a modulus $r \leq L^B$ is below W , so a principal character is one throughout the support of α . On a full cell the unweighted sum of $I - B_I$ is zero. For $f(t) = t^{iv-1}$, $|v| \leq L^B$, its variation on that cell is at most $O((1 + |v|)\Delta_L/Y)$. The absolute harmonic mass on a dyad is $O(1)$ (even the weaker $O(L^{1/2})$ would suffice). Consequently the full cells contribute $O(L^{B-S})$. Intersecting the test interval in (5.4) with K leaves at most two partial cells; their combined harmonic mass is $O(L^{-S} + x^{-b_1/4})$. These bounds imply the principal-character case with room to spare when $S > 2B + 3$.

For a nonprincipal character, first treat the actual roughness test. Each admitted integer has a unique nondecreasing list of at most $O(1/b_1)$ prime factors. Fix the first $l - 1$ factors and sum over the last prime. Its order restriction, its lower bound x^γ , and the interval restriction on the full product intersect in one prime interval. Its endpoints lie between x^γ and x^2 . By [Theorem 2.2](#) and partial summation, the harmonic sum of this prime against a nonprincipal character and the twist p^{iv} has arbitrarily strong logarithmic saving, uniformly for a prescribed logarithmic bound on the modulus and v . The reciprocal masses of the remaining factors are bounded in terms of b_1 , because $\sum_{x^\gamma < p \leq x^2} p^{-1} = O_{b_1}(1)$. Summing the bounded number of list lengths preserves the saving. Repeated primes are included by the nondecreasing enumeration. Imprimitive nonprincipal characters give the same estimate via their nonprincipal primitive characters; primes dividing the modulus are too small to occur. For I_{pr} this is the same argument with one prime.

It remains to treat the proxy. For every subinterval J' of a cell and every unit class $a \pmod{r}$, apply [Theorem 2.9](#) to that class, omitting primes dividing r from the sieve. The CRT gives an $O(1)$ remainder at each squarefree sieve product. Since all prime divisors of r are below W , the main term is

$$\frac{|J'|}{r} \prod_{\substack{p \leq W \\ p \nmid r}} (1 - 1/p) = \frac{|J'|V(W)}{\varphi(r)}.$$

Thus, also for very short J' as an absolute-error assertion,

$$\#\{m \in J' : m \equiv a \pmod{r}, P^-(m) > W\} = \frac{|J'|V(W)}{\varphi(r)}(1 + O(e^{-T^2})) + O(D_W). \quad (7.14)$$

Summing against a nonprincipal character cancels the common main term, and bounds the remaining sum by $O(|J'|V(W)e^{-T^2} + \varphi(r)D_W)$. Multiply by the cell constant $c_I \leq 1$ and use partial summation against m^{iv-1} . The number of cells is at most $O(L^{S+1})$; the moduli and twists cost fixed powers of L . The first error therefore remains smaller than every fixed negative power of L , and the second is $x^{-b_1/4+o(1)}$ times a fixed power of L . This proves the nonprincipal assertion, including all interval truncations. $\square$

For all later applications, prescribe the saving in [Theorem 6.1](#) large enough that its errors, after the required dyadic decompositions, are $o(X_A/L)$. Its coefficient exponent is fixed before S , by [Theorem 7.3](#); obtain its required discrepancy exponent and then choose S . There are only finitely many test types, so the same S works for all.

7.4 The local densities and their integral identity

The proxy discrepancy allows replacement of a factor test, while its local density controls the size of the replacement. We need ordinary rough-integer counts for the cofactor left after a least prime factor has been selected. For a product of l primes, their logarithms, divided by L , sum to the logarithmic size of that cofactor. This leads to the following finite sum of integrals over those logarithms.

For $w > \gamma > 0$ define

$$D_\gamma(w) = \frac{1}{w} + \sum_{l \geq 2} \frac{1}{l!} \int_{\substack{t_i \geq \gamma \ (1 \leq i < l) \\ \sum_{i < l} t_i \leq w - \gamma}} \frac{1}{w - \sum_{i < l} t_i} \prod_{i < l} \frac{dt_i}{t_i}. \quad (7.15)$$

Only finitely many terms are nonzero. On compact subsets of $w > \gamma > 0$ the functions are jointly continuous, including at thresholds $w = l\gamma$: the integrands have bounded denominators and the moving boundaries have measure zero.

These are Buchstab's rough-number densities in logarithmic coordinates, and the minimum-coordinate decomposition below is the corresponding Buchstab identity [4]. We derive the short-cell bounds and integral identity needed here directly.

Lemma 7.4 (Proxy density bounds). *For the prime proxy in the interval $x^{1/2-2\kappa} \leq m \leq x^{1/2+2\kappa}$,*

$$c_{\text{pr}}(m) \leq \frac{C}{LV(W)}, \quad (7.16)$$

where C is absolute for $0 < \kappa < 1/50$. If $mn = N_u$ on the support of $\Psi(u/x)$ and $\log n/L \in (\gamma, \gamma']$, then

$$c_\gamma(m) \leq \frac{\sup_{\gamma \leq a \leq \gamma'} D_\gamma(1-a) + o(1)}{LV(W)}. \quad (7.17)$$

At fixed b_1 one also has

$$\int_{b_1}^{1/2} D_\alpha(1-\alpha) \frac{d\alpha}{\alpha} = D_{b_1}(1) - 1, \quad (7.18)$$

$$D_{b_1}(1) \leq \frac{e^{-\gamma_E}}{b_1} (1 + O(e^{-c/b_1})). \quad (7.19)$$

The integrand at the endpoint $\alpha = 1/2$ is understood by its left limit; changing that endpoint value has no effect.

Proof. The prime number theorem with an arbitrarily large logarithmic error, applied at both endpoints of a cell, gives

$$\#\{p \in (Y, e^{\Delta_L Y}]\} = (1 + o(1)) \frac{\Delta_L Y}{\log Y}.$$

For $w = \log Y/L \in [0.46 + o(1), 0.54 + o(1)]$, divide by (7.13) to obtain (7.16) with an absolute eventual constant. Increasing fixed S changes the threshold for x , not that constant.

More generally, uniformly on compact sets with $\gamma \geq b_1$, $w - \gamma$ bounded below positively, and w bounded above, the number of integers rough above x^γ in the cell is at most

$$(D_\gamma(w) + o(1)) \frac{\Delta_L Y}{L}. \quad (7.20)$$

To see this, repeated prime factors contribute at most

$$\sum_{p > x^\gamma} \left\lfloor \frac{2Y}{p^2} \right\rfloor \ll Yx^{-b_1} = o(\Delta_L Y/L).$$

Count the squarefree remaining integers by ordered lists of l primes with weight $1/l!$, where l is bounded in terms of b_1 . Fix the first $l-1$ primes, write $Q = \prod_{i < l} p_i$ and $t_i = \log p_i/L$. A last prime can occur only when $\sum_{i < l} t_i \leq w - \gamma + O(\Delta_L/L)$. Dropping its lower cutoff, its count is at most

$$(1 + o(1)) \frac{\Delta_L Y}{LQ(w - \sum_{i < l} t_i)}.$$

The prime number theorem is uniform here because $Y/Q \geq x^\gamma/2$, and its logarithmic accuracy is chosen larger than $S + 3$. The reciprocal-prime measures for the remaining slots converge to dt_i/t_i . A finite grid approximation proves uniform convergence of the resulting integrals: denominators stay bounded away from zero, while strips around the moving hyperplane boundary have uniformly vanishing volume. This is exactly (7.15), proving (7.20). In the application $w = 1 - \log n/L + o(1)$ and $w - \gamma \geq 2\kappa + o(1)$. Continuity, followed by division by (7.13), proves (7.17).

For (7.18), write the l th term of $D_{b_1}(1)$, $l \geq 2$, on the simplex

$$t_1 + \cdots + t_l = 1, \quad t_i \geq b_1, \quad \frac{1}{l!} \frac{dt_1 \cdots dt_{l-1}}{t_1 \cdots t_l}.$$

This measure is invariant under every permutation of the l coordinates: exchanging a dependent and an independent coordinate has absolute Jacobian one. Except on a null set exactly one coordinate is the minimum, say α . Select that coordinate in l ways. Its measure is $d\alpha/\alpha$, and the remaining coordinates, with sum $1 - \alpha$ and lower bound α , give the $(l-1)$ -factor term of $D_\alpha(1 - \alpha)$. The coefficient $l/l!$ is precisely $1/(l-1)!$. Summing proves the identity; the one-prime term of $D_{b_1}(1)$ equals one.

For the inequality, first obtain a lower density for ordinary rough integers in $(x, 2x]$. Ordered prime lists with weight $1/l!$ assign at most unit mass to every integer, including those with repetitions. For $l \geq 2$ restrict the first $l-1$ logarithms to $\sum t_i < 1 - b_1 - \eta$, with $\eta > 0$ fixed, and sum the last prime over $(x/Q, 2x/Q]$. It is then wholly above the threshold. The prime number theorem and the same reciprocal-measure convergence give the corresponding integrals times x/L . Include the one-prime term and let η decrease to zero. Null boundaries give

$$\liminf_{x \rightarrow \infty} \frac{L}{x} \#\{x < m \leq 2x : P^-(m) > x^{b_1}\} \geq D_{b_1}(1).$$

On the other hand Theorem 2.9, with $g(p) = 1/p$, $O(1)$ remainders and $h = 2\lfloor 1/(40b_1) \rfloor$, bounds this count by

$$xV(x^{b_1})(1 + O(e^{-c/b_1})) + O(x^{b_1(4h+2)}).$$

Its remainder is $o(x/L)$ for small b_1 , and (7.9) proves (7.19). $\square$

The term 1 in (7.18) is the one-prime contribution to $D_{b_1}(1)$. Selecting a least prime factor accounts for all the other terms. Thus the integral provides the composite density to subtract from the preliminary sieve mass, with the one-prime term left over. These are densities for ordinary rough integers used in the proxies; their application to N_u uses Type II followed by Type I and the sieve.

7.5 Subtracting composites away from balance

Consider a composite N_u counted in (7.11), whose least prime factor is at most $x^{1/2-\kappa}$. Its least prime factor p lies in one bin $(x^\gamma, x^{\gamma'}]$, and $m = N_u/p$ satisfies $P^-(m) > x^\gamma$. Its contribution is therefore bounded by

$$\sum_{\substack{mp=N_u \\ x^\gamma < p \leq x^{\gamma'} \\ \text{prime}}} A(u)\Psi(u/x)I_\gamma(m).$$

All contributing dyads of m, p lie between x^{b_*} and x^{1-b_*} for large x , with $b_* = b_1/3$: the factors bounded by constants in $N_u \asymp x$ are absorbed by the strict exponent slack. Apply Theorem 6.1 with $\alpha = I_\gamma - B_\gamma$ restricted to each actual cofactor range, and β the prime indicator restricted to its bin and dyad. Theorem 7.3 supplies precisely its discrepancy hypothesis. Replacing I_γ by B_γ costs $o(X_A/L)$ in total.

For a prime $p > W$, the condition that $m = N_u/p$ be rough above W is equivalent to $P^-(N_u) > W$. By (7.17), the resulting upper bound reduces to a constant divided by $LV(W)$ times

$$\sum_{x^\gamma < p \leq x^{\gamma'}} \sum_{\substack{p|N_u \\ P^-(N_u) > W}} A(u)\Psi(u/x).$$

For each such p , sieve the odd primes up to W , using the base mass $X_A/(p-1)$, density $1/(l-1)$, and remainders $r(pd)$. Take $h = 2\lceil T^2 \rceil$ in Theorem 2.9. All products pd are at most $x^{1/2-\kappa+o(1)}$, which is below the Type I level $x^{1/2-\kappa/2}$. Moreover different pairs (p, d) give different products: $p > W$ is their unique prime factor above W . Thus (7.8) bounds the summed remainders without any divisor multiplicity loss. We obtain

$$\begin{aligned} & \sum_{x^\gamma < p \leq x^{\gamma'}} \sum_{\substack{p|N_u \\ P^-(N_u) > W}} A(u)\Psi(u/x) \\ &= X_A V_1(W) \sum_{x^\gamma < p \leq x^{\gamma'}} \frac{1}{p-1} + o(X_A V(W)). \end{aligned}$$

Since the reciprocal sum tends to $\log(\gamma'/\gamma)$, the composite contribution of the bin is at most

$$\frac{\mathfrak{S}X_A}{L} \left\{ \sup_{\gamma \leq a \leq \gamma'} D_\gamma(1-a) \log(\gamma'/\gamma) + o(1) \right\}. \quad (7.21)$$

At fixed b_1, κ , joint continuity in Theorem 7.4 permits a sufficiently fine fixed mesh such that the sum of these constants is at most

$$\int_{b_1}^{1/2-\kappa} D_\alpha(1-\alpha) \frac{d\alpha}{\alpha} + \frac{1}{10}.$$

Subtracting (7.21) from (7.11) and using Equations (7.18) and (7.19) leaves at least

$$\frac{\mathfrak{S}X_A}{L} \left\{ 1 - \frac{1}{10} - O(b_1^{-1}e^{-c/b_1}) + o(1) \right\} \geq \frac{\mathfrak{S}X_A}{2L} \quad (7.22)$$

on primes and composites with least prime factor exceeding $x^{1/2-\kappa}$, provided b_1 is sufficiently small.

7.6 A uniform upper bound for balanced composites

If a composite $N_u \ll x$ has least prime factor above $x^{1/2-\kappa}$, it has exactly two prime factors counted with multiplicity, because $3(1/2 - \kappa) > 1$. Both factors belong, for large x , to

$$\mathcal{B}_\kappa = [x^{1/2-2\kappa}, x^{1/2+2\kappa}].$$

We may overcount them by ordered pairs m, n of primes in that range with $mn = N_u$. Replace the first prime indicator by its proxy using [Theorem 6.1](#), then replace the second using the same theorem with the factor roles exchanged. The remaining coefficient in the second application is a proxy bounded by one. All required discrepancy and range hypotheses follow from [Theorem 7.3](#), as before. By (7.16), the resulting upper bound is $o(X_A/L)$ plus

$$\frac{C}{L^2 V(W)^2} \sum_{\substack{m, n \in \mathcal{B}_\kappa, mn=2u+1 \\ P^-(m) > W, P^-(n) > W}} A(u) \Psi(u/x). \quad (7.23)$$

We now bound this positive sum with a constant independent of κ, b_1, j_{**} and S .

More precisely, we will show that each dyadic pair with $mn \asymp x$ contributes $O(X_A V(W)^2)$. There are only $O(\kappa L + 1)$ such pairs in the indicated range, so the prefactor in (7.23) will give an $O((\kappa + L^{-1})X_A/L)$ bound. To prove the dyadic estimate, we retain a small divisor of u from its Q-slots and P-marks. After discarding the remaining large Q-prime restrictions, a two-variable sieve enforces roughness of m, n and the remaining small-prime restrictions on $u = (mn - 1)/2$.

Choose a fixed large index j' and then θ in the gap

$$c_2 s' 2^{-j'} < \theta < c_1 s' 2^{-j'+1}, \quad 10\theta \leq \frac{1}{20}, \quad \sum_{j \geq j'} r_0 c_2 s' 2^{-j} \leq \frac{1}{25}. \quad (7.24)$$

Such a gap exists because $c_2 < 2c_1$. These choices depend only on the candidate geometry. Increase the early lower bound j_0 to ensure $j_0 \geq j'$, so the rough slot is in a band $j \geq j'$. Call these the micro bands. Write H_Q^{mic} for the Q harmonic mass using just these bands. Omitting the finitely many fixed bands $j < j'$ gives

$$H_Q^{\text{mic}} \ll H_Q \quad (7.25)$$

with a constant independent of j_{**} .

An assignment consists of the complete ordered Q-lists in the micro bands, including the rough slot, and one marked prime from every P-group. Give it weight

$$\lambda_{\text{assgn}} = \prod_{j=j'}^{j_x} \frac{1}{r_0!} \prod_g \frac{1}{V_g},$$

and let D_1 be the product of all its entries, with multiplicity. The Q contribution has size at most $x^{1/25}$ by (7.24); the P marks have product $\exp(O(TL^{2/5})) = x^{o(1)}$. Thus

$$D_1 \leq x^{1/20}, \quad \sum_{\text{assgn}} \frac{\lambda_{\text{assgn}}}{D_1} = H_Q^{\text{mic}}, \quad \sum_{\text{assgn}} \lambda_{\text{assgn}} \leq x^{1/20} H_Q^{\text{mic}}. \quad (7.26)$$

In the middle identity the harmonic sum of the normalized P mark in each group is exactly $V_g^{-1} \sum p^{-1} = 1$.

The damping in W_1 can also be retained in this upper bound. Once one P-prime per group is marked in D_1 , every additional distinct P-prime divisor of u contributes a factor q . We represent

these factors by independent random permissions for those primes; averaging the permissions recovers exactly the damping.

For each assignment, ban every odd non-P prime $l \leq x^\theta$ not dividing D_1 . For each P prime not dividing D_1 , independently allow it with probability q and otherwise ban it. Primes dividing D_1 are never banned. All P primes are below x^θ and indeed below W for large x . We have the pointwise majorant

$$A(u) \leq \sum_{\text{assgn}} \lambda_{\text{assgn}} \mathbf{1}_{D_1|u} \mathbb{E}_{\text{mask}} \mathbf{1}_{\{l|u \text{ for every banned } l\}}. \quad (7.27)$$

To verify it, expand $A(u)$ into Q-lists and one P mark per group. For every genuine list the divisibility $D_1 | u$ holds, and every non-P prime divisor of u below x^θ occurs in D_1 : omitted Q-bands contain only pure primes above x^θ . The micro Q product has no P factors, so D_1 contains exactly one distinct P prime per group. For such a genuine list, the mask expectation is $q^{\omega(u)-s}$, precisely the damping in $W_1(u)$. Once the micro entries are fixed, the normalized number of omitted pure-prime lists with any fixed residual product is at most one. Explicitly, a band with prime multiplicities a_p contributes $\prod_p 1/a_p! \leq 1$; disjointness prevents alternative band assignments. Dropping these residual restrictions proves (7.27).

Fix now a pair of dyads $m \asymp M_1$, $n \asymp M_2$ meeting the product support $mn \asymp x$, an assignment, and a mask. Since D_1 is odd, $D_1 | u$ and $mn = 2u + 1$ imply $mn \equiv 1 \pmod{D_1}$. We may drop parity and use this congruence to bound the number of pairs in the full dyadic box. For an odd prime $l \leq z' := x^\theta$ with $l \nmid D_1$, the bad events are

$$mn \equiv 0 \pmod{l} \quad (l \leq W), \quad mn \equiv 1 \pmod{l} \quad (l \text{ banned}).$$

The two residue sets are disjoint and contain respectively $2l - 1$ and $l - 1$ pairs. Put

$$\nu(l) = (2l - 1)\mathbf{1}_{l \leq W} + (l - 1)\mathbf{1}_{l \text{ banned}}, \quad g(l) = \nu(l)/l^2.$$

These densities satisfy the hypotheses of [Theorem 2.9](#) uniformly in assignment and mask: $g(l) \leq 3/l$, and for $l \geq 3$ they are bounded away from one. The base mass is the product of the two real side lengths times $\varphi(D_1)/D_1^2$.

For a squarefree product d of the sieve primes, the CRT gives exactly $\varphi(D_1)\nu(d)$ residue pairs modulo $D_1 d$, including when D_1 has prime-power factors. Each pair of residue classes has lattice count equal to its area term with error

$$O\left(1 + \frac{M_1 + M_2}{D_1 d}\right).$$

As $\nu(d) \leq d 3^{\omega(d)}$ and $\varphi(D_1) \leq D_1$, the sum of absolute remainders up to $D = (z')^{10} = x^{10\theta}$ is

$$\ll L^{O(1)}((M_1 + M_2)D + D_1 D^2) \leq L^{O(1)}((M_1 + M_2)x^{1/20} + D_1 x^{1/10}). \quad (7.28)$$

Here the ordinary $\omega(d)$ occurs only in this elementary divisor bound; its sums follow, for example, from $3^{\omega(d)} \leq \tau_3(d)$, where $\tau_3(d)$ counts ordered triples of positive integers with product d . Apply the upper-bound version of [Theorem 2.9](#) with $h = 2$.

We record its Euler product explicitly. A banned prime below W has factor $1 - 3/l + 2/l^2$, and a banned prime above W has factor $1 - 1/l + 1/l^2$. Each is at most

$$(1 - 1/l)^{1+2\mathbf{1}_{l \leq W}}(1 + O(l^{-2})).$$

An allowed P prime has factor $(1 - 1/l)^2$, so it requires one compensating factor $(1 - 1/l)^{-1}$. Omitting the prime 2 changes only an absolute constant, as does the convergent product of the $1 + O(l^{-2})$ factors. Removed primes dividing D_1 require at most three compensating factors. Therefore

$$\prod_{\substack{3 \leq l \leq z' \\ l \nmid D_1}} (1 - g(l)) \leq CV(W)^2 V(z') \prod_{l|D_1} (1 - 1/l)^{-3} \prod_{\substack{p \text{ allowed} \\ p \nmid D_1}} (1 - 1/p)^{-1}. \quad (7.29)$$

Every prime dividing D_1 is at least L^{20} , so $D_1 \leq x^{1/20}$ makes its compensating product uniformly bounded. The expectation of the last product is

$$\prod_{\substack{p \text{ a P prime} \\ p \nmid D_1}} \left(1 - q + \frac{q}{1 - 1/p}\right) = \prod_{\substack{p \text{ a P prime} \\ p \nmid D_1}} \left(1 + \frac{q}{p - 1}\right) \leq H_P$$

by (7.6).

Since $\varphi(D_1)/D_1^2 \leq 1/D_1$, sum the main sieve terms over assignments using (7.26), and bound Ψ by its fixed supremum. Their contribution on this dyadic pair is

$$\ll xV(W)^2 V(x^\theta) H_Q^{\text{mic}} H_P \ll X_A V(W)^2,$$

by Equations (7.3) and (7.25) and $V(x^\theta) \ll 1/L$. To sum the error (7.28), use $M_i \ll x^{0.54}$ and (7.26). The result is

$$\ll x^{0.64} L^{O(1)} H_Q^{\text{mic}} = o(X_A V(W)^2).$$

The comparison is uniform in the later parameters: use $H_Q^{\text{mic}} \ll H_Q$, $H_P \geq 1$, and $X_A V(W)^2 \asymp x H_Q H_P / L^2$. We have proved the promised dyadic estimate

$$\sum_{\substack{m \asymp M_1, n \asymp M_2, mn = 2u+1 \\ P^-(m) > W, P^-(n) > W}} A(u) \Psi(u/x) \ll X_A V(W)^2. \quad (7.30)$$

There are $O(\kappa L + 1)$ possible dyads for m in $\mathcal{B}_\kappa$, and, for each, only $O(1)$ possible dyads for n because $mn \asymp x$. Combining (7.23) and (7.30) bounds the entire balanced composite contribution by

$$C_{\text{bal}}(\kappa + L^{-1}) \frac{X_A}{L} + o(X_A/L). \quad (7.31)$$

The constant C_{bal} depends only on $\delta, r_0, \Psi, j', \theta$ and the fixed P-group geometry. It does not depend on κ, b_1, j_{**} , the later analytic parameters, or S . Indeed the candidate comparison is uniform in j_{**} ; the omitted macro bands are fixed before the gap; the prime proxy bound is uniform on $[0.46, 0.54]$; and the sieve-density and lattice constants just used are absolute. Thresholds for x may depend on all parameters after they are fixed.

7.7 Closing the parameter choices and counting primes

For clarity, the choices in the preceding argument can be made in the following order.

- (i) Fix δ, Ψ, r_0 and the Q/P geometry. Choose the finite prefix for the candidate lower bound, then j' and θ in (7.24), and enlarge j_0 to cover both requirements. All these are early choices.
- (ii) Fix $\kappa \in (0, 1/50)$ so small that $C_{\text{bal}}\kappa < \mathfrak{S}/4$. This is possible because the constant in (7.31) is independent of the gap.

- (iii) Choose b_1 sufficiently small for (7.11), (7.19), and (7.22). Set $b_* = b_1/3$. Choose j_{**} to meet (7.10), and then choose the finite exponent mesh for (7.21).
- (iv) Prescribe the saving in Theorem 6.1 so that every replacement error, including dyadic sums, is $o(X_A/L)$. The coefficient exponents are already bounded independently of S . The proof of that theorem fixes its Cauchy splitting precision, shift-saving target, graph and analytic parameters, and finally a required discrepancy exponent in (5.4).
- (v) Choose S sufficiently large in Theorem 7.3 for that exponent, fix any prime-number-theorem accuracies needed for this S , and let x tend to infinity.

In particular the precision of the proxy cells causes no feedback into the Type II coefficient bound.

Subtract (7.31) from (7.22). For all sufficiently large x there remains a positive constant multiple of X_A/L on prime values of N_u :

$$\sum_{2u+1 \text{ prime}} A(u)\Psi(u/x) \gg X_A/L.$$

By Theorem 7.1 each summand is at most $\exp(O(\sqrt{L}))$ and $X_A \gg xL^{-C_A}$. The map $u \mapsto 2u+1$ is injective, so the number of distinct primes is at least

$$xL^{-C_A-1} \exp(-O(\sqrt{L})) = x^{1-o(1)}.$$

Their range is $2x < p \leq 4x+1 \leq 5x$. Finally $P^+(p-1) = P^+(2u) = \max(2, P^+(u)) \leq x^\delta$ for large x ; the strict inequality $c_2 s' < \delta$ in the candidate geometry leaves room for every subpower P prime, and the factor 2 is eventually below x^δ . This proves Theorem 1.2.

8 Large fibers of the totient function

We finish by deriving Theorem 1.1 from Theorem 1.2. This product-and-pigeonhole passage is the classical connection between smooth shifted primes and large totient fibers; see [19, Theorem B and its proof]. We first record finiteness of each fiber and the elementary upper bound mentioned in the Introduction.

Lemma 8.1. *For each positive integer n , the set of positive integers m with $\varphi(m) = n$ is finite. For every $\eta > 0$,*

$$g(n) \ll_\eta n^{1+\eta}.$$

Proof. If $p^a \parallel m$, then $p^{a-1}(p-1)$ divides $\varphi(m)$. For a fixed value n , this restricts p to the finite set with $p-1 \mid n$, and bounds a by $p^{a-1} \mid n$. Thus each fiber is finite.

For the quantitative bound, fix $0 < \theta < 1$. For all sufficiently large primes p , one has $1-1/p \geq p^{-\theta}$. The finitely many remaining primes contribute a fixed positive constant c_θ . Consequently, for every m ,

$$\varphi(m) = m \prod_{p \mid m} \left(1 - \frac{1}{p}\right) \geq c_\theta m \prod_{p \mid m} p^{-\theta} \geq c_\theta m^{1-\theta}.$$

Every preimage of n is therefore at most $(n/c_\theta)^{1/(1-\theta)}$. Choose θ so that $1/(1-\theta) \leq 1+\eta$, and count the possible positive integers. $\square$

Proof of Theorem 1.1. It suffices to consider $0 < \varepsilon < 1$. Choose $0 < \delta < 1/4$ with $4\delta < \varepsilon$. By Theorem 1.2, for every sufficiently large X there are more than $X^{1-\delta}$ primes $p \leq X$ such that $p-1$ is X^δ -smooth. Indeed, take $x = X/5$ in that theorem; its count $x^{1-o(1)}$ exceeds $X^{1-\delta}$ eventually, and $x^\delta \leq X^\delta$.

Let $\mathcal{P}$ be this set of primes, let $M = |\mathcal{P}|$, and put $k = \lfloor X^{2\delta} \rfloor$. Since $3\delta < 1$, we have $k \leq M$ for sufficiently large X . Different k -element subsets of $\mathcal{P}$ have different products by unique factorization. There are at least

$$\binom{M}{k} \geq \left(\frac{M}{k}\right)^k > X^{(1-3\delta)k} \quad (8.1)$$

such products. For completeness, the binomial inequality follows by writing $\binom{M}{k} = \prod_{j=0}^{k-1} (M-j)/(k-j)$ and observing that each factor is at least M/k .

For a squarefree product $m = \prod_{p \in \mathcal{Q}} p$ with $|\mathcal{Q}| = k$, multiplicativity of the totient gives

$$\varphi(m) = \prod_{p \in \mathcal{Q}} (p-1).$$

This value is X^δ -smooth and at most X^k . Every prime exponent in such a value is at most $k \log X / \log 2$, and there are at most X^δ primes available. Hence the total number of possible values is at most

$$\left(1 + \frac{k \log X}{\log 2}\right)^{X^\delta} = X^{o(k)}. \quad (8.2)$$

The last equality means that the logarithm of its left side, divided by $k \log X$, tends to zero: its numerator is $O_\delta(X^\delta \log X)$, whereas $k \asymp X^{2\delta}$.

By Equations (8.1) and (8.2), some $n \leq X^k$ therefore satisfies

$$g(n) \geq X^{(1-3\delta-o(1))k} > n^{1-\varepsilon}$$

for sufficiently large X . The strict final inequality follows from $3\delta + o(1) < \varepsilon$ and $1 - \varepsilon > 0$.

These lower bounds for $g(n)$ tend to infinity with X . By Theorem 8.1, the resulting values n cannot belong to a fixed finite set. They are therefore unbounded, which proves the required assertion for infinitely many n . If $\varepsilon \geq 1$, the assertion follows from any smaller positive choice of ε . $\square$

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